1.1 ./Link.Foundation.Links.Notation/Binary/ArityRange.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Diagnostics.CodeAnalysis;
using System.Globalization;

namespace Link.Foundation.Links.Notation.Binary;

/// <summary>
/// How many references the links of a <see cref="Section"/> hold:
/// <c>Min..=Max</c>, where a null <see cref="Max"/> means no upper bound.
/// </summary>
/// <param name="Min">Fewest references in a link, at least 1.</param>
/// <param name="Max">Most references in a link, null for no limit.</param>
public readonly record struct ArityRange(ulong Min, ulong? Max)
{
    /// <summary>The largest minimum arity a section shape can hold.</summary>
    public const ulong MaxMin = ulong.MaxValue >> 4;

    /// <summary>Links of exactly two references.</summary>
    public static ArityRange Doublets { get; } = Exactly(2);

    /// <summary>Links of exactly <paramref name="arity"/> references.</summary>
    public static ArityRange Exactly(ulong arity) => new(arity, arity);

    /// <summary>Links of at least <paramref name="min"/> references.</summary>
    public static ArityRange AtLeast(ulong min) => new(min, null);

    /// <summary>Links of <c>min..=max</c> references.</summary>
    public static ArityRange Between(ulong min, ulong max) => new(min, max);

    /// <summary>True when a link of <paramref name="length"/> references fits the
    ↪  range.</summary>
    public bool Contains(ulong length) => length >= Min && (Max is not { } max || length <= max);

    /// <summary>True when every link has the same number of references, so links need no length
    ↪  prefix.</summary>
    public bool IsFixed => Max == Min;

    /// <summary>0 for no maximum, otherwise <c>Max - Min</c>; only for variable
    ↪  arities.</summary>
    internal ulong Extra => Max is { } max ? max - Min : 0;

    /// <summary>Null for a usable range, otherwise why it is not.</summary>
    internal string? Problem()
    {
        if (Min == 0)
        {
            return "arity must be at least 1";
        }
        if (Min > MaxMin)
        {
            return $"arity {Min} is too large";
        }
        if (Max is { } max && max < Min)
        {
            return $"arity range {this} is empty";
        }
        return null;
    }

    /// <summary>The range a section header describes; <paramref name="extra"/> 0 means no
    ↪  maximum.</summary>
    internal static ArityRange FromShape(ulong min, ulong? extra)
    {
        ArityRange range;
        if (extra is not { } added)
        {
            range = Exactly(min);
        }
        else if (added == 0)
        {
            range = AtLeast(min);
        }
        else
        {
            if (min > ulong.MaxValue - added)
            {
```

```
                throw BinaryNotationException.Malformed("arity range overflows 64 bits");
            }
            range = Between(min, min + added);
        }
        if (range.Problem() is { } problem)
        {
            throw BinaryNotationException.Malformed(problem);
        }
        return range;
    }

    /// <summary>Formats the range as <c>n</c>, <c>min..max</c> or <c>min..</c>.</summary>
    public override string ToString() => Max switch
    {
        { } max when max == Min => max.ToString(CultureInfo.InvariantCulture),
        { } max => $"{Min}..{max}",
        null => $"{Min}..",
    };

    /// <summary>Parses <c>n</c>, <c>min..max</c> (inclusive) or <c>min..</c> (no maximum).</summary>
    /// <exception cref="FormatException">The text is not a usable arity range.</exception>
    public static ArityRange Parse(string text) =>
        TryParse(text, out var range, out var error) ? range : throw new FormatException(error);

    /// <summary>Parses <c>n</c>, <c>min..max</c> (inclusive) or <c>min..</c> (no maximum).</summary>
    public static bool TryParse([NotNullWhen(true)] string? text, out ArityRange range) =>
        TryParse(text, out range, out var ignored);

    /// <summary>Parses an arity range, explaining in <paramref name="error"/> why it failed.</summary>
    public static bool TryParse([NotNullWhen(true)] string? text, out ArityRange range,
        [NotNullWhen(false)] out string? error)
    {
        range = default;
        error = $"invalid arity '{text}': expected n, min..max or min..";
        if (text is null)
        {
            return false;
        }
        var separator = text.IndexOf("..", StringComparison.Ordinal);
        if (separator < 0)
        {
            if (!TryParseNumber(text, out var arity))
            {
                return false;
            }
            range = Exactly(arity);
        }
        else
        {
            var maxText = text[(separator + 2)..];
            if (!TryParseNumber(text[..separator], out var min))
            {
                return false;
            }
            if (maxText.Length == 0)
            {
                range = AtLeast(min);
            }
            else if (TryParseNumber(maxText, out var max))
            {
                range = Between(min, max);
            }
            else
            {
                return false;
            }
        }
        error = range.Problem();
        return error is null;
    }

    // Like Rust's u64::from_str: ASCII digits with an optional leading '+'.
    private static bool TryParseNumber(string text, out ulong value)
    {
        var digits = text.StartsWith('+') ? text[1..] : text;
        return ulong.TryParse(digits, NumberStyles.None, CultureInfo.InvariantCulture, out
            value);
```

```
    }
}
```

1.2 ./Link.Foundation.Links.Notation/Binary/BinaryLinoCodec.cs

```
using System.Collections.Generic;
using LinoLink = Link.Foundation.Links.Notation.Link<string>;

namespace Link.Foundation.Links.Notation.Binary;

/// <summary>Convenience codec for complete LiNo packets.</summary>
public sealed record BinaryLinoCodec
{
    public BinaryLinoCodec() { }

    public BinaryLinoCodec(BinaryLinoOptions options) => Options = options;

    /// <summary>Options used when writing.</summary>
    public BinaryLinoOptions Options { get; init; }

    /// <summary>Limits applied when reading and writing.</summary>
    public DecodeLimits Limits { get; init; } = DecodeLimits.Default;

    /// <summary>Encodes the supplied model without canonicalizing its groups.</summary>
    public byte[] Encode(IReadOnlyList<LinoLink> document) =>
        LinoMapping.EncodeDocument(document, Options, Limits).ToBytes();

    /// <summary>Decodes exactly one complete packet, rejecting trailing bytes.</summary>
    public IReadOnlyList<LinoLink> Decode(byte[] bytes) =>
        LinoMapping.DecodeDocument(LinksPacket.FromBytes(bytes, Limits), Limits);
}
```

1.3 ./Link.Foundation.Links.Notation/Binary/BinaryNotationException.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
namespace Link.Foundation.Links.Notation.Binary;

/// <summary>What went wrong while encoding, decoding or exchanging a LiNo message.</summary>
public enum BinaryErrorKind
{
    /// <summary>The underlying transport failed.</summary>
    Io,
    /// <summary>The peer sent bytes that are not a valid message.</summary>
    Malformed,
    /// <summary>The message text is not valid LiNo.</summary>
    InvalidLino,
    /// <summary>The message exceeds one of the configured <see cref="DecodeLimits"/>.</summary>
    LimitExceeded,
    /// <summary>The document cannot be represented with the chosen options.</summary>
    Unencodable,
}

/// <summary>Raised by the binary links notation codecs.</summary>
public sealed class BinaryNotationException : Exception
{
    public BinaryNotationException()
        : this(BinaryErrorKind.Malformed, "unknown protocol error")
    {
    }

    public BinaryNotationException(string message)
        : this(BinaryErrorKind.Malformed, message)
    {
    }

    public BinaryNotationException(string message, Exception innerException)
        : this(BinaryErrorKind.Io, message, innerException)
    {
    }

    public BinaryNotationException(BinaryErrorKind kind, string detail, Exception? innerException = null)
        : base($"{Describe(kind)}: {detail}", innerException)
    {
        Kind = kind;
        Detail = detail;
    }

    /// <summary>The category of the failure.</summary>
    public BinaryErrorKind Kind { get; }
```

```

    /// <summary>The message without the category prefix.</summary>
    public string Detail { get; }

    internal static BinaryNotationException Malformed(string detail) =>
    ↪  new(BinaryErrorKind.Malformed, detail);

    internal static BinaryNotationException Limit(string detail) =>
    ↪  new(BinaryErrorKind.LimitExceeded, detail);

    internal static BinaryNotationException Unencodable(string detail) =>
    ↪  new(BinaryErrorKind.Unencodable, detail);

    private static string Describe(BinaryErrorKind kind) => kind switch
    {
        BinaryErrorKind.Io => "I/O error",
        BinaryErrorKind.Malformed => "malformed message",
        BinaryErrorKind.InvalidLino => "invalid LiNo",
        BinaryErrorKind.LimitExceeded => "limit exceeded",
        BinaryErrorKind.Unencodable => "cannot encode",
          => kind.ToString(),
    };
}
```

1.4 ./Link.Foundation.Links.Notation/Binary/LinksPacket.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Buffers.Binary;

namespace Link.Foundation.Links.Notation.Binary;

/// <summary>One reference inside a <see cref="LinksPacket"/>.</summary>
/// <param name="IsExternal">True for an external value (a number or a code point).</param>
/// <param name="Value">The address inside the packet, or the external value.</param>
public readonly record struct PacketReference(bool IsExternal, ulong Value)
{
    /// <summary>The null reference.</summary>
    public static readonly PacketReference Null = Internal(LinksPacket.NullAddress);

    /// <summary>An address inside the packet (0 null, 1..5 markers, then links).</summary>
    public static PacketReference Internal(ulong address) => new(false, address);

    /// <summary>An external value, e.g. a number or a Unicode code point.</summary>
    public static PacketReference External(ulong value) => new(true, value);

    public override string ToString() => IsExternal ? $"External({Value})" :
    ↪  $"Internal({Value})";
}

/// <summary>Safety limits applied while decoding untrusted input.</summary>
public sealed record DecodeLimits
{
    /// <summary>Maximum number of links in a packet.</summary>
    public ulong MaxLinks { get; init; } = 1UL << 22;

    /// <summary>Maximum number of references in all links of a packet.</summary>
    public ulong MaxReferences { get; init; } = 1UL << 24;

    /// <summary>Maximum number of LiNo nodes a packet may expand to.</summary>
    public long MaxNodes { get; init; } = 1L << 22;

    /// <summary>Maximum total UTF-8 bytes in expanded references, including link ids.</summary>
    public long MaxStringBytes { get; init; } = 64L << 20;

    /// <summary>Maximum LiNo nesting depth.</summary>
    public int MaxDepth { get; init; } = 64;

    internal void Validate()
    {
        if (MaxNodes < 0 || MaxStringBytes < 0 || MaxDepth < 0)
            throw new ArgumentOutOfRangeException(nameof(DecodeLimits), "Limits must be
            ↪  non-negative.");
    }

    /// <summary>The default limits.</summary>
    public static DecodeLimits Default { get; } = new();

    /// <summary>Limits for trusted input such as a store archive: only the address space bounds
    ↪  the packet.</summary>
```

```
    public static DecodeLimits Unlimited { get; } = new()
    {
        MaxLinks = ulong.MaxValue,
        MaxReferences = ulong.MaxValue,
        MaxNodes = long.MaxValue,
        MaxStringBytes = long.MaxValue,
        MaxDepth = int.MaxValue,
    };
}

/// <summary>A run of links at consecutive addresses sharing an arity range and a reference width.</summary>
public sealed class Section
{
    /// <summary>Addresses skipped before the first link of the section.</summary>
    public ulong Gap { get; set; }

    /// <summary>The number of references each link may hold.</summary>
    public ArityRange Arity { get; set; } = ArityRange.Doublets;

    /// <summary>Bytes per reference: 1, 2, 4 or 8.</summary>
    public byte Width { get; set; } = 1;

    /// <summary>The links, in address order.</summary>
    public List<PacketReference[]> Links { get; } = new();

    internal void WriteHeader(List<byte> output)
    {
        var shape = LinksPacket.WidthCode(Width) | (Arity.Min << LinksPacket.ShapeMinAritityShift);
        if (Gap != 0)
        {
            shape |= LinksPacket.ShapeHasGap;
        }
        if (!Arity.IsFixed)
        {
            shape |= LinksPacket.ShapeVariableArity;
        }
        LinksPacket.WriteLeb128(output, shape);
        if (Gap != 0)
        {
            LinksPacket.WriteLeb128(output, Gap);
        }
        if (!Arity.IsFixed)
        {
            LinksPacket.WriteLeb128(output, Arity.Extra);
        }
        LinksPacket.WriteLeb128(output, (ulong)Links.Count);
    }

    /// <summary>Reads a section header, returning the still empty section and its link count.</summary>
    internal static (Section Section, ulong Count) ReadHeader(PacketReader reader)
    {
        var shape = LinksPacket.ReadLeb128(reader);
        var width = LinksPacket.WidthFromCode(shape & LinksPacket.ShapeWidthBits);
        var gap = (shape & LinksPacket.ShapeHasGap) != 0 ? LinksPacket.ReadLeb128(reader) : 0;
        ulong? extra = (shape & LinksPacket.ShapeVariableArity) != 0 ? LinksPacket.ReadLeb128(reader) : null;
        var arity = ArityRange.FromShape(shape >> LinksPacket.ShapeMinAritityShift, extra);
        var count = LinksPacket.ReadLeb128(reader);
        return (new Section { Gap = gap, Arity = arity, Width = width }, count);
    }

    internal void Validate()
    {
        LinksPacket.WidthCode(Width);
        if (Arity.Problem() is { } problem)
        {
            throw BinaryNotationException.Unencodable(problem);
        }
        foreach (var link in Links)
        {
            if (!Arity.Contains((ulong)link.Length))
            {
                throw BinaryNotationException.Unencodable(
                    $"a link of {link.Length} references in a section of arity {Arity}");
            }
        }
    }
}
```

```
/// <summary>
/// Binary links notation: a self-delimiting packet of links.
/// </summary>
/// <remarks>
/// A packet stores links, each a tuple of one or more references, at
/// implicit consecutive addresses. It knows nothing about LiNo; the LiNo
/// protocol (<see cref="LinoMapping"/>) is one use of it.
/// <code>
/// byte 0     0x10 | flags      high nibble 1 = format version 1
///                              bit 0     external references (Hybrid encoding)
///                              bit 1     explicit layout
///                              bits 2-3  log2 of the width (compact layout only)
///
/// compact layout (bit 1 clear): one section of doublets right after the markers
/// LEB128     N                 number of links, each `source target`
///
/// explicit layout (bit 1 set):
/// LEB128     S                 number of sections, then S section headers:
/// LEB128     shape             bits 0-1  log2 of the reference width in bytes
///                              bit 2     a gap follows
///                              bit 3     variable arity (else every link
///                                        holds exactly min arity references)
///                              bits 4+   min arity, at least 1
/// LEB128     gap               addresses skipped before the section (if bit 2)
/// LEB128     extra arity       0 = no maximum, else max - min (if bit 3)
/// LEB128     count             number of links in the section
///
/// links, section by section; a link in a variable-arity section starts
/// with LEB128 (length - min arity); every reference is `width` bytes,
/// little-endian
/// </code>
/// Address 0 is null. The first section starts at <c>1 + gap</c> and every
/// other section at <c>previous end + gap</c>, so gaps leave holes. The
/// compact layout is exactly one section with gap 5 (the LiNo marker points
/// 1..5), arity 2 and the header width. Each section has its own width: the
/// narrowest of 1, 2, 4 and 8 bytes that holds every reference in it;
/// <see cref="Pack"/> chooses the sections. With external references
/// enabled the top bit of a reference marks it as external, exactly like
/// <c>Platform.Data.Hybrid&lt;T&gt;</c>, which halves every internal range.
/// The byte layout is identical to the Rust <c>links notation::binary::LinksPacket</c>.
/// </remarks>
public sealed class LinksPacket
{
    /// <summary>The high nibble of the header byte; text never starts with 0x10..0x1F.</summary>
    public const byte BinaryVersion1 = 0x10;

    /// <summary>Null link address.</summary>
    public const ulong NullAddress = 0;
    /// <summary>Marker 1: the unary one; 2^k = (2^(k-1) 2^(k-1)).</summary>
    public const ulong One = 1;
    /// <summary>Marker 2: (Number unary) is a non-negative integer.</summary>
    public const ulong Number = 2;
    /// <summary>Marker 3: (String code points...) is a Unicode string.</summary>
    public const ulong String = 3;
    /// <summary>Marker 4: (List elements...) is a list of links.</summary>
    public const ulong List = 4;
    /// <summary>Marker 5: (Identified id values...) is a link with an id.</summary>
    public const ulong Identified = 5;
    /// <summary>Address of the first link after the marker points.</summary>
    public const ulong FirstLinkAddress = 6;

    private const byte FlagExternalReferences = 0b0001;
    private const byte FlagExplicitLayout = 0b0010;
    private const int WidthShift = 2;
    private const byte WidthBits = 0b1100;

    internal const ulong ShapeWidthBits = 0b0011;
    internal const ulong ShapeHasGap = 0b0100;
    internal const ulong ShapeVariableArity = 0b1000;
    internal const int ShapeMinAritityShift = 4;

    /// <summary>The addresses the compact layout skips: the marker points.</summary>
    private const ulong CompactGap = FirstLinkAddress - 1;

    private static readonly byte[] WidthValues = { 1, 2, 4, 8 };

    /// <summary>The reference widths, in bytes, that a packet may use.</summary>
    public static IReadOnlyList<byte> Widths => WidthValues;

    /// <summary>An empty packet.</summary>
```

```
    public LinksPacket(bool externalReferences = false) => ExternalReferences = externalReferences;

    /// <summary>Header bit 0: references may be external (Hybrid encoding).</summary>
    public bool ExternalReferences { get; set; }

    /// <summary>The links, section by section.</summary>
    public List<Section> Sections { get; } = new();

    /// <summary>Largest internal address that fits in <paramref name="width"/> bytes.</summary>
    public static ulong InternalCapacity(byte width, bool externalReferences)
    {
        var bits = width * 8 - (externalReferences ? 1 : 0);
        return bits >= 64 ? ulong.MaxValue : (1UL << bits) - 1;
    }

    /// <summary>Largest external value that fits in <paramref name="width"/> bytes.</summary>
    public static ulong ExternalCapacity(byte width) => (1UL << (width * 8 - 1)) - 1;

    /// <summary>The narrowest width able to hold the internal address.</summary>
    public static byte AddressTier(ulong address, bool externalReferences)
    {
        foreach (var width in WidthValues)
        {
            if (InternalCapacity(width, externalReferences) >= address)
            {
                return width;
            }
        }
        throw BinaryNotationException.Unencodable($"address {address} exceeds the internal range");
    }

    private static ulong WidthMask(byte width) => InternalCapacity(width, false);

    /// <summary>Encodes an external value at <paramref name="width"/> the way Hybrid&lt;T&gt; does.</summary>
    public static ulong EncodeExternal(ulong value, byte width) =>
        value == 0 ? 1UL << (width * 8 - 1) : (0UL - value) & WidthMask(width);

    /// <summary>Decodes a raw value; returns true and the value for externals.</summary>
    public static bool TryDecodeExternal(ulong raw, byte width, out ulong value)
    {
        var externalZero = 1UL << (width * 8 - 1);
        value = 0;
        if (raw == externalZero)
        {
            return true;
        }
        if (raw > externalZero)
        {
            value = (0UL - raw) & WidthMask(width);
            return true;
        }
        return false;
    }

    internal static ulong WidthCode(byte width)
    {
        var code = Array.IndexOf(WidthValues, width);
        return code >= 0 ? (ulong)code : throw BinaryNotationException.Unencodable($"invalid width {width}");
    }

    /// <summary>The width of a two-bit width code; every code names a width.</summary>
    internal static byte WidthFromCode(ulong code) => WidthValues[code & 0b11];

    /// <summary>The narrowest width able to hold <paramref name="reference"/>.</summary>
    public static byte ReferenceWidth(PacketReference reference, bool externalReferences)
    {
        if (!reference.IsExternal)
        {
            return AddressTier(reference.Value, externalReferences);
        }
        if (!externalReferences)
        {
            throw BinaryNotationException.Unencodable("external reference in a packet without external references");
        }
        foreach (var width in WidthValues)
```

```
        {
            if (ExternalCapacity(width) >= reference.Value)
            {
                return width;
            }
        }
        throw BinaryNotationException.Unencodable($"external value {reference.Value} exceeds 63 bits");
    }

    /// <summary>
    /// Lays out <paramref name="links"/> - addresses with their references, in
    /// ascending address order - using the version 1 section planner.
    /// </summary>
    /// <remarks>
    /// Holes between addresses start new sections. Without
    /// <paramref name="packedWidths"/> every section uses the width of the
    /// widest reference, so all references have the same size. With it each
    /// section gets the narrowest width its links need and sections split
    /// wherever that saves bytes; the result is never larger than the uniform one.
    /// </remarks>
    public static LinksPacket Pack(
        bool externalReferences,
        IReadOnlyList<(ulong Address, PacketReference[] References)> links,
        bool packedWidths)
    {
        ArgumentNullException.ThrowIfNull(links);
        var planner = new SectionPlanner(externalReferences, links);
        var uniformLayout = planner.Plan(false);
        var uniform = LayOut(externalReferences, links, uniformLayout);
        if (!packedWidths)
        {
            return uniform;
        }
        var packedLayout = planner.Plan(true);
        if (packedLayout.SequenceEqual(uniformLayout))
        {
            return uniform;
        }
        var packed = LayOut(externalReferences, links, packedLayout);
        return packed.ToBytes().Length < uniform.ToBytes().Length ? packed : uniform;
    }

    /// <summary>Splits <paramref name="links"/> into sections of <c>(link count, width)</c>.</summary>
    private static LinksPacket LayOut(
        bool externalReferences,
        IReadOnlyList<(ulong Address, PacketReference[] References)> links,
        IReadOnlyList<(int Count, byte Width)> layout)
    {
        var packet = new LinksPacket(externalReferences);
        var nextAddress = 1UL;
        var first = 0;
        foreach (var (count, width) in layout)
        {
            var start = links[first].Address;
            var shortest = ulong.MaxValue;
            var longest = 0UL;
            var section = new Section { Gap = start - nextAddress, Width = width };
            for (var index = first; index < first + count; index++)
            {
                var length = (ulong)links[index].References.Length;
                shortest = Math.Min(shortest, length);
                longest = Math.Max(longest, length);
                section.Links.Add(links[index].References);
            }
            section.Arity = ArityRange.Between(shortest, longest);
            packet.Sections.Add(section);
            nextAddress = start + (ulong)count;
            first += count;
        }
        return packet;
    }

    /// <summary>Every link with its address, in address order.</summary>
    public IEnumerable<(ulong Address, PacketReference[] References)> Links()
    {
        var nextAddress = 1UL;
        foreach (var section in Sections)
        {
            var start = SaturatingAdd(nextAddress, section.Gap);
```

```
            nextAddress = SaturatingAdd(start, (ulong)section.Links.Count);
            for (var index = 0; index < section.Links.Count; index++)
            {
                yield return (start + (ulong)index, section.Links[index]);
            }
        }
    }

    private static ulong SaturatingAdd(ulong left, ulong right) =>
        left > ulong.MaxValue - right ? ulong.MaxValue : left + right;

    /// <summary>The number of links in the packet.</summary>
    public ulong LinkCount => Sections.Aggregate(0UL, (total, section) => total +
    ↪  (ulong)section.Links.Count);

    private bool IsCompact => Sections.Count switch
    {
        0 => true,
        1 => Sections[0].Gap == CompactGap && Sections[0].Arity == ArityRange.Doublets &&
        ↪  Sections[0].Links.Count > 0,
          => false,
    };

    internal void Validate()
    {
        var nextAddress = 1UL;
        foreach (var section in Sections)
        {
            section.Validate();
            var count = (ulong)section.Links.Count;
            if (section.Gap > ulong.MaxValue - nextAddress || count > ulong.MaxValue -
            ↪  nextAddress - section.Gap)
            {
                throw BinaryNotationException.Unencodable("addresses overflow 64 bits");
            }
            nextAddress += section.Gap + count;
            foreach (var link in section.Links)
            {
                foreach (var reference in link)
                {
                    RawReference(reference, section.Width);
                }
            }
        }
    }

    /// <summary>Serializes the packet.</summary>
    public byte[] ToBytes()
    {
        Validate();
        var header = BinaryVersion1;
        if (ExternalReferences)
        {
            header |= FlagExternalReferences;
        }
        var output = new List<byte>();
        if (IsCompact)
        {
            var width = Sections.Count > 0 ? Sections[0].Width : (byte)1;
            output.Add((byte)(header | (WidthCode(width) << WidthShift)));
            WriteLeb128(output, LinkCount);
        }
        else
        {
            output.Add((byte)(header | FlagExplicitLayout));
            WriteLeb128(output, (ulong)Sections.Count);
            foreach (var section in Sections)
            {
                section.WriteHeader(output);
            }
        }
        foreach (var section in Sections)
        {
            foreach (var link in section.Links)
            {
                if (!section.Arity.IsFixed)
                {
                    WriteLeb128(output, (ulong)link.Length - section.Arity.Min);
                }
                foreach (var reference in link)
```

```
                {
                    WriteRaw(output, RawReference(reference, section.Width), section.Width);
                }
            }
        }
        return output.ToArray();
    }

    private ulong RawReference(PacketReference reference, byte width)
    {
        if (ReferenceWidth(reference, ExternalReferences) > width)
        {
            throw BinaryNotationException.Unencodable($"{reference} does not fit {width} byte(s)");
        }
        return reference.IsExternal ? EncodeExternal(reference.Value, width) : reference.Value;
    }

    /// <summary>Parses a complete packet; trailing bytes are an error.</summary>
    public static LinksPacket FromBytes(byte[] bytes, DecodeLimits? limits = null)
    {
        var reader = PacketReader.FromBytes(bytes);
        var packet = ReadFrom(reader, limits) ?? throw BinaryNotationException.Malformed("empty input");
        if (!reader.AtEnd)
        {
            throw BinaryNotationException.Malformed("trailing bytes after the packet");
        }
        return packet;
    }

    /// <summary>Reads one packet; null on a clean end of stream.</summary>
    public static LinksPacket? ReadFrom(PacketReader reader, DecodeLimits? limits = null)
    {
        ArgumentNullException.ThrowIfNull(reader);
        limits ??= DecodeLimits.Default;
        limits.Validate();
        var first = reader.ReadByte();
        if (first < 0)
        {
            return null;
        }
        var header = (byte)first;
        if ((header & 0xF0) != BinaryVersion1)
        {
            throw BinaryNotationException.Malformed($"unsupported binary header byte 0x{header:X2}");
        }
        var packet = new LinksPacket((header & FlagExternalReferences) != 0);
        var counts = new List<ulong>();
        if ((header & FlagExplicitLayout) == 0)
        {
            var width = WidthFromCode((ulong)((header & WidthBits) >> WidthShift));
            var count = ReadLeb128(reader);
            if (count > ulong.MaxValue - FirstLinkAddress)
            {
                throw BinaryNotationException.Malformed("addresses overflow 64 bits");
            }
            if (count > 0)
            {
                packet.Sections.Add(new Section { Gap = CompactGap, Arity = ArityRange.Doublets, Width = width });
                counts.Add(count);
            }
        }
        else
        {
            if ((header & WidthBits) != 0)
            {
                throw BinaryNotationException.Malformed("the explicit layout keeps the header width bits clear");
            }
            var sectionCount = ReadLeb128(reader);
            if (sectionCount > limits.MaxLinks)
            {
                throw TooManyLinks(limits);
            }
            var nextAddress = 1UL;
            for (ulong index = 0; index < sectionCount; index++)
            {
```

```
                var (section, count) = Section.ReadHeader(reader);
                if (section.Gap > ulong.MaxValue - nextAddress || count > ulong.MaxValue -
                ↪  nextAddress - section.Gap)
                {
                    throw BinaryNotationException.Malformed("addresses overflow 64 bits");
                }
                nextAddress += section.Gap + count;
                packet.Sections.Add(section);
                counts.Add(count);
            }
        }
        var total = 0UL;
        foreach (var count in counts)
        {
            if (count > limits.MaxLinks - total)
            {
                throw TooManyLinks(limits);
            }
            total += count;
        }
        var referencesLeft = limits.MaxReferences;
        for (var index = 0; index < packet.Sections.Count; index++)
        {
            var section = packet.Sections[index];
            section.Links.Capacity = (int)Math.Min(counts[index], 4096);
            for (ulong link = 0; link < counts[index]; link++)
            {
                var length = section.Arity.Min;
                if (!section.Arity.IsFixed)
                {
                    var extra = ReadLeb128(reader);
                    if (extra > ulong.MaxValue - length || !section.Arity.Contains(extra +
                    ↪  length))
                    {
                        throw BinaryNotationException.Malformed($"link length outside the section
                        ↪  arity {section.Arity}");
                    }
                    length += extra;
                }
                if (length > referencesLeft)
                {
                    throw BinaryNotationException.Limit($"references exceed the limit of
                    ↪  {limits.MaxReferences}");
                }
                referencesLeft -= length;
                var references = new List<PacketReference>((int)Math.Min(length, 4096));
                for (ulong item = 0; item < length; item++)
                {
                    var raw = ReadRaw(reader, section.Width);
                    references.Add(packet.ExternalReferences && TryDecodeExternal(raw,
                    ↪  section.Width, out var value)
                        ? PacketReference.External(value)
                        : PacketReference.Internal(raw));
                }
                section.Links.Add(references.ToArray());
            }
        }
        return packet;
    }

    private static BinaryNotationException TooManyLinks(DecodeLimits limits) =>
        BinaryNotationException.Limit($"packet declares more than {limits.MaxLinks} links");

    private static void WriteRaw(List<byte> output, ulong value, byte width)
    {
        for (var index = 0; index < width; index++)
        {
            output.Add((byte)(value >> (8 * index)));
        }
    }

    private static ulong ReadRaw(PacketReader reader, byte width)
    {
        Span<byte> bytes = stackalloc byte[8];
        bytes.Clear();
        reader.ReadExactly(bytes[..width]);
        return BinaryPrimitives.ReadUInt64LittleEndian(bytes);
    }

    /// <summary>Appends <paramref name="value"/> as unsigned LEB128.</summary>
```

```
    public static void WriteLeb128(List<byte> output, ulong value)
    {
        ArgumentNullException.ThrowIfNull(output);
        while (true)
        {
            var current = (byte)(value & 0x7F);
            value >>= 7;
            if (value == 0)
            {
                output.Add(current);
                return;
            }
            output.Add((byte)(current | 0x80));
        }
    }

    /// <summary>Reads an unsigned LEB128 value of at most 64 bits.</summary>
    public static ulong ReadLeb128(PacketReader reader)
    {
        ArgumentNullException.ThrowIfNull(reader);
        ulong value = 0;
        for (var shift = 0; shift < 64; shift += 7)
        {
            var next = reader.ReadByte();
            if (next < 0)
            {
                throw BinaryNotationException.Malformed("unexpected end of packet");
            }
            var payload = (ulong)(next & 0x7F);
            if (shift == 63 && payload > 1)
            {
                throw BinaryNotationException.Malformed("LEB128 value overflows 64 bits");
            }
            value |= payload << shift;
            if ((next & 0x80) == 0)
            {
                return value;
            }
        }
        throw BinaryNotationException.Malformed("LEB128 value overflows 64 bits");
    }
}
```

1.5 ./Link.Foundation.Links.Notation/Binary/LinoFormat.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using Link.Foundation.Links.Notation;

using LinoLink = Link.Foundation.Links.Notation.Link<string>;

namespace Link.Foundation.Links.Notation.Binary;

/// <summary>
/// Canonical LiNo text for documents, shared by every protocol.
/// </summary>
/// <remarks>
/// A document is the list of top-level links of a message. A reference is a
/// <see cref="LinoLink"/> whose <c>Values</c> is null. links-notation keeps
/// one extra wrapper around single-reference groups (<c>a</c> parses as
/// <c>(a)</c>); <see cref="ParseDocument"/> removes it, exactly like the Rust
/// port, so documents decoded from the same bytes compare equal in both
/// ports. Formatting preserves the canonical model of parsed documents.
/// Manually constructed ids without values cannot round trip through text.
/// </remarks>
public static class LinoFormat
{

    /// <summary>A reference: a link without values.</summary>
    public static LinoLink Reference(string text) => new(text);

    /// <summary>A link with an optional id and the given values.</summary>
    public static LinoLink Link(string? id, IList<LinoLink> values) => new(id!, values);

    /// <summary>Parses LiNo text into a canonical document. Blank input is the empty
    ↪  document.</summary>
    public static IReadOnlyList<LinoLink> ParseDocument(string text)
    {
        ArgumentNullException.ThrowIfNull(text);
        if (text.All(character => character is ' ' or '\t' or '\r' or '\n'))
```

```
        {
            return Array.Empty<LinoLink>();
        }
        IList<LinoLink> parsed;
        try
        {
            parsed = new Parser().Parse(text);
        }
        catch (Exception exception) when (exception is not OutOfMemoryException)
        {
            throw new BinaryNotationException(BinaryErrorKind.InvalidLino, exception.Message,
            ↪  exception);
        }
        return parsed.Select(Canonical).ToList();
    }

    /// <summary>
    /// Converts a link produced by the C# parser into the canonical model:
    /// an unnamed group holding exactly one reference is that reference.
    /// </summary>
    public static LinoLink Canonical(LinoLink link)
    {
        if (link.Values is null)
        {
            return link;
        }
        if (link.Id is null && link.Values.Count == 1 && link.Values[0].Values is null)
        {
            return link.Values[0];
        }
        return Link(link.Id, link.Values.Select(Canonical).ToList());
    }

    /// <summary>
    /// Formats a document as canonical LiNo text, one top-level link per line.
    /// A top-level link without an id and with at least two values is written
    /// without its outer parentheses: <c>() ((1 1))</c>.
    /// </summary>
    public static string FormatDocument(IEnumerable<LinoLink> document)
    {
        ArgumentNullException.ThrowIfNull(document);
        return string.Join("\n", document.Select(FormatTopLevel));
    }

    private static string FormatTopLevel(LinoLink link) =>
        link.Values is { Count: >= 2 } values && link.Id is null ? JoinValues(values) :
        ↪  FormatLink(link);

    /// <summary>Formats one link as it appears nested inside another link.</summary>
    public static string FormatLink(LinoLink link)
    {
        if (link.Values is not { } values)
        {
            return FormatReference(link.Id ?? string.Empty);
        }
        if (link.Id is null)
        {
            // `(a)` parses back as the reference `a`, so a one-reference
            // link needs a second pair of parentheses.
            return values.Count == 1 && values[0].Values is null
                ? $"(({FormatReference(values[0].Id ?? string.Empty)}))"
                : $"({JoinValues(values)})";
        }
        return values.Count == 0
            ? $"({FormatReference(link.Id)}:)"
            : $"({FormatReference(link.Id)}: {JoinValues(values)})";
    }

    private static string JoinValues(IEnumerable<LinoLink> values) => string.Join(" ",
    ↪  values.Select(FormatLink));

    /// <summary>Quotes a reference when it would not survive parsing as a bare word.</summary>
    /// <remarks>
    /// links-notation opens a quoted reference with a run of N equal quote
    /// characters, closes it with the next run of exactly N, and reads 2N
    /// quotes inside as N literal ones. The opening run is counted greedily,
    /// so the chosen quote must differ from the first character; N is one
    /// more than the longest run of that quote inside, and odd, because an
    /// even delimiter run may be read as an empty reference.
    /// </remarks>
```

```
    public static string FormatReference(string reference) => ReferenceLiteral.Format(reference);
}
```

1.6 ./Link.Foundation.Links.Notation/Binary/LinoMapping.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Globalization;
using System.Text;

using LinoLink = Link.Foundation.Links.Notation.Link<string>;

namespace Link.Foundation.Links.Notation.Binary;

/// <summary>
/// Optional features of the binary LiNo protocol. Every feature is off by
/// default; each one can be switched on independently, like stacking a decorator.
/// </summary>
public readonly record struct BinaryLinoOptions
{
    // Doublets are stored as null so that default(BinaryLinoOptions) means doublets.
    private readonly ArityRange?  arity;

    /// <summary>
    /// Send numbers and code points as Hybrid external references instead of
    /// in-band unary links. Halves the internal address range of each width.
    /// </summary>
    public bool ExternalReferences { get; init; }

    /// <summary>
    /// The link lengths the encoder may use. The default, exactly 2, sends
    /// only doublets; <c>2..3</c> adds triplets and <c>1..</c> any length.
    /// The range must include 2.
    /// </summary>
    public ArityRange Arity
    {
        get =>  arity ?? ArityRange.Doublets;
        init =>  arity = value == ArityRange.Doublets ? null : value;
    }

    /// <summary>
    /// Give every section of the packet the narrowest width its links need
    /// instead of one width for the whole packet.
    /// </summary>
    public bool PackedWidths { get; init; }

    /// <summary>Enables or disables external references.</summary>
    public BinaryLinoOptions WithExternalReferences(bool enabled = true) => this with {
    ↪  ExternalReferences = enabled };

    /// <summary>Sets the link lengths the encoder may use.</summary>
    public BinaryLinoOptions WithArity(ArityRange arity) => this with { Arity = arity };

    /// <summary>Enables or disables packed widths.</summary>
    public BinaryLinoOptions WithPackedWidths(bool enabled = true) => this with { PackedWidths =
    ↪  enabled };

    /// <summary>The options a peer most likely used to write <paramref name="packet"/>, so a
    ↪  reply can be written in the same style.</summary>
    public static BinaryLinoOptions OfPacket(LinksPacket packet)
    {
        ArgumentNullException.ThrowIfNull(packet);
        ulong shortest = 2;
        ulong longest = 2;
        foreach (var ( , link) in packet.Links())
        {
            shortest = Math.Min(shortest, (ulong)link.Length);
            longest = Math.Max(longest, (ulong)link.Length);
        }
        return new BinaryLinoOptions
        {
            ExternalReferences = packet.ExternalReferences,
            Arity = ArityRange.Between(shortest, longest),
            PackedWidths = packet.Sections.Select(section =>
            ↪  section.Width).Distinct().Skip(1).Any(),
        };
    }

    /// <summary>Lists every option, for test and error messages.</summary>
    public override string ToString() =>
```

```
        $"BinaryLinoOptions {{ ExternalReferences = {ExternalReferences}, Arity = {Arity},
        ↪  PackedWidths = {PackedWidths} }}";
}

/// <summary>
/// Lossless mapping between LiNo documents and <see cref="LinksPacket"/>s,
/// byte-for-byte compatible with the Rust port.
/// </summary>
/// <remarks>
/// <list type="bullet">
/// <item><c>()</c> is the null link 0.</item>
/// <item>A numeric reference n is <c>(Number unary(n))</c>, or an external reference when
↪  enabled.</item>
/// <item>Any other reference is <c>(String code points...)</c>; code points are unary numbers or
↪  externals.</item>
/// <item>A link without an id and with exactly two values is a plain doublet.</item>
/// <item>A link without an id and any other number of values is a list.</item>
/// <item>A link with an id is <c>(Identified id values...)</c>.</item>
/// <item>The document is a list of its top-level links, stored last (the root).</item>
/// </list>
/// A list or typed value with any number of values other than two is a
/// single link of that many references when <see cref="BinaryLinoOptions.Arity"/>
/// allows it (<c>[marker, elements...]</c> for typed values, the bare elements
/// for lists), and otherwise the doublet <c>(marker chain)</c>, where chain is
/// the nil-terminated cons list <c>(e1 (e2 (... (en 0))))</c>. Identical
/// sub-links are emitted once and shared. Links are numbered so that every
/// link only refers to earlier ones: doublets of plain doublets first, then
/// the rest in creation order.
/// </remarks>
public static class LinoMapping
{
    /// <summary>Converts a document into a packet.</summary>
    public static LinksPacket EncodeDocument(IReadOnlyList<LinoLink> document, BinaryLinoOptions
    ↪  options = default, DecodeLimits? limits = null)
    {
        ArgumentNullException.ThrowIfNull(document);
        if (options.Arity.Problem() is { } problem)
        {
            throw BinaryNotationException.Unencodable(problem);
        }
        if (!options.Arity.Contains(2))
        {
            throw BinaryNotationException.Unencodable($"arity {options.Arity} does not include
            ↪  doublets (2)");
        }
        // Check the model iteratively before entering the recursive encoder.
        limits ??= DecodeLimits.Default;
        limits.Validate();
        var pending = new Stack<(LinoLink Link, int Depth)>(document.Select(link => (link, 0)));
        var budget = limits.MaxNodes;
        var stringsLeft = limits.MaxStringBytes;
        while (pending.TryPop(out var item))
        {
            if (item.Depth >= limits.MaxDepth)
            {
                throw BinaryNotationException.Unencodable($"nesting deeper than
                ↪  {limits.MaxDepth}");
            }
            if (--budget < 0)
            {
                throw BinaryNotationException.Unencodable("too many LiNo nodes");
            }
            if (item.Link.Id is { } text)
            {
                if (item.Link.Values is not null && --budget < 0)
                    throw BinaryNotationException.Unencodable("too many LiNo nodes");
                var bytes = Encoding.UTF8.GetByteCount(text);
                if (bytes > stringsLeft) throw BinaryNotationException.Unencodable("too many
                ↪  string bytes");
                stringsLeft -= bytes;
            }
            if (item.Link.Values is { } children)
            {
                foreach (var child in children)
                {
                    pending.Push((child, item.Depth + 1));
                }
            }
        }
```

```
        var encoder = new Encoder(options);
        if (document.Count > 0)
        {
            encoder.List(document.Select(encoder.Encode).ToList());
        }
        var packet = encoder.Finish();
        if ((ulong)packet.Sections.Count > limits.MaxLinks || packet.LinkCount > limits.MaxLinks)
            throw BinaryNotationException.Unencodable("too many packet links");
        var referencesLeft = limits.MaxReferences;
        foreach (var ( , link) in packet.Links())
        {
            if ((ulong)link.Length > referencesLeft) throw
            ↪  BinaryNotationException.Unencodable("too many packet references");
            referencesLeft -= (ulong)link.Length;
        }
        return packet;
    }

    /// <summary>Converts a packet back into a document.</summary>
    public static IReadOnlyList<LinoLink> DecodeDocument(LinksPacket packet, DecodeLimits?
    ↪  limits = null)
    {
        ArgumentNullException.ThrowIfNull(packet);
        limits ??= DecodeLimits.Default;
        limits.Validate();
        var decoder = new Decoder(packet, limits);
        if (decoder.LinkCount == 0)
        {
            return Array.Empty<LinoLink>();
        }
        var root = decoder.View(PacketReference.Internal(LinksPacket.FirstLinkAddress +
        ↪  (ulong)decoder.LinkCount - 1));
        IReadOnlyList<PacketReference> items = root switch
        {
            { Kind: ViewKind.Link, Items: [{ IsExternal: false, Value: LinksPacket.List }, var
            ↪  chain] } => decoder.Chain(chain),
            { Kind: ViewKind.Link } when !StartsWithMarker(root.Items!) => root.Items!,
              => throw BinaryNotationException.Malformed("the root link is not a list"),
        };
        var budget = limits.MaxNodes;
        return items.Select(item => decoder.Decode(item, 0, ref budget)).ToList();
    }

    /// <summary>Parses a canonical unsigned decimal number (no sign, no leading
    ↪  zeros).</summary>
    public static bool TryParseCanonicalNumber(string text, out ulong value)
    {
        value = 0;
        var canonical = !string.IsNullOrEmpty(text)
            && text.All(character => character is >= '0' and <= '9')
            && (text == "0" || text[0] != '0');
        return canonical && ulong.TryParse(text, NumberStyles.None,
        ↪  CultureInfo.InvariantCulture, out value);
    }

    private static bool IsMarker(PacketReference reference) =>
        !reference.IsExternal && reference.Value is >= LinksPacket.One and <
        ↪  LinksPacket.FirstLinkAddress;

    private static bool StartsWithMarker(IReadOnlyList<PacketReference> items) => items.Count >
    ↪  0 && IsMarker(items[0]);

    private enum NodeKind
    {
        Internal,
        External,
        /// <summary>A doublet of plain doublets, markers and externals.</summary>
        Doublet,
        /// <summary>Any other link; numbered after every doublet.</summary>
        Tuple,
    }

    private readonly record struct Node(NodeKind Kind, ulong Value)
    {
        public static Node Internal(ulong address) => new(NodeKind.Internal, address);

        public static Node External(ulong value) => new(NodeKind.External, value);
    }

    private sealed class NodeListComparer : IEqualityComparer<List<Node>>
```

```
    {
        public static readonly NodeListComparer Instance = new();

        public bool Equals(List<Node>? x, List<Node>? y) =>
            ReferenceEquals(x, y) || (x is not null && y is not null && x.SequenceEqual(y));

        public int GetHashCode(List<Node> obj)
        {
            var hash = new HashCode();
            foreach (var node in obj)
            {
                hash.Add(node);
            }
            return hash.ToHashCode();
        }
    }

    private sealed class Encoder
    {
        private readonly BinaryLinoOptions  options;
        private readonly List<List<Node>>  doublets = new();
        private readonly List<List<Node>>  tuples = new();
        private readonly Dictionary<List<Node>, Node>  created = new(NodeListComparer.Instance);
        private readonly List<Node>  powers = new() { Node.Internal(LinksPacket.One) };

        public Encoder(BinaryLinoOptions options) =>  options = options;

        private Node Link(List<Node> items)
        {
            if ( created.TryGetValue(items, out var existing))
            {
                return existing;
            }
            Node node;
            if (items.Count == 2 && items.All(item => item.Kind != NodeKind.Tuple))
            {
                 doublets.Add(items);
                node = new Node(NodeKind.Doublet, (ulong) doublets.Count - 1);
            }
            else
            {
                 tuples.Add(items);
                node = new Node(NodeKind.Tuple, (ulong) tuples.Count - 1);
            }
             created[items] = node;
            return node;
        }

        private Node Pair(Node first, Node second) => Link(new List<Node> { first, second });

        private Node Chain(IReadOnlyList<Node> items)
        {
            var tail = Node.Internal(LinksPacket.NullAddress);
            for (var index = items.Count - 1; index >= 0; index--)
            {
                tail = Pair(items[index], tail);
            }
            return tail;
        }

        // One link of `items` when the arity allows it, except that two items
        // are always a doublet.
        private bool FitsOneLink(int items) => items != 2 &&
        ↪   options.Arity.Contains((ulong)items);

        private Node Typed(ulong marker, List<Node> elements)
        {
            if (FitsOneLink(elements.Count + 1))
            {
                elements.Insert(0, Node.Internal(marker));
                return Link(elements);
            }
            return Pair(Node.Internal(marker), Chain(elements));
        }

        public Node List(List<Node> elements) => elements.Count switch
        {
            0 => Node.Internal(LinksPacket.NullAddress),
            2 => Pair(elements[0], elements[1]),
              when FitsOneLink(elements.Count) => Link(elements),
              => Typed(LinksPacket.List, elements),
```

```
        };

        private Node Power(int exponent)
        {
            while ( powers.Count <= exponent)
            {
                var previous =  powers[^1];
                 powers.Add(Pair(previous, previous));
            }
            return  powers[exponent];
        }

        private Node Unary(ulong value)
        {
            var powers = new List<Node>();
            for (var bit = 63; bit >= 0; bit--)
            {
                if ((value & (1UL << bit)) != 0)
                {
                    powers.Add(Power(bit));
                }
            }
            if (powers.Count == 0)
            {
                return Node.Internal(LinksPacket.NullAddress);
            }
            var sum = powers[^1];
            for (var index = powers.Count - 2; index >= 0; index--)
            {
                sum = Pair(powers[index], sum);
            }
            return sum;
        }

        private bool IsExternal(ulong value) =>  options.ExternalReferences && value <=
        ↪  LinksPacket.ExternalCapacity(8);

        private Node Scalar(ulong value) => IsExternal(value) ? Node.External(value) :
        ↪  Unary(value);

        private Node Reference(string text)
        {
            var remaining = text.AsSpan();
            while (!remaining.IsEmpty)
            {
                if (Rune.DecodeFromUtf16(remaining, out var rune, out var consumed) !=
                ↪  System.Buffers.OperationStatus.Done)
                {
                    throw BinaryNotationException.Unencodable("reference contains invalid
                    ↪  UTF-16");
                }
                remaining = remaining[consumed..];
            }
            if (TryParseCanonicalNumber(text, out var value))
            {
                return IsExternal(value) ? Node.External(value) :
                ↪  Pair(Node.Internal(LinksPacket.Number), Unary(value));
            }
            var codePoints = text.EnumerateRunes().Select(rune =>
            ↪  Scalar((ulong)rune.Value)).ToList();
            return Typed(LinksPacket.String, codePoints);
        }

        public Node Encode(LinoLink link)
        {
            if (link.Values is not { } values)
            {
                return Reference(link.Id ?? string.Empty);
            }
            if (link.Id is { } id)
            {
                var elements = new List<Node> { Reference(id) };
                elements.AddRange(values.Select(Encode));
                return Typed(LinksPacket.Identified, elements);
            }
            return List(values.Select(Encode).ToList());
        }

        public LinksPacket Finish()
```

```
371         {
372             var doubletCount = (ulong) doublets.Count;
373             PacketReference Resolve(Node node) => node.Kind switch
374             {
375                 NodeKind.Internal => PacketReference.Internal(node.Value),
376                 NodeKind.External => PacketReference.External(node.Value),
377                 NodeKind.Doublet => PacketReference.Internal(LinksPacket.FirstLinkAddress +
                    ↪  node.Value),
378                   => PacketReference.Internal(LinksPacket.FirstLinkAddress + doubletCount +
                    ↪  node.Value),
379             };
380             var links =  doublets
381                 .Concat( tuples)
382                 .Select((items, index) => (LinksPacket.FirstLinkAddress + (ulong)index,
                    ↪  items.Select(Resolve).ToArray()))
383                 .ToList();
384             return LinksPacket.Pack( options.ExternalReferences, links,  options.PackedWidths);
385         }
386     }
387
388     private enum ViewKind
389     {
390         Null,
391         Marker,
392         External,
393         Link,
394     }
395
396     private readonly record struct View(ViewKind Kind, ulong Value = 0, PacketReference[]? Items
        ↪  = null);
397
398     private sealed class Decoder
399     {
400         private readonly DecodeLimits  limits;
401         private long  stringBytesLeft;
402         //  links[i] is the link at address FirstLinkAddress + i.
403         private readonly List<PacketReference[]>  links = new();
404         //  unary[i] is the number link i denotes, if it is a unary number.
405         private readonly List<ulong?>  unary = new();
406
407         public Decoder(LinksPacket packet, DecodeLimits limits)
408         {
409              limits = limits;
410              stringBytesLeft = limits.MaxStringBytes;
411             if ((ulong)packet.Sections.Count > limits.MaxLinks || packet.LinkCount >
                ↪  limits.MaxLinks)
412             {
413                 throw BinaryNotationException.Limit($"packet declares more than {limits.MaxLinks}
                    ↪  links");
414             }
415             var referencesLeft = limits.MaxReferences;
416             try
417             {
418                 packet.Validate();
419             }
420             catch (BinaryNotationException exception)
421             {
422                 throw BinaryNotationException.Malformed(exception.Detail);
423             }
424             foreach (var (address, link) in packet.Links())
425             {
426                 if ((ulong)link.Length > referencesLeft)
427                 {
428                     throw BinaryNotationException.Limit($"references exceed the limit of
                        ↪  {limits.MaxReferences}");
429                 }
430                 referencesLeft -= (ulong)link.Length;
431                 var expected = LinksPacket.FirstLinkAddress + (ulong) links.Count;
432                 if (address != expected)
433                 {
434                     throw BinaryNotationException.Malformed(
435                         $"a LiNo packet stores its links contiguously from address
                            ↪  {LinksPacket.FirstLinkAddress}, " +
436                         $"found link {address} where {expected} belongs");
437                 }
438                 foreach (var reference in link)
439                 {
440                     if (!reference.IsExternal && reference.Value >= address)
441                     {
442                         throw BinaryNotationException.Malformed(
```

```
                        $"link {address} refers to {reference}, which is not an earlier link");
                }
            }
            // Links only refer backwards, so one forward pass evaluates every
            // unary number without recursion.
            ulong? value = null;
            if (link.Length == 2
                && UnaryValue(link[0]) is { } source
                && UnaryValue(link[1]) is { } target
                && source <= ulong.MaxValue - target)
            {
                value = source + target;
            }
            unary.Add(value);
            links.Add(link);
        }
    }

    public int LinkCount => links.Count;

    private ulong? UnaryValue(PacketReference reference) => reference switch
    {
        { IsExternal: true } => null,
        { Value: LinksPacket.NullAddress } => 0,
        { Value: LinksPacket.One } => 1,
        { Value: >= LinksPacket.FirstLinkAddress } => unary[(int)(reference.Value - LinksPacket.FirstLinkAddress)],
        => null,
    };

    public View View(PacketReference reference) => reference switch
    {
        { IsExternal: true } => new View(ViewKind.External, reference.Value),
        { Value: LinksPacket.NullAddress } => new View(ViewKind.Null),
        { Value: < LinksPacket.FirstLinkAddress } => new View(ViewKind.Marker, reference.Value),
        => new View(ViewKind.Link, Items: links[(int)(reference.Value - LinksPacket.FirstLinkAddress)]),
    };

    private ulong Number(PacketReference reference) =>
        reference.IsExternal
            ? reference.Value
            : UnaryValue(reference) ?? throw BinaryNotationException.Malformed("expected a unary number");

    /// <summary>The elements of the cons list <c>(e1 (e2 (... (en 0))))</c>.</summary>
    public List<PacketReference> Chain(PacketReference tail)
    {
        var elements = new List<PacketReference>();
        while (true)
        {
            var view = View(tail);
            switch (view)
            {
                case { Kind: ViewKind.Null }:
                    return elements;
                case { Kind: ViewKind.Link, Items: [var head, var next] }:
                    if (elements.Count >= limits.MaxNodes)
                    {
                        throw BinaryNotationException.Limit("chain too long");
                    }
                    elements.Add(head);
                    tail = next;
                    break;
                default:
                    throw BinaryNotationException.Malformed("broken element chain");
            }
        }
    }

    private LinoLink Typed(ulong marker, IReadOnlyList<PacketReference> elements, int depth, ref long budget)
    {
        switch (marker)
        {
            case LinksPacket.Number:
                if (elements.Count != 1)
                {
```

```
                        throw BinaryNotationException.Malformed("a number needs exactly one value");
                    }
                    return ReferenceText(Number(elements[0]).ToString(CultureInfo.InvariantCulture));
                case LinksPacket.String:
                    var text = new StringBuilder((int)Math.Min(elements.Count, stringBytesLeft));
                    long byteCount = 0;
                    foreach (var element in elements)
                    {
                        var codePoint = Number(element);
                        if (codePoint > int.MaxValue || !Rune.IsValid((int)codePoint))
                        {
                            throw BinaryNotationException.Malformed($"invalid code point {codePoint}");
                        }
                        var rune = new Rune((int)codePoint);
                        if (rune.Utf8SequenceLength > stringBytesLeft - byteCount)
                        {
                            throw BinaryNotationException.Limit("too many expanded string bytes");
                        }
                        byteCount += rune.Utf8SequenceLength;
                        text.Append(rune.ToString());
                    }
                    return ReferenceText(text.ToString());
                case LinksPacket.List:
                    return List(elements, 0, depth, ref budget);
                case LinksPacket.Identified:
                    if (elements.Count == 0)
                    {
                        throw BinaryNotationException.Malformed("an identified link needs an id");
                    }
                    var id = Decode(elements[0], depth, ref budget);
                    if (id.Values is not null)
                    {
                        throw BinaryNotationException.Malformed("a link id must be a reference");
                    }
                    var values = List(elements, 1, depth, ref budget).Values!;
                    return LinoFormat.Link(id.Id ?? string.Empty, values);
                default:
                    throw BinaryNotationException.Malformed($"marker {marker} cannot start a typed value");
            }
        }

        private LinoLink List(IReadOnlyList<PacketReference> elements, int skip, int depth, ref long budget)
        {
            var values = new List<LinoLink>(Math.Max(0, elements.Count - skip));
            for (var index = skip; index < elements.Count; index++)
            {
                values.Add(Decode(elements[index], depth + 1, ref budget));
            }
            return LinoFormat.Link(null, values);
        }

        private LinoLink ReferenceText(string text)
        {
            var bytes = Encoding.UTF8.GetByteCount(text);
            if (bytes > stringBytesLeft)
            {
                throw BinaryNotationException.Limit("too many expanded string bytes");
            }
            stringBytesLeft -= bytes;
            return LinoFormat.Reference(text);
        }

        public LinoLink Decode(PacketReference reference, int depth, ref long budget)
        {
            if (depth >= limits.MaxDepth)
            {
                throw BinaryNotationException.Limit($"nesting deeper than {limits.MaxDepth}");
            }
            if (budget <= 0)
            {
                throw BinaryNotationException.Limit("too many LiNo nodes");
```

```
            }
            budget--;
            var view = View(reference);
            switch (view)
            {
                case { Kind: ViewKind.Null }:
                    return LinoFormat.Link(null, new List<LinoLink>());
                case { Kind: ViewKind.External }:
                    return ReferenceText(view.Value.ToString(CultureInfo.InvariantCulture));
                case { Kind: ViewKind.Marker }:
                    throw BinaryNotationException.Malformed($"marker {view.Value} used as a
                    ↪  value");
                case { Items: [{ IsExternal: false, Value: LinksPacket.Number }, var value] }:
                    return Typed(LinksPacket.Number, new[] { value }, depth, ref budget);
                case { Items: [var marker, var chain] } when IsMarker(marker):
                    return Typed(marker.Value, Chain(chain), depth, ref budget);
                default:
                    var items = view.Items!;
                    return StartsWithMarker(items)
                        ? Typed(items[0].Value, items[1..], depth, ref budget)
                        : List(items, 0, depth, ref budget);
            }
        }
    }
}
```

1.7 ./Link.Foundation.Links.Notation/Binary/PacketReader.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
namespace Link.Foundation.Links.Notation.Binary;

/// <summary>
/// A buffered reader for consecutive binary packets. Reuse it across reads
/// so bytes buffered from the following packet remain available.
/// </summary>
public sealed class PacketReader
{
    private readonly Stream  stream;
    private readonly byte[]  buffer;
    private int  position;
    private int  length;

    public PacketReader(Stream stream, int bufferSize = 8192)
    {
        ArgumentNullException.ThrowIfNull(stream);
        ArgumentOutOfRangeException.ThrowIfNegativeOrZero(bufferSize);
         stream = stream;
         buffer = new byte[bufferSize];
    }

    /// <summary>Wraps a byte array, e.g. a whole encoded message.</summary>
    public static PacketReader FromBytes(byte[] bytes) => new(new MemoryStream(bytes, writable:
    ↪  false));

    /// <summary>The next byte without consuming it, or -1 at the end of the stream.</summary>
    public int PeekByte() => Fill() ?  buffer[ position] : -1;

    /// <summary>The next byte, or -1 at the end of the stream.</summary>
    public int ReadByte() => Fill() ?  buffer[ position++] : -1;

    /// <summary>True when every byte of the stream has been consumed.</summary>
    public bool AtEnd => !Fill();

    /// <summary>Fills <paramref name="destination"/>; a short stream is a malformed
    ↪  message.</summary>
    public void ReadExactly(Span<byte> destination)
    {
        while (!destination.IsEmpty)
        {
            if (!Fill())
            {
                throw BinaryNotationException.Malformed("unexpected end of packet");
            }
            var count = Math.Min(destination.Length,  length -  position);
             buffer.AsSpan( position, count).CopyTo(destination);
             position += count;
            destination = destination[count..];
        }
    }
```

```

    private bool Fill()
    {
        if ( position <  length)
        {
            return true;
        }
        try
        {
             length =  stream.Read( buffer, 0,  buffer.Length);
        }
        catch (IOException exception)
        {
            throw new BinaryNotationException(BinaryErrorKind.Io, exception.Message, exception);
        }
        catch (ObjectDisposedException exception)
        {
            throw new BinaryNotationException(BinaryErrorKind.Io, exception.Message, exception);
        }
         position = 0;
        return  length > 0;
    }
}
```

1.8 ./Link.Foundation.Links.Notation/Binary/SectionPlanner.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
namespace Link.Foundation.Links.Notation.Binary;

/// <summary>Splits links into <see cref="Section"/>s with a linear dynamic program.</summary>
/// <remarks>
/// After link k, <c>costs[s]</c> is the fewest estimated bytes for links
/// 0..=k with link k in a section of state s. A link either continues the
/// section of the previous link (same state, no hole between them and, for a
/// fixed arity, the same length) or opens a new section after the cheapest
/// previous state, paying for a section header. The tie-breaking matches the
/// Rust <c>SectionPlanner</c>, so both ports produce identical bytes.
/// </remarks>
internal sealed class SectionPlanner
{
    /// <summary>Estimated bytes of a section header (shape and count), used to weigh a split.</summary>
    private const ulong SectionHeaderEstimate = 2;
    /// <summary>Estimated extra bytes of a variable-arity section: its extra arity field and each length prefix.</summary>
    private const ulong VariableArityEstimate = 1;
    private const ulong Unreachable = ulong.MaxValue / 4;
    /// <summary>Planner states: a width code (0..4) times fixed (0) or variable (1) arity.</summary>
    private static readonly int States = LinksPacket.Widths.Count * 2;

    private readonly IReadOnlyList<(ulong Address, PacketReference[] References)>  links;
    private readonly byte[]  needs;

    public SectionPlanner(bool externalReferences, IReadOnlyList<(ulong Address, PacketReference[] References)> links)
    {
         links = links;
         needs = new byte[links.Count];
        var previousAddress = 0UL;
        for (var index = 0; index < links.Count; index++)
        {
            var (address, references) = links[index];
            if (address <= previousAddress)
            {
                throw BinaryNotationException.Unencodable(
                    $"link addresses must ascend from 1, got {address} after {previousAddress}");
            }
            if (address == ulong.MaxValue)
            {
                throw BinaryNotationException.Unencodable("addresses overflow 64 bits");
            }
            if (references.Length == 0)
            {
                throw BinaryNotationException.Unencodable($"link {address} has no references");
            }
            previousAddress = address;
            byte need = 1;
            foreach (var reference in references)
```

```
            {
                need = Math.Max(need, LinksPacket.ReferenceWidth(reference, externalReferences));
            }
             needs[index] = need;
        }
    }

    private static int FirstCheapest(ulong[] costs)
    {
        var best = 0;
        for (var state = 1; state < costs.Length; state++)
        {
            if (costs[state] < costs[best])
            {
                best = state;
            }
        }
        return best;
    }

    /// <summary>
    /// The sections as <c>(link count, width)</c> pairs; with
    /// <paramref name="packedWidths"/> every width may be used, otherwise only
    /// the widest one needed.
    /// </summary>
    public List<(int Count, byte Width)> Plan(bool packedWidths)
    {
        var sections = new List<(int Count, byte Width)>();
        if ( links.Count == 0)
        {
            return sections;
        }
        var widest =  needs.Max();
        var opensSection = new byte[ links.Count];
        var previousBest = new byte[ links.Count];
        var costs = Enumerable.Repeat(Unreachable, States).ToArray();
        for (var index = 0; index <  links.Count; index++)
        {
            var (address, link) =  links[index];
            var best = FirstCheapest(costs);
            previousBest[index] = (byte)best;
            var cheapestBefore = index == 0 ? 0 : costs[best];
            var continues = index > 0 &&  links[index - 1].Address + 1 == address;
            var sameLength = index > 0 &&  links[index - 1].References.Length == link.Length;
            var next = Enumerable.Repeat(Unreachable, States).ToArray();
            for (var state = 0; state < States; state++)
            {
                var width = LinksPacket.Widths[state / 2];
                var variable = state % 2 == 1;
                if (!(packedWidths || width == widest) || width <  needs[index])
                {
                    continue;
                }
                var variableEstimate = variable ? VariableArityEstimate : 0;
                var body = (ulong)link.Length * width + variableEstimate;
                var openingCost = cheapestBefore + SectionHeaderEstimate + variableEstimate;
                var continuingCost = continues && (variable || sameLength) ? costs[state] :
                ↪  Unreachable;
                if (continuingCost <= openingCost)
                {
                    next[state] = continuingCost + body;
                }
                else
                {
                    next[state] = openingCost + body;
                    opensSection[index] |= (byte)(1 << state);
                }
            }
            costs = next;
        }
        var current = FirstCheapest(costs);
        var end =  links.Count;
        for (var index =  links.Count - 1; index >= 0; index--)
        {
            if ((opensSection[index] & (1 << current)) != 0)
            {
                sections.Add((end - index, LinksPacket.Widths[current / 2]));
                end = index;
                current = previousBest[index];
            }
        }
```

```
        sections.Reverse();
        return sections;
    }
}
```

1.9 ./Link.Foundation.Links.Notation/Comments.cs

```
namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Comments, and the rule that decides where one starts.
    /// </summary>
    /// <remarks>
    /// <c>#</c> starts a comment when it opens a token: at the start of the document or
    /// after a space, a tab or a line break. A <c>#</c> written inside a token
    /// (<c>issue#1047</c>) or inside a delimited reference (<c>"# not a comment"</c>) is
    /// an ordinary character.
    /// </remarks>
    public static class Comments
    {
        /// <summary>
        /// The character that starts a comment.
        /// </summary>
        public const char Comment = '#';

        /// <summary>
        /// What ends a bare reference. A token starts after one of these, at the start of
        /// the document, and right after a delimited reference or a versioned literal: the
        /// places the grammar reads a reference from.
        /// </summary>
        internal const string Separators = " \t\n\r(:)";

        /// <summary>
        /// What a comment may start after.
        /// </summary>
        private const string BeforeComment = " \t\n\r";

        /// <summary>
        /// Blanks every comment of a document.
        /// </summary>
        /// <param name="document">The document to read.</param>
        /// <returns>
        /// The same document with the characters of every comment replaced by spaces.
        /// </returns>
        /// <remarks>
        /// Every character of a comment is replaced rather than removed, so every later
        /// character keeps the position it had in the document the caller wrote, and
        /// anything the parser reports about a position still points at what the reader
        /// can see. The document is read token by token, the way the grammar reads it, so
        /// a <c>#</c> inside a delimited reference that starts right after a group,
        /// another reference or a literal is content, as it is in one that starts after a
        /// space (issue #377).
        /// </remarks>
        public static string StripComments(string document)
        {
            char[]? blanked = null;
            int position = 0;
            // Whether a token may start at the position.
            bool tokenStart = true;

            while (position < document.Length)
            {
                char character = document[position];

                if (character == Comment && Follows(document, position, BeforeComment))
                {
                    blanked ??= document.ToCharArray();
                    while (position < document.Length &&
                           document[position] != '\n' &&
                           document[position] != '\r')
                    {
                        blanked[position] = ' ';
                        position++;
                    }

                    continue;
                }

                if (Separators.IndexOf(character) >= 0)
                {
                    tokenStart = true;
                    position++;
```

```
                    continue;
                }

                if (tokenStart)
                {
                    int end = DelimitedTokenEnd(document, position);
                    if (end >= 0)
                    {
                        position = end;
                        continue;
                    }
                }

                // Anything else is part of a bare reference, which runs to a separator.
                tokenStart = false;
                position++;
            }

            return blanked == null ? document : new string(blanked);
        }

        /// <summary>
        /// The position just past the versioned literal or delimited reference that starts
        /// the token at <paramref name="start"/>, or -1 when the token there is a bare
        /// reference: a delimiter run that never closes is read as one, as the grammar
        /// reads it.
        /// </summary>
        private static int DelimitedTokenEnd(string document, int start)
        {
            int literal = ReferenceLiteral.End(document, start);
            if (literal >= 0) return literal;
            return Quotes.IsQuote(document[start]) ? Quotes.ReferenceEnd(document, start) : -1;
        }

        /// <summary>
        /// Reports whether the character before the position is one of the allowed ones,
        /// the start of the document counting as allowed.
        /// </summary>
        private static bool Follows(string document, int position, string allowed) =>
            position == 0 || allowed.IndexOf(document[position - 1]) >= 0;
    }
}
```

1.10 ./Link.Foundation.Links.Notation/CstBuilder.cs

```
using System.Collections.Generic;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Builds the concrete syntax tree of a document that has already been parsed.
    /// </summary>
    /// <remarks>
    /// The tokens are read with the rules the grammar uses: comments are found by
    /// <see cref="Comments.StripComments"/>, delimited references by <see cref="Quotes"/>
    /// and versioned literals by <see cref="ReferenceLiteral"/>, so the tree agrees with
    /// what <see cref="Parser.Parse"/> reads. Every token's text is taken from the document
    /// as written, never from the copy with its comments blanked.
    /// </remarks>
    internal static class CstBuilder
    {
        /// <summary>What ends a bare reference.</summary>
        private const string BareReferenceEnd = " \t\r\n(:)";

        /// <summary>An open container (the document or a group) and the line being filled in
        ↪  it.</summary>
        private sealed class Container
        {
            public Container(CstNode node) => Node = node;

            public CstNode Node { get; }

            public CstNode? Line { get; set; }
        }

        /// <summary>
        /// Reads <paramref name="input"/> into its concrete syntax tree.
        /// </summary>
        /// <param name="input">The document as written.</param>
        /// <param name="comments">Whether <c>#</c> starts a comment.</param>
        /// <returns>The <see cref="CstKind.Document"/> node.</returns>
```

```
        /// <exception cref="ParseException">The parentheses of the document are unbalanced.</exception>
        internal static CstNode Build(string input, bool comments)
        {
            // Bare references end where they end in the document the grammar reads: the
            // one with its comments blanked, which keeps every position of the one
            // written. Delimited references and literals are read, as the grammar reads
            // them, in the document as written, so a comment never changes what a
            // reference holds (issue #377).
            var prepared = comments ? Notation.Comments.StripComments(input) : input;
            var document = new CstNode(CstKind.Document);
            // The open containers, innermost last.
            var containers = new List<Container> { new Container(document) };

            void Add(CstNode node)
            {
                var container = containers[^1];
                if (container.Line == null)
                {
                    container.Line = new CstNode(CstKind.Line);
                    container.Node.Add(container.Line);
                }
                container.Line.Add(node);
            }

            var position = 0;
            while (position < input.Length)
            {
                var character = input[position];
                var end = position + 1;

                if (comments && character == Notation.Comments.Comment && prepared[position] != Notation.Comments.Comment)
                {
                    // A blanked '#' opens a comment, which runs to the line ending.
                    while (end < input.Length && input[end] != '\n' && input[end] != '\r')
                    {
                        end++;
                    }
                    Add(Token(CstKind.Comment, input, position, end));
                }
                else if (character == '\r' || character == '\n')
                {
                    if (character == '\r' && end < input.Length && input[end] == '\n')
                    {
                        end++;
                    }
                    Add(Token(CstKind.Newline, input, position, end));
                    containers[^1].Line = null;
                }
                else if (character == ' ' || character == '\t')
                {
                    while (end < input.Length && (input[end] == ' ' || input[end] == '\t'))
                    {
                        end++;
                    }
                    Add(Token(CstKind.Whitespace, input, position, end));
                }
                else if (character == '(')
                {
                    var group = new CstNode(CstKind.Group);
                    group.Add(Token(CstKind.OpenParen, input, position, end));
                    Add(group);
                    containers.Add(new Container(group));
                }
                else if (character == ')')
                {
                    if (containers.Count == 1)
                    {
                        throw new ParseException(input, position);
                    }
                    containers[^1].Node.Add(Token(CstKind.CloseParen, input, position, end));
                    containers.RemoveAt(containers.Count - 1);
                }
                else if (character == ':')
                {
                    Add(Token(CstKind.Colon, input, position, end));
                }
                else
                {
```

```
                    Add(ReadReference(input, prepared, position, out end));
                }
                position = end;
            }

            if (containers.Count != 1)
            {
                throw new ParseException(input, input.Length);
            }
            return document;
        }

        private static CstNode Token(CstKind kind, string input, int start, int end) =>
            new(kind, input.Substring(start, end - start));

        /// <summary>
        /// Reads the reference that starts at <paramref name="start"/> the way the grammar
        /// reads it: a versioned literal, a delimited reference, or else a bare one.
        /// </summary>
        private static CstNode ReadReference(string input, string prepared, int start, out int
        ↪  end)
        {
            var literal = ReferenceLiteral.End(input, start);
            if (literal >= 0)
            {
                end = literal;
                var text = input.Substring(start, end - start);
                return new CstNode(CstKind.Reference, text, ReferenceLiteral.Decode(text));
            }
            if (Quotes.IsQuote(input[start]) && Quotes.TryParse(input, start, out var value, out
            ↪  var length))
            {
                end = start + length;
                return new CstNode(CstKind.Reference, input.Substring(start, length), value,
                ↪  input[start]);
            }
            end = start + 1;
            while (end < prepared.Length && BareReferenceEnd.IndexOf(prepared[end]) < 0)
            {
                end++;
            }
            var bare = input.Substring(start, end - start);
            return new CstNode(CstKind.Reference, bare, bare);
        }
    }
}
```

1.11 ./Link.Foundation.Links.Notation/CstKind.cs

```
using System;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// The kinds of nodes and tokens of a concrete syntax tree; see <see cref="CstNode"/>.
    /// </summary>
    public enum CstKind
    {
        /// <summary>The whole document: its lines.</summary>
        Document,

        /// <summary>One line: its tokens and groups, up to and including its line
        ↪  ending.</summary>
        Line,

        /// <summary>A parenthesized group: <c>(</c>, the lines inside it, and
        ↪  <c>)</c>.</summary>
        Group,

        /// <summary>A run of spaces and tabs.</summary>
        Whitespace,

        /// <summary>One line ending: <c>\n</c>, <c>\r\n</c> or <c>\r</c>.</summary>
        Newline,

        /// <summary>A comment, from <c>#</c> up to the line ending.</summary>
        Comment,

        /// <summary><c>(</c></summary>
        OpenParen,

        /// <summary><c>)</c></summary>
```

```
        CloseParen,

        /// <summary><c>:</c></summary>
        Colon,

        /// <summary>A bare, quoted or versioned literal reference.</summary>
        Reference,
    }

    /// <summary>
    /// Provides the names every Links Notation implementation gives the kinds of
    /// <see cref="CstKind"/>.
    /// </summary>
    public static class CstKindExtensions
    {
        /// <summary>                    -
        /// The name of the kind-as <see cref="CstNode.ToSExpression"/> writes it, the same in
        /// every implementation: <c>document</c>, <c>line</c>, <c>group</c>,
        /// <c>whitespace</c>, <c>newline</c>, <c>comment</c>, <c>open paren</c>,
        /// <c>close paren</c>, <c>colon</c> or <c>reference</c>.
        /// </summary>
        /// <param name="kind">The kind to name.</param>
        /// <returns>The shared name of the kind.</returns>
        public static string ToName(this CstKind kind) => kind switch
        {
            CstKind.Document => "document",
            CstKind.Line => "line",
            CstKind.Group => "group",
            CstKind.Whitespace => "whitespace",
            CstKind.Newline => "newline",
            CstKind.Comment => "comment",
            CstKind.OpenParen => "open paren",
            CstKind.CloseParen => "close paren",
            CstKind.Colon => "colon",
            CstKind.Reference => "reference",
              => throw new ArgumentOutOfRangeException(nameof(kind), kind, "Unknown concrete
            ↪  syntax tree kind."),
        };

        /// <summary>
        /// Whether nodes of the kind hold children (<see cref="CstKind.Document"/>,
        /// <see cref="CstKind.Line"/> and <see cref="CstKind.Group"/>) rather than text.
        /// </summary>
        /// <param name="kind">The kind to test.</param>
        /// <returns><see langword="true"/> for a node kind, <see langword="false"/> for a token
        ↪  kind.</returns>
        public static bool IsNode(this CstKind kind) =>
            kind is CstKind.Document or CstKind.Line or CstKind.Group;
    }
}
```

1.12 ./Link.Foundation.Links.Notation/CstNode.cs

```
using System;
using System.Collections.Generic;
using System.Text;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// One node or token of the concrete syntax tree of a Links Notation document.
    /// </summary>
    /// <remarks>
    /// Every character of a document belongs to exactly one token, so the text of the
    /// tree is the document itself, comments, blank lines, indentation, quotes and line
    /// endings included. A <see cref="CstKind.Document"/> holds <see cref="CstKind.Line"/>
    /// nodes; a line holds the tokens up to and including its line ending, and a
    /// parenthesized <see cref="CstKind.Group"/> holds its own parentheses and the lines
    /// written between them. A group is part of the line it opens on, however many lines
    /// it spans. Read a tree with <see cref="Parser.ParseCst"/>.
    /// </remarks>
    public sealed class CstNode
    {
        private static readonly IReadOnlyList<CstNode> NoChildren = Array.Empty<CstNode>();

        private readonly List<CstNode>?  children;
        private readonly string  text;

        /// <summary>
        /// Creates a token.
        /// </summary>
        /// <param name="kind">A token kind.</param>
        /// <param name="text">The token's text exactly as written.</param>
```

```
        /// <param name="value">What a reference token means; <see langword="null"/>
        ↪  otherwise.</param>
        /// <param name="quote">The delimiter of a quoted reference token; <see langword="null"/>
        ↪  otherwise.</param>
        /// <exception cref="ArgumentException"><paramref name="kind"/> is a node
        ↪  kind.</exception>
        public CstNode(CstKind kind, string text, string? value = null, char? quote = null)
        {
            if (kind.IsNode())
            {
                _
                throw new ArgumentException($"{kind.ToName()} is a node, not a token.",
                ↪  nameof(kind));
            }
            Kind = kind;
             text = text ?? throw new ArgumentNullException(nameof(text));
            Value = value;
            Quote = quote;
        }

        /// <summary>
        /// Creates a node.
        /// </summary>
        /// <param name="kind">A node kind.</param>
        /// <param name="children">The node's children, in document order.</param>
        /// <exception cref="ArgumentException"><paramref name="kind"/> is a token
        ↪  kind.</exception>
        public CstNode(CstKind kind, IEnumerable<CstNode>? children = null)
        {
            if (!kind.IsNode())
            {
                throw new ArgumentException($"{kind.ToName()} is a token, not a node.",
                ↪  nameof(kind));
            }
            Kind = kind;
             text = string.Empty;
             children = children == null ? new List<CstNode>() : new List<CstNode>(children);
        }

        /// <summary>The kind of this node or token.</summary>
        public CstKind Kind { get; }

        /// <summary>The children of a node, in document order; empty for a token.</summary>
        public IReadOnlyList<CstNode> Children =>  children ?? NoChildren;

        /// <summary>
        /// What a reference token means, its delimiters and escapes resolved: the ID a
        /// parsed link has. <see langword="null"/> for every other node and token.
        /// </summary>
        public string? Value { get; }

        /// <summary>
        /// The delimiter a quoted reference token was written between (<c>'</c>, <c>"</c> or
        /// <c>`</c>); <see langword="null"/> for a bare or literal reference and every other
        ↪  kind.
        /// </summary>
        public char? Quote { get; }

        /// <summary>Whether this is a token rather than a node with children.</summary>
        public bool IsToken =>  children == null;

        /// <summary>
        /// The exact text this node or token was read from: a token's own text, and the
        /// text of its children one after another for a node.
        /// </summary>
        public string Text
        {
            get
            {
                if ( children == null)
                {
                    return  text;
                }
                var text = new StringBuilder();
                AppendText(text);
                return text.ToString();
            }
        }

        /// <summary>The exact text this node or token was read from; see <see
        ↪  cref="Text"/>.</summary>
```

```
        /// <returns>The text of this node or token.</returns>
        public override string ToString() => Text;

        /// <summary>
        /// Every token of this subtree, in document order; a token yields just itself.
        /// </summary>
        /// <returns>The tokens of this subtree.</returns>
        public IReadOnlyList<CstNode> Tokens()
        {
            var tokens = new List<CstNode>();
            AppendTokens(tokens);
            return tokens;
        }

        /// <summary>
        /// The tree as an S-expression: <c>(kind child ...)</c> for a node and
        /// <c>kind:"text"</c> for a token, the text with <c>\</c>, <c>"</c>, line feeds,
        /// carriage returns and tabs escaped. Every Links Notation implementation writes the
        /// same S-expression for the same document, and two trees are the same exactly when
        /// their S-expressions are.
        /// </summary>
        /// <returns>The S-expression of this subtree.</returns>
        /// <example>
        /// <c>new Parser().ParseCst("a '1'").ToSExpression()</c> is
        /// <c>(document (line reference:"a" whitespace:" " reference:"'1'"))</c>.
        /// </example>
        public string ToSExpression()
        {
            var expression = new StringBuilder();
            AppendSExpression(expression);
            return expression.ToString();
        }

        /// <summary>Appends a child; only the builder adds children once a tree is
        ↪  made.</summary>
        internal void Add(CstNode child) =>  children!.Add(child);

        private void AppendText(StringBuilder text)
        {
            if ( children == null)
            {
                text.Append( text);
                return;
            }
            foreach (var child in  children)
            {
                child.AppendText(text);
            }
        }

        private void AppendTokens(List<CstNode> tokens)
        {
            if ( children == null)
            {
                tokens.Add(this);
                return;
            }
            foreach (var child in  children)
            {
                child.AppendTokens(tokens);
            }
        }

        private void AppendSExpression(StringBuilder expression)
        {
            if ( children == null)
            {
                expression.Append(Kind.ToName()).Append(":\"");
                foreach (var character in  text)
                {
                    switch (character)
                    {
                        case '\\': expression.Append("\\\\"); break;
                        case '"': expression.Append("\\\""); break;
                        case '\n': expression.Append("\\n"); break;
                        case '\r': expression.Append("\\r"); break;
                        case '\t': expression.Append("\\t"); break;
                        default: expression.Append(character); break;
                    }
                }
```

```
                expression.Append('"');
                return;
            }
            expression.Append('(').Append(Kind.ToName());
            foreach (var child in  children)
            {
                expression.Append(' ');
                child.AppendSExpression(expression);
            }
            expression.Append(')');
        }
    }
}
```

1.13 ./Link.Foundation.Links.Notation/FormatConfig.cs

```
namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// FormatConfig is an alias for FormatOptions to maintain API consistency
    /// with the Python implementation.
    ///
    /// C# uses 'FormatOptions' internally, but provides 'FormatConfig'
    /// as an alias to match Python's naming convention.
    /// </summary>
    public class FormatConfig : FormatOptions
    {
        // FormatConfig inherits all functionality from FormatOptions
        // No additional implementation needed - this is purely for naming consistency
    }
}
```

1.14 ./Link.Foundation.Links.Notation/FormatOptions.cs

```
using System;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// FormatOptions for Lino notation formatting.
    /// Provides configuration options for controlling how Link objects are formatted.
    /// </summary>
    public class FormatOptions
    {
        /// <summary>
        /// If true, omit parentheses where safe (default: false)
        /// </summary>
        public bool LessParentheses { get; set; } = false;

        /// <summary>
        /// Maximum line length before auto-indenting (default: 80)
        /// </summary>
        public int MaxLineLength { get; set; } = 80;

        /// <summary>
        /// If true, indent lines exceeding MaxLineLength (default: false)
        /// </summary>
        public bool IndentLongLines { get; set; } = false;

        /// <summary>
        /// Maximum number of references before auto-indenting (default: null = unlimited)
        /// </summary>
        public int? MaxInlineRefs { get; set; } = null;

        /// <summary>
        /// If true, group consecutive links with same ID (default: false)
        /// </summary>
        public bool GroupConsecutive { get; set; } = false;

        /// <summary>
        /// String to use for indentation (default: "  " = two spaces)
        /// </summary>
        public string IndentString { get; set; } = "  ";

        /// <summary>
        /// If true, prefer inline format when under thresholds (default: true)
        /// </summary>
        public bool PreferInline { get; set; } = true;

        /// <summary>
        /// Check if line should be indented based on length
        /// </summary>
```

```
        /// <param name="line">The line to check</param>
        /// <returns>True if line should be indented based on length threshold</returns>
        public bool ShouldIndentByLength(string line)
        {
            if (!IndentLongLines)
            {
                return false;
            }
            // Count printable unicode characters
            return line.Length > MaxLineLength;
        }

        /// <summary>
        /// Check if link should be indented based on reference count
        /// </summary>
        /// <param name="refCount">Number of references in the link</param>
        /// <returns>True if link should be indented based on reference count threshold</returns>
        public bool ShouldIndentByRefCount(int refCount)
        {
            if (MaxInlineRefs == null)
            {
                return false;
            }
            return refCount > MaxInlineRefs.Value;
        }
    }
}
```

1.15 ./Link.Foundation.Links.Notation/FurthestFailureTracer.cs

```
using Pegasus.Common;
using Pegasus.Common.Tracing;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Remembers the furthest position a parse reached.
    /// </summary>
    /// <remarks>
    /// A parsing expression grammar backtracks, so the cursor the generated parser raises
    /// its error with is the start of the document rather than the place the document
    /// stopped making sense. The furthest position any rule reached is that place, and the
    /// tracer is the only hook the generated parser offers to observe it.
    /// </remarks>
    internal sealed class FurthestFailureTracer : ITracer
    {
        /// <summary>Offset of the furthest position the parse reached.</summary>
        public int Furthest { get; private set; }

        public void TraceRuleEnter(string ruleName, Cursor cursor) => Reach(cursor.Location);

        public void TraceRuleExit<T>(string ruleName, Cursor cursor, IParseResult<T> result)
        {
            if (result != null) Reach(result.EndCursor.Location);
        }

        public void TraceCacheHit<T>(string ruleName, Cursor cursor, CacheKey key,
        ↪  IParseResult<T> result)
        {
            if (result != null) Reach(result.EndCursor.Location);
        }

        public void TraceCacheMiss(string ruleName, Cursor cursor, CacheKey key)
        {
        }

        public void TraceInfo(string ruleName, Cursor cursor, string info)
        {
        }

        private void Reach(int location)
        {
            if (location > Furthest) Furthest = location;
        }
    }
}
```

1.16 ./Link.Foundation.Links.Notation/ILinksGroupListExtensions.cs

```
using System.Collections.Generic;
using System.Runtime.CompilerServices;

namespace Link.Foundation.Links.Notation
```

```
{
    /// <summary>
    /// Provides extension methods for collections of <see cref="LinksGroup{TLinkAddress}"/>
    ↪ instances.
    /// </summary>
    public static class ILinksGroupListExtensions
    {
        /// <summary>
        /// Converts a collection of links groups to a flat list of links by processing each
        ↪ group hierarchically.
        /// </summary>
        /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can
        ↪ be any type that uniquely identifies a link, such as string, int, or
        ↪ Guid.</typeparam>
        /// <param name="groups">The collection of links groups to convert.</param>
        /// <returns>A flat list of links representing all the links from the input
        ↪ groups.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static List<Link<TLinkAddress>> ToLinksList<TLinkAddress>(this
        ↪ IList<LinksGroup<TLinkAddress>> groups)
        {
            var list = new List<Link<TLinkAddress>>();
            for (var i = 0; i < groups.Count; i++)
            {
                CollectLinksWithIndentedIdSyntaxSupport(list, groups[i], null);
            }
            return list;
        }

        private static void
        ↪ CollectLinksWithIndentedIdSyntaxSupport<TLinkAddress>(List<Link<TLinkAddress>> list,
        ↪ LinksGroup<TLinkAddress> group, Link<TLinkAddress>? parentDependency)
        {
            var link = group.Link;
            var groups = group.Groups;

            if (groups != null && groups.Count > 0)
            {
                bool isIndentedIdSyntax = link.Id != null &&
                                          (link.Values == null || link.Values.Count == 0);

                if (isIndentedIdSyntax && !parentDependency.HasValue)
                {
                    var childValues = new List<Link<TLinkAddress>>();
                    for (int i = 0; i < groups.Count; i++)
                    {
                        var transformedLink = TransformIndentedIdLink(groups[i]);
                        childValues.Add(transformedLink);
                    }

                    var linkWithChildren = new Link<TLinkAddress>(link.Id, childValues) { RawId
                    ↪ = link.RawId };
                    list.Add(linkWithChildren);
                }
                else
                {
                    var transformedGroups = new List<LinksGroup<TLinkAddress>>();
                    for (int i = 0; i < groups.Count; i++)
                    {
                        var childGroup = groups[i];
                        var childLink = childGroup.Link;
                        var childGroups = childGroup.Groups;

                        if (childLink.Id != null &&
                            (childLink.Values == null || childLink.Values.Count == 0) &&
                            childGroups != null && childGroups.Count > 0)
                        {
                            var transformedLink = TransformIndentedIdLink(childGroup);
                            transformedGroups.Add(new LinksGroup<TLinkAddress>(transformedLink));
                        }
                        else
                        {
                            transformedGroups.Add(childGroup);
                        }
                    }

                    var currentDependency = parentDependency.HasValue ?
                    ↪ parentDependency.Value.Combine(link) : link;
                    list.Add(currentDependency);

```

```
                    for (int i = 0; i < transformedGroups.Count; i++)
                    {
                        CollectLinksWithIndentedIdSyntaxSupport(list, transformedGroups[i],
                        ↪ currentDependency);
                    }
                }
            }
            else
            {
                var currentLink = parentDependency.HasValue ?
                ↪ parentDependency.Value.Combine(link) : link;
                list.Add(currentLink);
            }
        }

        private static Link<TLinkAddress>
        ↪ TransformIndentedIdLink<TLinkAddress>(LinksGroup<TLinkAddress> group)
        {
            var link = group.Link;
            var groups = group.Groups;

            if (groups != null && groups.Count > 0 &&
                link.Id != null &&
                (link.Values == null || link.Values.Count == 0))
            {
                var childValues = new List<Link<TLinkAddress>>();
                for (int i = 0; i < groups.Count; i++)
                {
                    var transformedChild = TransformIndentedIdLink(groups[i]);
                    childValues.Add(transformedChild);
                }
                return new Link<TLinkAddress>(link.Id, childValues) { RawId = link.RawId };
            }
            else if (groups != null && groups.Count > 0)
            {
                var values = new List<Link<TLinkAddress>>();
                if (link.Values != null)
                {
                    values.AddRange(link.Values);
                }
                for (int i = 0; i < groups.Count; i++)
                {
                    values.Add(TransformIndentedIdLink(groups[i]));
                }
                return new Link<TLinkAddress>(link.Id, values) { RawId = link.RawId };
            }
            else if (!group.IsParenthesized && link.Id == null && link.Values != null &&
            ↪ link.Values.Count == 1)
            {
                return link.Values[0];
            }
            else
            {
                return link;
            }
        }
    }
}
```

1.17 ./Link.Foundation.Links.Notation/IListExtensions.cs

```
using Platform.Collections;
using System;
using System.Collections.Generic;
using System.Linq;
using System.Runtime.CompilerServices;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Provides extension methods for formatting collections of <see cref="Link{TLinkAddress}"/>
    ↪ instances.
    /// </summary>
    public static class IListExtensions
    {
        /// <summary>
        /// Formats a collection of links as a multi-line string with each link on a separate
        ↪ line.
        /// </summary>
        /// <typeparam name="TLinkAddress">The type used for link
        ↪ addresses/identifiers.</typeparam>
```

```
        /// <param name="links">The collection of links to format.</param>
        /// <returns>A multi-line string representation of the links.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links) =>
        ↪ string.Join(Environment.NewLine, links.Select(l => l.ToString()));

        /// <summary>
        /// Formats a collection of links as a multi-line string with optional parentheses
        ↪ trimming for cleaner output.
        /// </summary>
        /// <typeparam name="TLinkAddress">The type used for link
        ↪ addresses/identifiers.</typeparam>
        /// <param name="links">The collection of links to format.</param>
        /// <param name="lessParentheses">True to remove outer parentheses from each link for
        ↪ cleaner formatting; false to use standard formatting.</param>
        /// <returns>A multi-line string representation of the links.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links, bool
        ↪ lessParentheses)
        {
            if (lessParentheses == false)
            {
                return links.Format();
            }
            else
            {
                return string.Join(Environment.NewLine, links.Select(FormatWithLessParentheses));
            }
        }

        /// <summary>
        /// Writes a link without its outer parentheses, unless they are what the link is:
        /// an empty anonymous link, or an anonymous link that nests one link with values.
        /// </summary>
        private static string FormatWithLessParentheses<TLinkAddress>(Link<TLinkAddress> link)
        {
            var text = link.ToString();
            var isEmpty = link.Id == null && (link.Values == null || link.Values.Count == 0);
            return isEmpty || (link.Id == null && link.HasLoneNestedValue()) ? text :
            ↪ text.TrimSingle('(').TrimSingle(')');
        }

        /// <summary>
        /// Formats a collection of links using FormatOptions configuration.
        /// </summary>
        /// <typeparam name="TLinkAddress">The type used for link
        ↪ addresses/identifiers.</typeparam>
        /// <param name="links">The collection of links to format.</param>
        /// <param name="options">The FormatOptions to use for formatting.</param>
        /// <returns>A multi-line string representation of the links.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links,
        ↪ FormatOptions options)
        {
            if (links == null || links.Count == 0)
            {
                return string.Empty;
            }

            // Apply consecutive link grouping if enabled
            var linksToFormat = options.GroupConsecutive ? GroupConsecutiveLinks(links) : links;

            // Format each link with options
            var formattedLinks = linksToFormat.Select(l => l.FormatWithOptions(options));
            return string.Join(Environment.NewLine, formattedLinks);
        }

        /// <summary>
        /// Group consecutive links with the same ID.
        /// </summary>
        private static List<Link<TLinkAddress>>
        ↪ GroupConsecutiveLinks<TLinkAddress>(IList<Link<TLinkAddress>> links)
        {
            if (links == null || links.Count == 0)
            {
                return new List<Link<TLinkAddress>>();
            }

            var grouped = new List<Link<TLinkAddress>>();
```

```
            int i = 0;

            while (i < links.Count)
            {
                var current = links[i];

                // Look ahead for consecutive links with same ID
                if (current.Id != null && current.Values != null && current.Values.Count > 0)
                {
                    // Collect all values with same ID
                    var sameIdValues = new List<Link<TLinkAddress>>(current.Values);
                    int j = i + 1;

                    while (j < links.Count)
                    {
                        var nextLink = links[j];
                        if (EqualityComparer<TLinkAddress>.Default.Equals(nextLink.Id,
                        ↪  current.Id) &&
                            nextLink.Values != null && nextLink.Values.Count > 0)
                        {
                            sameIdValues.AddRange(nextLink.Values);
                            j++;
                        }
                        else
                        {
                            break;
                        }
                    }

                    // If we found consecutive links, create grouped link
                    if (j > i + 1)
                    {
                        var groupedLink = new Link<TLinkAddress>(current.Id, sameIdValues);
                        grouped.Add(groupedLink);
                        i = j;
                        continue;
                    }
                }

                grouped.Add(current);
                i++;
            }

            return grouped;
        }
    }
}
```

1.18 ./Link.Foundation.Links.Notation/Link.cs

```
using System;
using System.Collections.Generic;
using System.Runtime.CompilerServices;
using System.Text;
using Platform.Collections;
using Platform.Collections.Lists;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Represents a link in the Links Notation with an optional identifier and nested values.
    /// Links can represent simple references, complex nested structures, or relationships
    ↪  between references to links.
    /// </summary>
    /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
    ↪  any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
    public struct Link<TLinkAddress> : IEquatable<Link<TLinkAddress>>
    {
        private static readonly EqualityComparer<TLinkAddress> EqualityComparerInstance =
        ↪  EqualityComparer<TLinkAddress>.Default;

        /// <summary>
        /// Gets the identifier or address of this link. Can be null for anonymous links.
        /// </summary>
        public readonly TLinkAddress? Id;

        /// <summary>
        /// Gets the collection of nested link values. Can be null or empty for simple reference
        ↪  links.
        /// </summary>
        public readonly IList<Link<TLinkAddress>>? Values;

        /// <summary>
```

```
        /// Gets the identifier exactly as the parsed document wrote it: <c>'1'</c> for a
        /// quoted <c>1</c>, <c>1</c> for a bare one, <c>~1{31}</c> for a versioned literal.
        /// </summary>
        /// <remarks>
        /// <see langword="null"/> for a link without an identifier, a link built in code, a
        /// link made by combining the lines of an indented document, and every link read by
        /// a parser whose <see cref="Parser.Syntax"/> is <see langword="false"/>. It takes no
        /// part in <see cref="Equals(Link{TLinkAddress})"/>, <see cref="GetHashCode"/>,
        /// <see cref="ToString"/> or formatting, so <c>1</c> and <c>'1'</c> stay equal links.
        /// </remarks>
        public string? RawId { get; init; }

        /// <summary>
        /// Gets the delimiter the identifier was written between: <c>'</c>, <c>"</c> or
        /// <c>`</c> when it was quoted, and <see langword="null"/> when it was bare, a
        /// versioned literal, or not read from a document (see <see cref="RawId"/>).
        /// </summary>
        public char? Quote => ReferenceQuote(RawId);

        /// <summary>
        /// Initializes a new link with the specified identifier and values.
        /// </summary>
        /// <param name="id">The link identifier or address.</param>
        /// <param name="values">The nested link values.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public Link(TLinkAddress? id, IList<Link<TLinkAddress>>? values) => (Id, Values) = (id,
        ↪  values);

        /// <summary>
        /// Initializes a new anonymous link with the specified values.
        /// </summary>
        /// <param name="values">The nested link values.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public Link(IList<Link<TLinkAddress>> values) : this(default!, values) { }

        /// <summary>
        /// Initializes a new anonymous link with the specified array of values.
        /// </summary>
        /// <param name="values">The nested link values as a parameter array.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public Link(params Link<TLinkAddress>[] values) : this(default!, values) { }

        /// <summary>
        /// Initializes a new simple reference link with the specified identifier.
        /// </summary>
        /// <param name="id">The link identifier or address.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public Link(TLinkAddress id) : this(id, default!) { }

        /// <summary>
        /// Returns the string representation of this link in Links Notation format.
        /// </summary>
        /// <returns>A string representation of the link with proper escaping and
        ↪  formatting.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public override string ToString() => Id == null ?
         $"({GetValuesString()})" :
            (Values == null || Values.Count == 0) ?
                $"({EscapeReference(Id.ToString())})" :
                $"({EscapeReference(Id.ToString())}: {GetValuesString()})";

        /// <summary>
        /// Gets the string representation of the link's values.
        /// </summary>
        /// <returns>A space-separated string of the values, or empty string if no values
        ↪  exist.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public string GetValuesString()
        {
            if (Values == null || Values.Count == 0)
            {
                return "";
            }
            var sb = new StringBuilder();
            for (int i = 0; i < Values.Count; i++)
            {
                if (i > 0)
                {
                    sb.Append(' ');
                }
```

```
                sb.Append(GetValueString(Values[i]));
            }
            return sb.ToString();
        }

        /// <summary>
        /// Simplifies the link structure by unwrapping single-value containers and recursively
        ↪  simplifying nested values.
        /// </summary>
        /// <returns>A simplified version of this link.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public Link<TLinkAddress> Simplify()
        {
            if (Values == null || Values.Count == 0)
            {
                return this;
            }
            else if (Values.Count == 1)
            {
                return Values[0];
            }
            else
            {
                var newValues = new Link<TLinkAddress>[Values.Count];
                for (int i = 0; i < Values.Count; i++)
                {
                    newValues[i] = Values[i].Simplify();
                }
                return new Link<TLinkAddress>(Id, newValues);
            }
        }

        /// <summary>
        /// Combines this link with another link to create a new link containing both as values.
        /// </summary>
        /// <param name="other">The other link to combine with.</param>
        /// <returns>A new link containing both links as values.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public Link<TLinkAddress> Combine(Link<TLinkAddress> other) => new
        ↪  Link<TLinkAddress>(this, other);

        /// <summary>
        /// Gets the string representation of a link value, choosing the appropriate format.
        /// </summary>
        /// <param name="value">The link value to convert to string.</param>
        /// <returns>The string representation of the link value.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static string GetValueString(Link<TLinkAddress> value) =>
        ↪  value.ToLinkOrIdString();

        /// <summary>
        /// Escapes a reference string for safe use in Links Notation format by adding quotes if
        ↪  necessary.
        /// </summary>
        /// <param name="reference">The reference string to escape.</param>
        /// <returns>The escaped reference string with appropriate quoting.</returns>
        public static string EscapeReference(string? reference)
        {
            if (reference == null)
            {
                return "";
            }
            return ReferenceLiteral.Format(reference);
        }

        /// <summary>
        /// Reads the delimiter a reference was written between from the reference as it was
        /// written.
        /// </summary>
        /// <param name="raw">One reference exactly as written.</param>
        /// <returns>
        /// <c>'</c>, <c>"</c> or <c>`</c> when <paramref name="raw"/> is one complete quoted
        /// reference, however many delimiters open it; <see langword="null"/> for a bare
        /// reference (<c>'unclosed</c> is one), a versioned literal, more than one reference
        /// (<c>'a' b</c>) or no text.
        /// </returns>
        public static char? ReferenceQuote(string? raw) => Quotes.ReferenceQuote(raw);

        /// <summary>
        /// Returns a copy of this link that records <paramref name="rawId"/> as the way its
```

```
        /// identifier was written.
        /// </summary>
        /// <param name="rawId">The identifier exactly as written, or <see
        ↪ langword="null"/>.</param>
        /// <returns>An equal link whose <see cref="RawId"/> is <paramref
        ↪ name="rawId"/>.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public Link<TLinkAddress> WithRawId(string? rawId) => this with { RawId = rawId };

        /// <summary>
        /// Converts the link to its string representation, choosing between simple reference
        ↪ format or full link format.
        /// </summary>
        /// <returns>A string representation optimized for the link's content.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public string ToLinkOrIdString() => (Values == null || Values.Count == 0) ? (Id == null
        ↪ ? "" : EscapeReference(Id?.ToString())) : ToString();

        /// <summary>
        /// Implicitly converts a value to a simple reference link.
        /// </summary>
        /// <param name="value">The value to convert.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator Link<TLinkAddress>(TLinkAddress value) => new
        ↪ Link<TLinkAddress>(value);

        /// <summary>
        /// Implicitly converts a tuple of identifier and values to a link.
        /// </summary>
        /// <param name="value">The tuple containing identifier and values.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator Link<TLinkAddress>((TLinkAddress,
        ↪ IList<Link<TLinkAddress>>) value) => new(value.Item1, value.Item2);

        /// <summary>
        /// Implicitly converts a source-target pair to an anonymous link.
        /// </summary>
        /// <param name="value">The tuple containing source and target links.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator Link<TLinkAddress>((Link<TLinkAddress> source,
        ↪ Link<TLinkAddress> target) value) => new(value.source, value.target);

        /// <summary>
        /// Implicitly converts a tuple of identifier, source, and target to a link.
        /// </summary>
        /// <param name="value">The tuple containing id, source and target.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator Link<TLinkAddress>((id<TLinkAddress> id,
        ↪ Link<TLinkAddress> source, Link<TLinkAddress> target) value) => new(value.id.Id,
        ↪ new[] { value.source, value.target });

        /// <summary>
        /// Implicitly converts a source-linker-target triplet to an anonymous link.
        /// </summary>
        /// <param name="value">The tuple containing source, linker, and target links.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator Link<TLinkAddress>((Link<TLinkAddress> source,
        ↪ Link<TLinkAddress> linker, Link<TLinkAddress> target) value) => new(value.source,
        ↪ value.linker, value.target);

        /// <summary>
        /// Implicitly converts a tuple of identifier, source, linker, and target to a link.
        /// </summary>
        /// <param name="value">The tuple containing id, source, linker, and target.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator Link<TLinkAddress>((id<TLinkAddress> id,
        ↪ Link<TLinkAddress> source, Link<TLinkAddress> linker, Link<TLinkAddress> target)
        ↪ value) => new(value.id.Id, new[] { value.source, value.linker, value.target });

        /// <summary>
        /// Determines whether the specified object is equal to this link.
        /// </summary>
        /// <param name="obj">The object to compare with this link.</param>
        /// <returns>True if the specified object is equal to this link; otherwise,
        ↪ false.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public override bool Equals(object? obj) => obj is Link<TLinkAddress> link &&
        ↪ Equals(link);

```

```
247         /// <summary>
248         /// Returns the hash code for this link.
249         /// </summary>
250         /// <returns>A hash code for this link.</returns>
251         [MethodImpl(MethodImplOptions.AggressiveInlining)]
252         public override int GetHashCode() => (Id, (Values ??
            ↪ Array.Empty<Link<TLinkAddress>>()).GenerateHashCode()).GetHashCode();
253
254         /// <summary>
255         /// Indicates whether the current link is equal to another link.
256         /// Two anonymous links (both with null Ids) are considered equal if their values are
            ↪ equal.
257         /// </summary>
258         /// <param name="other">The link to compare with this link.</param>
259         /// <returns>True if the current link is equal to the other parameter; otherwise,
            ↪ false.</returns>
260         [MethodImpl(MethodImplOptions.AggressiveInlining)]
261         public bool Equals(Link<TLinkAddress> other)
262         {
263             // Both have null IDs - compare only values
264             if (Id == null && other.Id == null)
265             {
266                 return (Values ?? Array.Empty<Link<TLinkAddress>>()).EqualTo(other.Values ??
                    ↪ Array.Empty<Link<TLinkAddress>>());
267             }
268
269             // Only one has null ID - not equal
270             if (Id == null || other.Id == null)
271             {
272                 return false;
273             }
274
275             // Both have IDs - compare IDs and values
276             return EqualityComparerInstance.Equals(Id, other.Id) &&
277                    (Values ?? Array.Empty<Link<TLinkAddress>>()).EqualTo(other.Values ??
                        ↪ Array.Empty<Link<TLinkAddress>>());
278         }
279
280         /// <summary>
281         /// Determines whether two link instances are equal.
282         /// </summary>
283         /// <param name="left">The first link to compare.</param>
284         /// <param name="right">The second link to compare.</param>
285         /// <returns>True if the links are equal; otherwise, false.</returns>
286         [MethodImpl(MethodImplOptions.AggressiveInlining)]
287         public static bool operator ==(Link<TLinkAddress> left, Link<TLinkAddress> right) =>
            ↪ left.Equals(right);
288
289         /// <summary>
290         /// Determines whether two link instances are not equal.
291         /// </summary>
292         /// <param name="left">The first link to compare.</param>
293         /// <param name="right">The second link to compare.</param>
294         /// <returns>True if the links are not equal; otherwise, false.</returns>
295         [MethodImpl(MethodImplOptions.AggressiveInlining)]
296         public static bool operator !=(Link<TLinkAddress> left, Link<TLinkAddress> right) =>
            ↪ !(left == right);
297     }
298 }
```

1.19 ./Link.Foundation.Links.Notation/LinkFormatExtensions.cs

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Runtime.CompilerServices;
5  using System.Text;
6
7  namespace Link.Foundation.Links.Notation
8  {
9      /// <summary>
10     /// Provides extension methods for formatting <see cref="Link{TLinkAddress}"/> instances with
       ↪ FormatOptions.
11     /// </summary>
12     public static class LinkFormatExtensions
13     {
14         /// <summary>
15         /// Format a link using FormatOptions configuration.
16         /// </summary>
17         /// <typeparam name="TLinkAddress">The type used for link
           ↪ addresses/identifiers.</typeparam>
```

```
        /// <param name="link">The link to format.</param>
        /// <param name="options">The FormatOptions to use.</param>
        /// <returns>Formatted string representation.</returns>
        public static string FormatWithOptions<TLinkAddress>(this Link<TLinkAddress> link,
        ↪  FormatOptions options)
        {
            // Empty link
            if (link.Id == null && (link.Values == null || link.Values.Count == 0))
            {
                return "()";
            }

            // Link with only ID, no values
            if (link.Values == null || link.Values.Count == 0)
            {
                var escapedId = Link<TLinkAddress>.EscapeReference(link.Id?.ToString());
                return options.LessParentheses && !NeedsParentheses(link.Id?.ToString()) ?
                    escapedId :
                    $"({escapedId})";
            }

            // Check if we should use indented format
            bool shouldIndent = false;
            if (options.ShouldIndentByRefCount(link.Values.Count))
            {
                shouldIndent = true;
            }
            else
            {
                // Try inline format first
                var valuesStr = string.Join(" ", link.Values.Select(v => GetValueString(v)));
                string testLine;
                if (link.Id != null)
                {
                    var idStr = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
                    testLine = options.LessParentheses ? $"{idStr}: {valuesStr}" : $"({idStr}:
                    ↪  {valuesStr})";
                }
                else
                {
                    testLine = options.LessParentheses && !link.HasLoneNestedValue() ? valuesStr
                    ↪  : $"({valuesStr})";
                }

                if (options.ShouldIndentByLength(testLine))
                {
                    shouldIndent = true;
                }
            }

            // Format with indentation if needed
            if (shouldIndent && options.PreferInline == false)
            {
                return FormatIndented(link, options);
            }

            // Standard inline formatting
            var values = string.Join(" ", link.Values.Select(v => GetValueString(v)));

            // Link with values only (null id)
            if (link.Id == null)
            {
                if (options.LessParentheses && !link.HasLoneNestedValue())
                {
                    var allSimple = link.Values.All(v => v.Values == null || v.Values.Count ==
                    ↪  0);
                    if (allSimple)
                    {
                        return string.Join(" ", link.Values.Select(v =>
                        ↪  Link<TLinkAddress>.EscapeReference(v.Id?.ToString())));
                    }
                    return values;
                }
                return $"({values})";
            }

            // Link with ID and values
            var id = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
            var withColon = $"{id}: {values}";
            return options.LessParentheses && !NeedsParentheses(link.Id?.ToString()) ?
```

```
                withColon :
                $"({withColon})";
        }

        /// <summary>
        /// Whether the link holds exactly one link that has values. The parentheses of an
        /// anonymous link are what nest such a value: ((b c)) without them is (b c), a
        /// different link, so less parentheses has to keep them.
        /// </summary>
        internal static bool HasLoneNestedValue<TLinkAddress>(this Link<TLinkAddress> link)
        {
            return link.Values != null && link.Values.Count == 1 && link.Values[0].Values !=
            ↪  null && link.Values[0].Values!.Count > 0;
        }

        /// <summary>
        /// Format a link with indentation.
        /// </summary>
        private static string FormatIndented<TLinkAddress>(Link<TLinkAddress> link,
        ↪  FormatOptions options)
        {
            // FormatIndented is only reached with a non-empty Values collection,
            // but the nullable annotation on Link<T>.Values does not know that.
            var values = link.Values ?? Array.Empty<Link<TLinkAddress>>();

            if (link.Id == null)
            {
                // Values only - format each on separate line
                var lines = values.Select(v => options.IndentString + GetValueString(v));
                return string.Join(Environment.NewLine, lines);
            }

            // Link with ID - format as id:\n  value1\n  value2
            var idStr = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
            var sb = new StringBuilder();
            sb.Append($"{idStr}:");

            foreach (var v in values)
            {
                sb.Append(Environment.NewLine);
                sb.Append(options.IndentString);
                sb.Append(GetValueString(v));
            }

            return sb.ToString();
        }

        /// <summary>
        /// Get the string representation of a value.
        /// </summary>
        private static string GetValueString<TLinkAddress>(Link<TLinkAddress> value)
        {
            return value.ToLinkOrIdString();
        }

        /// <summary>
        /// Check if a string needs to be wrapped in parentheses.
        /// </summary>
        private static bool NeedsParentheses(string? s)
        {
            return s != null && (s.Contains(" ") || s.Contains(":") || s.Contains("(") ||
            ↪  s.Contains(")"));
        }
    }
}
```

1.20 ./Link.Foundation.Links.Notation/LinksGroup.cs

```
using System;
using System.Collections.Generic;
using System.Runtime.CompilerServices;
using Platform.Collections;
using Platform.Collections.Lists;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Represents a group of links with hierarchical structure, where each group contains a
    ↪  primary link and optional nested groups.
    /// This structure supports the indentation-based syntax of the Links Notation.
    /// </summary>
```

```
13      /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
        ↪  any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
14      public struct LinksGroup<TLinkAddress> : IEquatable<LinksGroup<TLinkAddress>>
15      {
16          /// <summary>
17          /// Gets or sets the primary link of this group.
18          /// </summary>
19          public Link<TLinkAddress> Link
20          {
21              [MethodImpl(MethodImplOptions.AggressiveInlining)]
22              get;
23              [MethodImpl(MethodImplOptions.AggressiveInlining)]
24              set;
25          }
26
27          /// <summary>
28          /// Gets or sets the collection of nested groups within this group. Can be null if no
            ↪  nested groups exist.
29          /// </summary>
30          public IList<LinksGroup<TLinkAddress>>? Groups
31          {
32              [MethodImpl(MethodImplOptions.AggressiveInlining)]
33              get;
34              [MethodImpl(MethodImplOptions.AggressiveInlining)]
35              set;
36          }
37
38          /// <summary>
39          /// Gets or sets a value indicating whether this group was written as a parenthesized
            ↪  group.
40          /// A parenthesized group is a link in its own right, so it keeps its identifier and
            ↪  values
41          /// instead of being unwrapped when it is used as a child of the indented identifier
            ↪  syntax.
42          /// This is a parse-time hint and therefore takes no part in equality comparisons.
43          /// </summary>
44          internal bool IsParenthesized
45          {
46              [MethodImpl(MethodImplOptions.AggressiveInlining)]
47              get;
48              [MethodImpl(MethodImplOptions.AggressiveInlining)]
49              set;
50          }
51
52          /// <summary>
53          /// Initializes a new links group with the specified link and nested groups.
54          /// </summary>
55          /// <param name="link">The primary link of this group.</param>
56          /// <param name="groups">The nested groups within this group.</param>
57          [MethodImpl(MethodImplOptions.AggressiveInlining)]
58          public LinksGroup(Link<TLinkAddress> link, IList<LinksGroup<TLinkAddress>>? groups)
59          {
60              Link = link;
61              Groups = groups;
62          }
63
64          /// <summary>
65          /// Initializes a new links group with the specified link and no nested groups.
66          /// </summary>
67          /// <param name="link">The primary link of this group.</param>
68          [MethodImpl(MethodImplOptions.AggressiveInlining)]
69          public LinksGroup(Link<TLinkAddress> link) : this(link, null) { }
70
71          /// <summary>
72          /// Implicitly converts a links group to a list of links by flattening the hierarchical
            ↪  structure.
73          /// </summary>
74          /// <param name="value">The links group to convert.</param>
75          [MethodImpl(MethodImplOptions.AggressiveInlining)]
76          public static implicit operator List<Link<TLinkAddress>>(LinksGroup<TLinkAddress> value)
            ↪  => value.ToLinksList();
77
78          /// <summary>
79          /// Converts this links group to a flat list of links, preserving the hierarchical
            ↪  relationships.
80          /// </summary>
81          /// <returns>A list of links representing the flattened structure of this
            ↪  group.</returns>
82          [MethodImpl(MethodImplOptions.AggressiveInlining)]
83          public List<Link<TLinkAddress>> ToLinksList()
84          {
```

```
85              var list = new List<Link<TLinkAddress>>();
86              AppendToLinksList(list);
87              return list;
88          }
89
90          /// <summary>
91          /// Appends this links group to the specified list as flattened links.
92          /// </summary>
93          /// <param name="list">The list to append the links to.</param>
94          [MethodImpl(MethodImplOptions.AggressiveInlining)]
95          public void AppendToLinksList(List<Link<TLinkAddress>> list) => AppendToLinksList(list,
            ↪ Link, this);
96
97          /// <summary>
98          /// Recursively appends a links group to a list, combining dependencies with nested
            ↪ groups to create a flattened structure.
99          /// </summary>
100         /// <param name="list">The list to append the links to.</param>
101         /// <param name="dependency">The dependency link to combine with nested groups.</param>
102         /// <param name="group">The links group to process.</param>
103         [MethodImpl(MethodImplOptions.AggressiveInlining)]
104         public static void AppendToLinksList(List<Link<TLinkAddress>> list, Link<TLinkAddress>
            ↪ dependency, LinksGroup<TLinkAddress> group)
105         {
106             list.Add(dependency);
107             var groups = group.Groups;
108             if (groups != null && !groups.IsNullOrEmpty())
109             {
110                 for (int i = 0; i < groups.Count; i++)
111                 {
112                     var innerGroup = groups[i];
113                     AppendToLinksList(list, dependency.Combine(innerGroup.Link), innerGroup);
114                 }
115             }
116         }
117
118         /// <summary>
119         /// Determines whether the specified object is equal to this links group.
120         /// </summary>
121         /// <param name="obj">The object to compare with this links group.</param>
122         /// <returns>True if the specified object is equal to this links group; otherwise,
            ↪ false.</returns>
123         [MethodImpl(MethodImplOptions.AggressiveInlining)]
124         public override bool Equals(object? obj) => obj is LinksGroup<TLinkAddress> linksGroup ?
            ↪ Equals(linksGroup) : false;
125
126         /// <summary>
127         /// Returns the hash code for this links group.
128         /// </summary>
129         /// <returns>A hash code for this links group.</returns>
130         [MethodImpl(MethodImplOptions.AggressiveInlining)]
131         public override int GetHashCode() => (Link, (Groups ??
            ↪ Array.Empty<LinksGroup<TLinkAddress>>()).GenerateHashCode()).GetHashCode();
132
133         /// <summary>
134         /// Indicates whether the current links group is equal to another links group.
135         /// </summary>
136         /// <param name="other">The links group to compare with this links group.</param>
137         /// <returns>True if the current links group is equal to the other parameter; otherwise,
            ↪ false.</returns>
138         [MethodImpl(MethodImplOptions.AggressiveInlining)]
139         public bool Equals(LinksGroup<TLinkAddress> other) => Link.Equals(other.Link) && (Groups
            ↪ ?? Array.Empty<LinksGroup<TLinkAddress>>()).EqualTo(other.Groups ??
            ↪ Array.Empty<LinksGroup<TLinkAddress>>());
140
141         /// <summary>
142         /// Determines whether two links group instances are equal.
143         /// </summary>
144         /// <param name="left">The first links group to compare.</param>
145         /// <param name="right">The second links group to compare.</param>
146         /// <returns>True if the links groups are equal; otherwise, false.</returns>
147         [MethodImpl(MethodImplOptions.AggressiveInlining)]
148         public static bool operator ==(LinksGroup<TLinkAddress> left, LinksGroup<TLinkAddress>
            ↪ right) => left.Equals(right);
149
150         /// <summary>
151         /// Determines whether two links group instances are not equal.
152         /// </summary>
153         /// <param name="left">The first links group to compare.</param>
```

```
        /// <param name="right">The second links group to compare.</param>
        /// <returns>True if the links groups are not equal; otherwise, false.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static bool operator !=(LinksGroup<TLinkAddress> left, LinksGroup<TLinkAddress>
        ↪  right) => !(left == right);

        /// <summary>
        /// Returns a string representation of this links group.
        /// </summary>
        /// <returns>A string representation of this links group.</returns>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public override string ToString() => ToLinksList().Format();
    }
}
```

1.21 ./Link.Foundation.Links.Notation/ParseException.cs

```
using System;
using System.Text;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// The error raised when a document does not parse, carrying the position the parser
    /// stopped at.
    /// </summary>
    /// <remarks>
    /// The position is the furthest one the parser reached, which is the character the
    /// document stops making sense at rather than the point the last alternative gave up
    /// on. It derives from <see cref="FormatException"/>, the exception the generated
    /// parser raised on its own, so code that catches that keeps working.
    /// A document nested deeper than <see cref="Parser.MaxDepth"/> is refused with this
    /// exception too; then <see cref="MaxDepth"/> says how deep the nesting may go, and the
    /// position is the group or the line that is one level too deep.
    /// </remarks>
    public class ParseException : FormatException
    {
        /// <summary>The number of characters a quoted line is cut down to.</summary>
        private const int QuotedLineWidth = 80;

        /// <summary>What a message writes in place of the part of a long line it left
        ↪  out.</summary>
        private const string Ellipsis = "...";

        /// <summary>
        /// Creates an exception describing where <paramref name="subject"/> stopped parsing.
        /// </summary>
        /// <param name="subject">The document that failed to parse.</param>
        /// <param name="offset">Offset of the position the parser stopped at.</param>
        /// <param name="innerException">The error the generated parser raised, if any.</param>
        public ParseException(string subject, int offset, Exception? innerException = null)
            : this(Locate(subject ?? string.Empty, offset), null, innerException)
        {
        }

        /// <summary>
        /// Creates an exception saying that <paramref name="subject"/> nests deeper than
        /// <paramref name="maxDepth"/> at <paramref name="offset"/>.
        /// </summary>
        /// <param name="subject">The document that nests too deeply.</param>
        /// <param name="offset">Offset of the group or the line that is one level too
        ↪  deep.</param>
        /// <param name="maxDepth">The deepest nesting allowed.</param>
        /// <param name="innerException">The error the parser raised, if any.</param>
        public ParseException(string subject, int offset, int maxDepth, Exception?
        ↪  innerException = null)
            : this(Locate(subject ?? string.Empty, offset), maxDepth, innerException)
        {
        }

        private ParseException(Position position, int? maxDepth, Exception? innerException)
            : base(Describe(position, maxDepth), innerException)
        {
            Offset = position.Offset;
            Line = position.Line;
            Column = position.Column;
            Found = position.Found;
            LineText = position.LineText;
            MaxDepth = maxDepth;
        }

        /// <summary>Offset of the offending position from the start of the document.</summary>
```

```
63          public int Offset { get; }
64
65          /// <summary>Line the offending position is on, counted from 1.</summary>
66          public int Line { get; }
67
68          /// <summary>Column the offending position is at, counted from 1.</summary>
69          public int Column { get; }
70
71          /// <summary>The character found instead, or null at the end of the document.</summary>
72          public char? Found { get; }
73
74          /// <summary>The offending line, as written, without its line ending.</summary>
75          public string LineText { get; }
76
77          /// <summary>
78          /// The deepest nesting allowed, when the document is nested deeper than that;
79          /// <see langword="null"/> for any other error.
80          /// </summary>
81          public int? MaxDepth { get; }
82
83          /// <summary>
84          /// The one-line summary: where the parser stopped and what stands there, or that the
85          /// nesting is too deep there.
86          /// </summary>
87          public string Summary => Summarize(new Position(Offset, Line, Column, Found, LineText),
            ↪  MaxDepth);
88
89          /// <summary>
90          /// The offending line with a caret under the offending column, quoted the way a
91          /// compiler quotes source. A long line is shown as a window around the caret, so the
92          /// message stays the same size whether the document has ten lines or fifteen hundred.
93          /// </summary>
94          public string Snippet => Quote(new Position(Offset, Line, Column, Found, LineText));
95
96          /// <summary>Where the parser stopped, and what the document holds there.</summary>
97          private readonly record struct Position(
98              int Offset,
99              int Line,
100             int Column,
101             char? Found,
102             string LineText);
103
104         /// <summary>The message the exception carries: the summary and the quoted
            ↪  line.</summary>
105         private static string Describe(Position position, int? maxDepth) =>
106             $"{(maxDepth.HasValue ? "Nesting too deep" : "Syntax error")} at {Summarize(position,
                ↪  maxDepth)}\n{Quote(position)}";
107
108         private static string Summarize(Position position, int? maxDepth)
109         {
110             if (maxDepth.HasValue)
111             {
112                 return $"line {position.Line}, column {position.Column}: nesting depth exceeds
                    ↪  the maximum of {maxDepth.Value}";
113             }
114             var found = position.Found.HasValue
115                 ? $"\"{Escape(position.Found.Value)}\""
116                 : "end of input";
117             return $"line {position.Line}, column {position.Column}: unexpected {found}";
118         }
119
120         private static string Quote(Position position)
121         {
122             var (quoted, column) = QuoteLine(position.LineText, position.Column);
123             var number = position.Line.ToString();
124             var gutter = new string(' ', number.Length);
125             return $"{number} | {quoted}\n{gutter} | {new string(' ', column - 1)}^";
126         }
127
128         private static string Escape(char character) => character switch
129         {
130             '\n' => "\\n",
131             '\r' => "\\r",
132             '\t' => "\\t",
133             '"' => "\\\"",
134             '\\' => "\\\\",
135               => character.ToString(),
136         };
137
138         /// <summary>
139         /// Turns the position the parser stopped at into a line, a column and the line
```

```
        /// itself, so the message can point at the defect instead of quoting the rest of the
        /// document.
        /// </summary>
        private static Position Locate(string document, int offset)
        {
            offset = Math.Max(0, Math.Min(offset, document.Length));

            var line = 1;
            var lineStart = 0;
            for (var index = 0; index < offset; index++)
            {
                if (document[index] == '\n')
                {
                    line++;
                    lineStart = index + 1;
                }
            }

            var lineEnd = document.IndexOf('\n', lineStart);
            if (lineEnd < 0) lineEnd = document.Length;

            var lineText = document.Substring(lineStart, lineEnd - lineStart).TrimEnd('\r');
            var found = offset < document.Length ? document[offset] : (char?)null;
            return new Position(offset, line, offset - lineStart + 1, found, lineText);
        }

        /// <summary>
        /// Cuts <paramref name="line"/> down to a window around <paramref name="column"/>, and
        /// says which column the offending character sits at in that window. Both columns
        /// count from 1.
        /// </summary>
        private static (string Quoted, int Column) QuoteLine(string line, int column)
        {
            if (line.Length <= QuotedLineWidth) return (line, column);

            var target = column - 1;
            var lastStart = line.Length - QuotedLineWidth;
            var start = Math.Min(Math.Max(target - (QuotedLineWidth / 2), 0), lastStart);
            var end = start + QuotedLineWidth;

            var quoted = new StringBuilder();
            if (start > 0) quoted.Append(Ellipsis);
            quoted.Append(line, start, QuotedLineWidth);
            if (end < line.Length) quoted.Append(Ellipsis);

            var shift = start > 0 ? Ellipsis.Length : 0;
            return (quoted.ToString(), target - start + shift + 1);
        }
    }
}
```

1.22 ./Link.Foundation.Links.Notation/Parser.cs

```
using System;
using System.Collections.Generic;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Parses Links Notation documents into links.
    /// </summary>
    /// <remarks>
    /// The rules live in <c>Parser.peg</c> and are compiled into
    /// <see cref="GeneratedParser"/>; this class runs them and, when a document does not
    /// parse, reports the position the parser stopped at instead of the bare
    /// <see cref="FormatException"/> the generated parser raises.
    /// </remarks>
    public class Parser
    {
        /// <summary>
        /// How deep links may nest when nothing says otherwise.
        /// </summary>
        /// <remarks>
        /// The parser recurses once per level of nesting, and running out of stack ends the
        /// process where no caller can catch it. The limit is set well below what fits in a
        /// 1 MiB thread stack, and it is the same in every Links Notation implementation, so
        /// a document one of them reads is not too deep for another.
        /// </remarks>
        public const int DefaultMaxDepth = 64;

        /// <summary>
        /// Creates a parser.
```

```
        /// </summary>
        /// <param name="comments">
        /// When <see langword="false"/>, <c>#</c> is read as an ordinary character
        /// instead of the start of a comment.
        /// </param>
        public Parser(bool comments = true) : this(comments, DefaultMaxDepth)
        {
        }

        /// <summary>
        /// Creates a parser that refuses links nested deeper than <paramref name="maxDepth"/>.
        /// </summary>
        /// <param name="comments">
        /// When <see langword="false"/>, <c>#</c> is read as an ordinary character
        /// instead of the start of a comment.
        /// </param>
        /// <param name="maxDepth">How deep links may nest; see <see cref="MaxDepth"/>.</param>
        /// <exception cref="ArgumentOutOfRangeException"><paramref name="maxDepth"/> is
        ↪  negative.</exception>
        public Parser(bool comments, int maxDepth)
        {
            Comments = comments;
            MaxDepth = maxDepth;
        }

        /// <summary>
        /// Whether <c>#</c> starts a comment that runs to the end of its line.
        /// </summary>
        public bool Comments { get; }

        /// <summary>
        /// How deep links may nest. Every parenthesized group and every indentation level
        /// is one level, and the lines of a document are at level 0, so with a limit of 1
        /// <c>(a)</c> parses while <c>((a))</c>, <c>(a (b))</c> and a group on an indented
        /// line do not. A document nested deeper is refused with a
        /// <see cref="ParseException"/> whose <see cref="ParseException.MaxDepth"/> is set,
        /// rather than recursed into until the stack runs out.
        /// </summary>
        /// <exception cref="ArgumentOutOfRangeException">The value is negative.</exception>
        public int MaxDepth
        {
            get =>  maxDepth;
            init =>  maxDepth = value >= 0
                ? value
                : throw new ArgumentOutOfRangeException(nameof(MaxDepth), value, "Maximum nesting
                ↪  depth cannot be negative.");
        }

        private readonly int  maxDepth;

        /// <summary>
        /// Whether parsed links record how their identifiers were written, in
        /// <see cref="Link{TLinkAddress}.RawId"/> and <see cref="Link{TLinkAddress}.Quote"/>.
        /// <see langword="true"/> by default; set it to <see langword="false"/> to read links
        /// as a bare abstract syntax tree (AST mode), in which both are always
        /// <see langword="null"/>. Equality and formatting are the same either way.
        /// </summary>
        public bool Syntax { get; init; } = true;

        /// <summary>
        /// Parses a Links Notation document.
        /// </summary>
        /// <param name="subject">The document to parse.</param>
        /// <param name="fileName">The optional file name to use in error messages.</param>
        /// <returns>The links the document holds.</returns>
        /// <exception cref="ParseException">
        /// Thrown when the document does not parse, or nests deeper than
        /// <see cref="MaxDepth"/>. The exception carries the line, the column and the
        /// offending line of the position the parser stopped at.
        /// </exception>
        public IList<Link<string>> Parse(string subject, string? fileName = null)
        {
            // Comments are blanked rather than removed, so a position reported for the
            // prepared document is the same position in the document the caller wrote.
            var prepared = Comments ? Notation.Comments.StripComments(subject) : subject;
            var tracer = new FurthestFailureTracer();
            var parser = new GeneratedParser { Tracer = tracer, MaxDepth = MaxDepth, Syntax =
            ↪  Syntax, Source = subject };
            try
            {
```

```
                return parser.Parse(prepared, fileName);
            }
            catch (GeneratedParser.NestingTooDeepException error)
            {
                throw new ParseException(subject, error.Offset, error.MaxDepth, error);
            }
            catch (FormatException error) when (error is not ParseException)
            {
                throw new ParseException(subject, tracer.Furthest, error);
            }
        }

        /// <summary>
        /// Reads a Links Notation document as a lossless concrete syntax tree.
        /// </summary>
        /// <param name="subject">The document to read.</param>
        /// <returns>
        /// The <see cref="CstKind.Document"/> node. Every character of the document,
        /// comments, whitespace and line endings included, belongs to exactly one token, so
        /// its <see cref="CstNode.Text"/> is <paramref name="subject"/>.
        /// </returns>
        /// <exception cref="ParseException">
        /// Thrown exactly when <see cref="Parse"/> refuses the document: it is parsed with
        /// this parser's options first.
        /// </exception>
        public CstNode ParseCst(string subject)
        {
            ArgumentNullException.ThrowIfNull(subject);
            Parse(subject);
            return CstBuilder.Build(subject, Comments);
        }
    }
}
```

1.23 ./Link.Foundation.Links.Notation/Quotes.cs

```
using System.Text;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// Reads a delimited reference, the shape shared by the grammar and by
    /// <see cref="Comments"/>.
    /// </summary>
    /// <remarks>
    /// Any number N of delimiters opens and closes a reference, and 2N delimiters stand
    /// for N written as content. The grammar reads references through this class, so
    /// blanking comments reaches a reference exactly as far as the parser does.
    /// </remarks>
    internal static class Quotes
    {
        /// <summary>
        /// Reports whether the character is one of the delimiters a reference can be
        /// written between.
        /// </summary>
        /// <param name="character">The character to test.</param>
        /// <returns><see langword="true"/> when the character opens a reference.</returns>
        internal static bool IsQuote(char character) =>
            character == '"' || character == '\'' || character == '`';

        /// <summary>
        /// A body written between an even run of delimiters is substantive when it
        /// holds at least one visible character and does not straddle a parenthesis.
        /// An even run can always be read as delimiter pairs enclosing nothing, so the
        /// n-quote reading is only taken when it carries something the pairs cannot.
        /// </summary>
        /// <param name="content">The body to weigh.</param>
        /// <returns><see langword="true"/> when the body carries something a pair
        ↪  cannot.</returns>
        internal static bool IsSubstantiveBody(string content)
        {
            int depth = 0;
            bool hasVisible = false;

            foreach (var c in content)
            {
                if (c == '(')
                {
                    depth++;
                }
                else if (c == ')')
```

```
                {
                    depth--;
                    if (depth < 0) return false;
                }

                if (!char.IsWhiteSpace(c)) hasVisible = true;
            }

            return hasVisible && depth == 0;
        }

        /// <summary>
        /// Reads the reference starting at the given position.
        /// </summary>
        /// <param name="subject">The whole document being read.</param>
        /// <param name="start">The position of the opening delimiter.</param>
        /// <param name="value">The content of the reference, without its delimiters.</param>
        /// <param name="length">How many characters the reference spans.</param>
        /// <returns><see langword="true"/> when a reference was read at that position.</returns>
        internal static bool TryParse(string? subject, int start, out string? value, out int
        ↪  length)
        {
            value = null;
            length = 0;
            if (subject == null || start >= subject.Length || !IsQuote(subject[start])) return
            ↪  false;

            char quoteChar = subject[start];

            // Count opening quotes
            int quoteCount = 0;
            int pos = start;
            while (pos < subject.Length && subject[pos] == quoteChar)
            {
                quoteCount++;
                pos++;
            }

            bool isEvenRun = quoteCount % 2 == 0;

            string openClose = new string(quoteChar, quoteCount);
            string escapeSeq = new string(quoteChar, quoteCount * 2);

            var content = new StringBuilder();
            while (pos < subject.Length)
            {
                // Check for escape sequence (2*N quotes)
                if (pos + escapeSeq.Length <= subject.Length &&
                    string.CompareOrdinal(subject, pos, escapeSeq, 0, escapeSeq.Length) == 0)
                {
                    content.Append(openClose); // 2*N quotes become N quotes
                    pos += escapeSeq.Length;
                    continue;
                }

                // Check for closing quotes (exactly N quotes, not more)
                if (pos + quoteCount <= subject.Length &&
                    string.CompareOrdinal(subject, pos, openClose, 0, quoteCount) == 0)
                {
                    int afterClose = pos + quoteCount;
                    if (afterClose >= subject.Length || subject[afterClose] != quoteChar)
                    {
                        // Found valid closing
                        if (isEvenRun && !IsSubstantiveBody(content.ToString()))
                        {
                            return SetEmptyReference(isEvenRun, quoteCount, out value, out
                            ↪  length);
                        }

                        value = content.ToString();
                        length = afterClose - start;
                        return true;
                    }
                }

                // Take next character
                content.Append(subject[pos]);
                pos++;
            }

            // No valid closing found
```

```
            return SetEmptyReference(isEvenRun, quoteCount, out value, out length);
        }

        /// <summary>
        /// Finds where the reference starting at the given position ends.
        /// </summary>
        /// <param name="subject">The whole document being read.</param>
        /// <param name="start">The position of the opening delimiter.</param>
        /// <returns>
        /// The position right after the closing delimiters, or -1 when no reference
        /// starts there.
        /// </returns>
        internal static int ReferenceEnd(string subject, int start) =>
            TryParse(subject, start, out var ignored, out var length) ? start + length : -1;

        /// <summary>
        /// Reports whether a run of delimiters after the one opening at
        /// <paramref name="start"/> closes it: a run of R closes a reference opened by N
        /// exactly when R / N is odd. An even run nothing closes is the empty reference,
        /// until a closing run is written after it.
        /// </summary>
        /// <param name="subject">The whole document being read.</param>
        /// <param name="start">The position of the opening delimiter.</param>
        /// <returns><see langword="true"/> when a later run closes the reference.</returns>
        internal static bool ClosingRunFollows(string subject, int start)
        {
            char quote = subject[start];
            int count = 0;
            while (start + count < subject.Length && subject[start + count] == quote) count++;
            int position = start + count;
            while (true)
            {
                position = subject.IndexOf(quote, position);
                if (position < 0) return false;
                int length = 0;
                while (position + length < subject.Length && subject[position + length] ==
                ↪  quote) length++;
                if (length / count % 2 == 1) return true;
                position += length;
            }
        }

        /// <summary>
        /// The delimiter of a reference written as <paramref name="raw"/>, when
        /// <paramref name="raw"/> is exactly one delimited reference.
        /// </summary>
        /// <param name="raw">One reference exactly as written.</param>
        /// <returns>The first character when reading a reference there consumes all of
        /// <paramref name="raw"/>; otherwise <see langword="null"/>.</returns>
        internal static char? ReferenceQuote(string? raw) =>
            raw != null && raw.Length > 0 && IsQuote(raw[0]) && ReferenceEnd(raw, 0) ==
            ↪  raw.Length
                ? raw[0]
                : null;

        /// <summary>
        /// A run of an even number of delimiters that does not open a reference with a
        /// substantive body is the empty reference: the shortest reading, a bare
        /// delimiter pair enclosing nothing, wins over a longer n-quote delimiter.
        /// </summary>
        private static bool SetEmptyReference(bool isEvenRun, int quoteCount, out string? value,
        ↪  out int length)
        {
            if (!isEvenRun)
            {
                value = null;
                length = 0;
                return false;
            }

            value = string.Empty;
            length = quoteCount;
            return true;
        }
    }
}
```

1.24 ./Link.Foundation.Links.Notation/ReferenceLiteral.cs

```
using System;
using System.Linq;
using System.Text;
```

```cs
using System.Text.RegularExpressions;

namespace Link.Foundation.Links.Notation;

/// <summary>Version 1 lossless Unicode reference literals.</summary>
public static class ReferenceLiteral
{
    private static readonly UTF8Encoding Utf8 = new(false, true);
    private static readonly Regex Literal = new("\\A~([0-9]+)\\{([0-9a-fA-F]*)\\}\\z");

    /// <summary>Encodes exact Unicode as a version 1 UTF-8 hexadecimal literal.</summary>
    public static string Encode(string text) =>
        $"~1{{{Convert.ToHexString(Utf8.GetBytes(text)).ToLowerInvariant()}}}";

    /// <summary>Decodes a complete literal; malformed UTF-8 and versions are refused.</summary>
    public static string Decode(string literal)
    {
        var match = Literal.Match(literal);
        if (!match.Success || match.Groups[1].Value != "1" || match.Groups[2].Length % 2 != 0)
        {
            throw new FormatException("Invalid or unsupported reference literal (expected ~1{UTF-8 hex})");
        }
        try { return Utf8.GetString(Convert.FromHexString(match.Groups[2].Value)); }
        catch (DecoderFallbackException error) { throw new FormatException("Reference literal must contain valid UTF-8", error); }
    }

    internal static bool HasPrefix(string subject, int start)
    {
        if (start >= subject.Length || subject[start] != '~') return false;
        var position = start + 1;
        while (position < subject.Length && subject[position] is >= '0' and <= '9') position++;
        return position > start + 1 && position < subject.Length && subject[position] == '{';
    }

    /// <summary>
    /// Where the versioned literal <c>~&lt;digits&gt;{&lt;payload&gt;}</c> that starts at
    /// <paramref name="start"/> ends, or -1 if none starts there. The payload holds no
    /// <c>}</c>, space, tab, line break or parenthesis, as the grammar reads it.
    /// </summary>
    internal static int End(string text, int start)
    {
        if (start >= text.Length || text[start] != '~') return -1;
        var index = start + 1;
        while (index < text.Length && text[index] is >= '0' and <= '9') index++;
        if (index == start + 1 || index >= text.Length || text[index] != '{') return -1;
        index++;
        while (index < text.Length && text[index] is not ('}' or ' ' or '\t' or '\r' or '\n' or '(' or ')')) index++;
        return index < text.Length && text[index] == '}' ? index + 1 : -1;
    }

    internal static bool TryRead(string subject, int start, out string value, out int length)
    {
        value = "";
        length = 0;
        if (!HasPrefix(subject, start)) return false;
        var opening = subject.IndexOf('{', start);
        var closing = subject.IndexOf('}', opening + 1);
        if (closing < 0) throw new FormatException("Unclosed reference literal");
        length = closing + 1 - start;
        value = Decode(subject.Substring(start, length));
        return true;
    }

    internal static string Format(string text)
    {
        Utf8.GetByteCount(text); // Refuse unpaired surrogates before formatting.
        if (text.Length == 0 || text.Any(c => c < 32 || c == 127)) return Encode(text);
        var needsQuotes = HasPrefix(text, 0) || text.StartsWith('#') || text.Any(c => char.IsWhiteSpace(c) || c is '\ufeff' or '(' or ')' or ':' or '\'' or '"' or '`');
        if (!needsQuotes) return text;
        var (quote, count) = new[] { '\'', '"', '`' }
            .Where(quote => text[0] != quote)
            .Select(quote => (Quote: quote, Count: (LongestRun(text, quote) + 1) | 1))
            .MinBy(candidate => candidate.Count);
        var delimiter = new string(quote, count);
        return $"{delimiter}{text}{delimiter}";
    }
```

```

    private static int LongestRun(string text, char quote)
    {
        var (longest, current) = (0, 0);
        foreach (var character in text)
        {
            current = character == quote ? current + 1 : 0;
            longest = Math.Max(longest, current);
        }
        return longest;
    }
}
```

1.25 ./Link.Foundation.Links.Notation/StreamParseException.cs

```
using System;

namespace Link.Foundation.Links.Notation
{
    /// <summary>A parser error located relative to the complete input stream.</summary>
    public sealed class StreamParseException : FormatException
    {
        internal StreamParseException(ParseException error, int offset, int line, int column)
            : base($"Stream parse error at line {line}, column {column}: {error.Message}", error)
        {
            Offset = offset;
            Line = line;
            Column = column;
            ParseError = error;
        }

        /// <summary>The canonical parser error raised for the buffered record.</summary>
        public ParseException ParseError { get; }

        /// <summary>Absolute character offset from the start of the stream.</summary>
        public int Offset { get; }

        /// <summary>One-based line number in the complete stream.</summary>
        public int Line { get; }

        /// <summary>One-based column number in the complete stream.</summary>
        public int Column { get; }
    }
}
```

1.26 ./Link.Foundation.Links.Notation/StreamParser.cs

```
using System;
using System.Collections.Generic;
using System.Runtime.CompilerServices;
using System.Text;
using System.Threading;
using System.Threading.Tasks;

namespace Link.Foundation.Links.Notation
{
    /// <summary>Incrementally emits complete top-level Links Notation records.</summary>
    public sealed class StreamParser
    {
        private const int DefaultMaxBufferSize = 10 * 1024 * 1024;
        private const string BeforeComment = " \t\n\r";

        private readonly Parser  parser;
        private readonly StringBuilder  buffer = new();
        private readonly StringBuilder  currentLine = new();
        private readonly List<Link<string>>  links = new();
        private int?  baseIndentation;
        private bool  lineClassified;
        private int  offset;
        private int  line;
        private int  column;
        private int  segmentOffset;
        private int  segmentLine;
        private bool  ended;
        private int  maxBufferSize = DefaultMaxBufferSize;

        /// <summary>Creates a stream backed by a default canonical parser.</summary>
        public StreamParser() : this(new Parser()) { }

        /// <summary>Creates a stream backed by <paramref name="parser"/>.</summary>
        public StreamParser(Parser parser)
        {
             parser = parser ?? throw new ArgumentNullException(nameof(parser));
            Reset();
```

```
        }
                  -                  -
        /// <summary>Raised once for every completed link.</summary>
        public event Action<Link<string>>? LinkParsed;

        /// <summary>Whether emitted links are retained for <see cref="Finish()"/> and <see
        ↪  cref="Drain"/>.</summary>
        public bool Collect { get; set; } = true;

        /// <summary>The largest unresolved record accepted by the stream.</summary>
        public int MaxBufferSize
        {
            get =>  maxBufferSize;
            set =>  maxBufferSize = value > 0
                ? value
                : throw new ArgumentOutOfRangeException(nameof(value), "Maximum buffer size must
                ↪  be positive.");
        }       -

        /// <summary>The point immediately after the last character written.</summary>
        public StreamPosition Position => new(
             offset,              -
             line,
             column,
             buffer.Length +  currentLine.Length);

        /// <summary>Consumes a chunk and returns links made complete by it.</summary>
        public IReadOnlyList<Link<string>> Write(string chunk)
        {
            ArgumentNullException.ThrowIfNull(chunk);
            if ( ended) throw new InvalidOperationException("Cannot write after Finish().");

            var emitted = new List<Link<string>>();
            foreach (var character in chunk)
            {
                 currentLine.Append(character);
                 offset++;

                if (character == '\n')
                {
                     buffer.Append( currentLine);
                     currentLine.Clear();
                     lineClassified = false;
                     line++;
                     column = 1;
                }
                else
                {
                    if (! lineClassified && character != ' ' && character != '\t' && character
                    ↪  != '\r')
                    {
                         lineClassified = true;
                        if (!( parser.Comments && character == Comments.Comment))
                        {
                            StartContentLine(LeadingSpaces( currentLine), emitted);
                        }
                    }
                     column++;
                }

                if ( buffer.Length +  currentLine.Length > MaxBufferSize)
                {
                    throw new ArgumentException(
                        $"Buffered record exceeds maximum size of {MaxBufferSize} characters.",
                        nameof(chunk));
                }
            }
            return emitted;
        }

        /// <summary>Finishes the stream and returns all undrained links.</summary>
        public IReadOnlyList<Link<string>> Finish() => Finish(string.Empty);

        /// <summary>Consumes a final chunk and finishes the stream.</summary>
        public IReadOnlyList<Link<string>> Finish(string chunk)
        {
            if ( ended)
            {
                if (chunk.Length > 0) throw new InvalidOperationException("Cannot write after
                ↪  Finish().");
                return Collect ?  links.ToArray() : Array.Empty<Link<string>>();
```

```
            }

            var emitted = chunk.Length == 0
                ? new List<Link<string>>()
                : new List<Link<string>>(Write(chunk));
            var document =  buffer.ToString() +  currentLine;
            if (document.Length > 0)
            {
                IList<Link<string>> parsed;
                try
                {
                    parsed =  parser.Parse(document);
                }
                catch (ParseException error)
                {
                    throw new StreamParseException(
                        error,
                         segmentOffset + error.Offset,
                         segmentLine + error.Line - 1,
                        error.Column);
                }
                Publish(parsed, emitted);
                AdvanceSegment(document);
            }

             buffer.Clear();
             currentLine.Clear();
             baseIndentation = null;
             ended = true;
            return Collect ?  links.ToArray() : emitted;
        }

        /// <summary>Returns and forgets links retained since the previous call.</summary>
        public IReadOnlyList<Link<string>> Drain()
        {
            var links =  links.ToArray();
             links.Clear();
            return links;
        }

        /// <summary>Reuses the parser while preserving its options and event handlers.</summary>
        public StreamParser Reset()
        {
             buffer.Clear();
             currentLine.Clear();
             baseIndentation = null;
             lineClassified = false;
             links.Clear();
             offset = 0;
             line = 1;
             column = 1;
             segmentOffset = 0;
             segmentLine = 1;
             ended = false;
            return this;
        }

        /// <summary>Lazily parses a sequence of chunks as a native enumerable.</summary>
        public static IEnumerable<Link<string>> ParseChunks(
            IEnumerable<string> chunks,
            Parser? parser = null)
        {
            var stream = new StreamParser(parser ?? new Parser()) { Collect = false };
            foreach (var chunk in chunks)
            {
                foreach (var link in stream.Write(chunk)) yield return link;
            }
            foreach (var link in stream.Finish()) yield return link;
        }

        /// <summary>Lazily parses an asynchronous sequence of chunks.</summary>
        public static async IAsyncEnumerable<Link<string>> ParseChunksAsync(
            IAsyncEnumerable<string> chunks,
            Parser? parser = null,
            [EnumeratorCancellation] CancellationToken cancellationToken = default)
        {
            var stream = new StreamParser(parser ?? new Parser()) { Collect = false };
            await foreach (var chunk in chunks.WithCancellation(cancellationToken))
            {
                foreach (var link in stream.Write(chunk)) yield return link;
            }
            foreach (var link in stream.Finish()) yield return link;
```

```
        }

        private void StartContentLine(int indentation, List<Link<string>> emitted)
        {
            if ( buffer.Length > 0 &&
                 baseIndentation.HasValue &&
                indentation <=  baseIndentation.Value &&
                StructurallyComplete( buffer.ToString(),  parser.Comments))
            {
                try
                {
                    var document =  buffer.ToString();
                    var parsed =  parser.Parse(document);
                    Publish(parsed, emitted);
                    AdvanceSegment(document);
                     buffer.Clear();
                     baseIndentation = null;
                }
                catch (ParseException)
                {
                    // A rejected prefix can become valid when more input arrives.
                }
            }

             baseIndentation ??= indentation;
        }

        private void Publish(IEnumerable<Link<string>> links, List<Link<string>> emitted)
        {
            foreach (var link in links)
            {
                emitted.Add(link);
                if (Collect)  links.Add(link);
                LinkParsed?.Invoke(link);
            }
        }

        private void AdvanceSegment(string document)
        {
             segmentOffset += document.Length;
            foreach (var character in document)
            {
                if (character == '\n')  segmentLine++;
            }
        }

        private static int LeadingSpaces(StringBuilder line)
        {
            var indentation = 0;
            while (indentation < line.Length && line[indentation] == ' ') indentation++;
            return indentation;
        }

        private static bool StructurallyComplete(string document, bool comments)
        {
            // Tokens are read as the comment reader and the grammar read them: a reference
            // may start after a separator, and right after a delimited reference or a
            // literal (issue #377).
            var depth = 0;
            var tokenStart = true;
            for (var position = 0; position < document.Length; position++)
            {
                var character = document[position];
                if (comments && character == Comments.Comment && Follows(document, position,
                    ↪  BeforeComment))
                {
                    var newline = document.IndexOf('\n', position);
                    if (newline < 0) break;
                    position = newline;
                    tokenStart = true;
                    continue;
                }
                if (Comments.Separators.IndexOf(character) >= 0)
                {
                    if (character == '(') depth++;
                    else if (character == ')') depth--;
                    tokenStart = true;
                    continue;
                }
                if (tokenStart && Quotes.IsQuote(character))
                {
```

```
                    // A reference nothing closes yet may be closed by a later line, which
                    // changes what it holds: an even run is the empty reference only when
                    // no run in the whole document closes it.
                    var end = Quotes.ReferenceEnd(document, position);
                    if (end < 0 || !Quotes.ClosingRunFollows(document, position)) return false;
                    position = end - 1;
                    continue;
                }
                if (tokenStart)
                {
                    var end = ReferenceLiteral.End(document, position);
                    if (end >= 0)
                    {
                        position = end - 1;
                        continue;
                    }
                }
                tokenStart = false;
            }                       -
            return depth == 0;
        }

        private static bool Follows(string document, int position, string allowed) =>
            position == 0 || allowed.IndexOf(document[position - 1]) >= 0;
    }

    /// <summary>An absolute stream position and the unresolved record size.</summary>
    public readonly record struct StreamPosition(int Offset, int Line, int Column, int Buffered);
}
```

1.27 ./Link.Foundation.Links.Notation/ .cs

```
using System.Runtime.CompilerServices;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// A utility struct that provides convenient implicit conversions for creating links from
    ↪ various tuple combinations.
    /// The underscore name follows functional programming conventions for placeholder/utility
    ↪ types.
    /// </summary>
    /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
    ↪ any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
    public struct  <TLinkAddress>
    {
        /// <summary>
        /// Gets the underlying link represented by this utility struct.
        /// </summary>
        public readonly Link<TLinkAddress> Link;

        /// <summary>
        /// Initializes a new instance with the specified link.
        /// </summary>
        /// <param name="id">The link to wrap.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public  (Link<TLinkAddress> id) => Link = id;

        /// <summary>
        /// Implicitly converts a link to this utility struct.
        /// </summary>
        /// <param name="value">The link to convert.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator  <TLinkAddress>(Link<TLinkAddress> value) => new(value);

        /// <summary>
        /// Implicitly converts a link address to this utility struct.
        /// </summary>
        /// <param name="id">The link address to convert.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator  <TLinkAddress>(TLinkAddress id) => new(id);

        /// <summary>
        /// Implicitly converts a tuple of address and link to this utility struct.
        /// </summary>
        /// <param name="value">The tuple containing address and link.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator  <TLinkAddress>((TLinkAddress, Link<TLinkAddress>)
        ↪ value) => new Link<TLinkAddress>(value.Item1, new[] { value.Item2 });

        /// <summary>
        /// Implicitly converts a tuple of address and two links to this utility struct.
```

```
        /// </summary>
        /// <param name="value">The tuple containing address and two links.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator  <TLinkAddress>((TLinkAddress, Link<TLinkAddress>,
        ↪  Link<TLinkAddress>) value) => new Link<TLinkAddress>(value.Item1, new[] {
        ↪  value.Item2, value.Item3 });

        /// <summary>
        /// Implicitly converts a tuple of address and three links to this utility struct.
        /// </summary>
        /// <param name="value">The tuple containing address and three links.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator  <TLinkAddress>((TLinkAddress, Link<TLinkAddress>,
        ↪  Link<TLinkAddress>, Link<TLinkAddress>) value) => new
        ↪  Link<TLinkAddress>(value.Item1, new[] { value.Item2, value.Item3, value.Item4 });
                                                          -
        /// <summary>         -
        /// Implicitly converts this utility struct back to a link.
        /// </summary>
        /// <param name="value">The utility struct to convert.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static implicit operator Link<TLinkAddress>( <TLinkAddress> value) => value.Link;
    }
}
```

1.28 ./Link.Foundation.Links.Notation/id.cs

```
using System.Runtime.CompilerServices;

namespace Link.Foundation.Links.Notation
{
    /// <summary>
    /// A readonly struct that explicitly represents a link identifier/address.
    /// This type is used to distinguish between regular values and explicit identifiers in link
    ↪  construction.
    /// </summary>
    /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
    ↪  any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
    public readonly struct id<TLinkAddress>
    {
        /// <summary>
        /// Gets the identifier value.
        /// </summary>
        public readonly TLinkAddress Id;

        /// <summary>
        /// Initializes a new identifier with the specified value.
        /// </summary>
        /// <param name="id">The identifier value.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public id(TLinkAddress id) => Id = id;

        /// <summary>
        /// Explicitly converts a value to an identifier struct.
        /// </summary>
        /// <param name="id">The value to convert to an identifier.</param>
        [MethodImpl(MethodImplOptions.AggressiveInlining)]
        public static explicit operator id<TLinkAddress>(TLinkAddress id) => new(id);
    }
}
```

1.29 ./Link.Foundation.Links.Notation.Tests/ApiTests.cs

```
using System;
using System.Collections.Generic;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class ApiTests
    {
        [Fact]
        public static void IsRefTest()
        {
            // C# doesn't have separate Ref/Link types, but we can test simple link behavior
            var simpleLink = new Link<string>("some value", null);
            Assert.Equal("some value", simpleLink.Id);
            Assert.Null(simpleLink.Values);
        }

        [Fact]
        public static void IsLinkTest()
```

```
        {
            // Test link with values
            var values = new List<Link<string>> { new Link<string>("child", null) };
            var link = new Link<string>("id", values);
            Assert.Equal("id", link.Id);
            Assert.Single(link.Values!);
            Assert.Equal("child", link.Values?[0].Id);
        }

        [Fact]
        public static void IsRefEquivalentTest()
        {
            // C# doesn't have separate Ref/Link types, but we can test simple link behavior
            var simpleLink = new Link<string>("some value", null);
            Assert.Equal("some value", simpleLink.Id);
            Assert.Null(simpleLink.Values);
        }

        [Fact]
        public static void IsLinkEquivalentTest()
        {
            // Test link with values
            var values = new List<Link<string>> { new Link<string>("child", null) };
            var link = new Link<string>("id", values);
            Assert.Equal("id", link.Id);
            Assert.Single(link.Values!);
            Assert.Equal("child", link.Values?[0].Id);
        }

        [Fact]
        public static void EmptyLinkTest()
        {
            var link = new Link<string>(null, new List<Link<string>>());
            var output = link.ToString();
            Assert.Equal("()", output);
        }

        [Fact]
        public static void SimpleLinkTest()
        {
            var input = "(1: 1 1)";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // Validate regular formatting
            var output = parsed.Format();
            Assert.Contains("1:", output);
            Assert.Contains("1", output);
        }

        [Fact]
        public static void LinkWithSourceTargetTest()
        {
            var input = "(index: source target)";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // Validate regular formatting
            var output = parsed.Format();
            Assert.Equal(input, output);
        }

        [Fact]
        public static void LinkWithSourceTypeTargetTest()
        {
            var input = "(index: source type target)";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // Validate regular formatting
            var output = parsed.Format();
            Assert.Equal(input, output);
        }

        [Fact]
        public static void SingleLineFormatTest()
        {
            var input = "id: value1 value2";
            var parser = new Parser();
            var parsed = parser.Parse(input);

```

```
        // The parser should handle single-line format
        var output = parsed.Format();
        Assert.Contains("id", output);
        Assert.Contains("value1", output);
        Assert.Contains("value2", output);
    }

    [Fact]
    public static void QuotedReferencesTest()
    {
        var input = @"(""quoted id"": ""value with spaces"")";
        var parser = new Parser();
        var parsed = parser.Parse(input);

        var output = parsed.Format();
        Assert.Contains("quoted id", output);
        Assert.Contains("value with spaces", output);
    }

    [Fact]
    public static void QuotedReferencesParsingTest()
    {
        // Test that quoted references are parsed correctly
        var input = @"(""quoted id"": ""value with spaces"")";
        var parser = new Parser();
        var parsed = parser.Parse(input);

        // Verify parsing worked correctly
        var output = parsed.Format();
        Assert.Contains("quoted id", output);
        Assert.Contains("value with spaces", output);
    }

    [Fact]
    public static void IndentedIdSyntaxParsingTest()
    {
        // Test that indented ID syntax is parsed correctly
        var indented = "id:\n  value1\n  value2";
        var inline = "(id: value1 value2)";

        var parser = new Parser();
        var indentedParsed = parser.Parse(indented);
        var inlineParsed = parser.Parse(inline);

        // Both should produce equivalent structures
        var indentedOutput = indentedParsed.Format();
        var inlineOutput = inlineParsed.Format();
        Assert.Equal(indentedOutput, inlineOutput);
        Assert.Equal("(id: value1 value2)", indentedOutput);
    }

    [Fact]
    public static void IndentedIdSyntaxRoundtripTest()
    {
        var input = "id:\n  value1\n  value2";
        var parser = new Parser();
        var parsed = parser.Parse(input);

        // Validate that we can format with indented syntax using FormatOptions
        var options = new FormatOptions
        {
            MaxInlineRefs = 1,  // Force indentation with more than 1 ref
            PreferInline = false
        };
        var output = parsed.Format(options);
        Assert.Equal(input, output);
    }

    [Fact]
    public static void MultipleIndentedIdSyntaxParsingTest()
    {
        // Test that multiple indented ID links are parsed correctly
        var indented = "id1:\n  a\n  b\nid2:\n  c\n  d";
        var inline = "(id1: a b)\n(id2: c d)";

        var parser = new Parser();
        var indentedParsed = parser.Parse(indented);
        var inlineParsed = parser.Parse(inline);

        // Both should produce equivalent structures
        var indentedOutput = indentedParsed.Format();
```

```
            var inlineOutput = inlineParsed.Format();
            Assert.Equal(indentedOutput, inlineOutput);
            Assert.Equal("(id1: a b)\n(id2: c d)", indentedOutput);
        }

        [Fact]
        public static void MultipleIndentedIdSyntaxRoundtripTest()
        {
            var input = "id1:\n  a\n  b\nid2:\n  c\n  d";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // Validate that we can format with indented syntax using FormatOptions
            var options = new FormatOptions
            {
                MaxInlineRefs = 1,  // Force indentation with more than 1 ref
                PreferInline = false
            };
            var output = parsed.Format(options);
            Assert.Equal(input, output);
        }
    }
}
```

1.30 ./Link.Foundation.Links.Notation.Tests/BinaryLinksNotationTests.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using Xunit;
using Link.Foundation.Links.Notation.Binary;

using LinoLink = Link.Foundation.Links.Notation.Link<string>;

namespace Link.Foundation.Links.Notation.Tests.Binary;

/// <summary>
/// The binary links notation: packets, sections, arities, widths and the
/// LiNo mapping on top of them. Mirrors <c>rust/tests/protocol packet tests.rs</c>.
/// </summary>
public sealed class BinaryLinksNotationTests
{
    internal static string Hex(byte[] bytes) => string.Join(" ", bytes.Select(value =>
    ↪  value.ToString("x2")));

    internal static byte[] Unhex(string text) =>
        text.Split(' ', StringSplitOptions.RemoveEmptyEntries).Select(value =>
        ↪  Convert.ToByte(value, 16)).ToArray();

    private static IReadOnlyList<LinoLink> Parse(string text) => LinoFormat.ParseDocument(text);

    private static bool ReferencesOption(string field) => field switch
    {
        "plain" => false,
        "external" => true,
          => throw new ArgumentException($"unknown references option {field}"),
    };

    private static bool WidthsOption(string field) => field switch
    {
        "uniform" => false,
        "packed" => true,
          => throw new ArgumentException($"unknown widths option {field}"),
    };

    /// <summary><c>address:reference reference...</c> separated by <c>;</c>, <c>#v</c> being
    ↪  external.</summary>
    private static List<(ulong Address, PacketReference[] References)> ParseLinks(string text) =>
        text.Split(';').Select(link =>
        {
            var parts = link.Split(':', 2);
            var references = parts[1].Split(' ').Select(reference => reference.StartsWith('#')
                ? PacketReference.External(ulong.Parse(reference[1..]))
                : PacketReference.Internal(ulong.Parse(reference))).ToArray();
            return (ulong.Parse(parts[0]), references);
        }).ToList();

    /// <summary>The golden vectors shared with the Rust test suite: both ports must write and
    ↪  read exactly these bytes.</summary>
    private static IEnumerable<string[]> GoldenVectors() =>
        File.ReadAllLines(Path.Combine(AppContext.BaseDirectory, "Protocol",
        ↪  "binary-links-notation-vectors.txt"))
```

```
            .Where(line => line.Length > 0 && !line.StartsWith('#'))
            .Select(line => line.Split('\t'));

    private static string Describe(IEnumerable<(ulong Address, PacketReference[] References)>
    ↪  links) =>
        string.Join(";", links.Select(link => $"{link.Address}:{string.Join(" ",
        ↪  link.References)}"));

    private static string Describe(LinksPacket packet) =>
        $"{packet.ExternalReferences} " + string.Join(" | ", packet.Sections.Select(section =>
            $"gap {section.Gap} arity {section.Arity} width {section.Width}: "
            + string.Join(";", section.Links.Select(link => string.Join(" ", link)))));

    [Fact]
    public void GoldenDocumentsAreStable()
    {
        var checkedVectors = 0;
        foreach (var fields in GoldenVectors().Where(fields => fields[0] == "document"))
        {
            Assert.Equal(6, fields.Length);
            var text = fields[1].Replace("\\n", "\n");
            var options = new BinaryLinoOptions
            {
                ExternalReferences = ReferencesOption(fields[2]),
                Arity = ArityRange.Parse(fields[3]),
                PackedWidths = WidthsOption(fields[4]),
            };
            var document = Parse(text);
            var binary = new BinaryLinoCodec(options);
            Assert.True(fields[5] == Hex(binary.Encode(document)), $"{text} with {options}");
            Assert.Equal(document, binary.Decode(Unhex(fields[5])));
            checkedVectors++;
        }
        Assert.True(checkedVectors >= 80, $"only {checkedVectors} document vectors");
    }

    [Fact]
    public void GoldenPacketsAreStable()
    {
        var checkedVectors = 0;
        foreach (var fields in GoldenVectors().Where(fields => fields[0] == "links"))
        {
            Assert.Equal(5, fields.Length);
            var links = ParseLinks(fields[1]);
            var packet = LinksPacket.Pack(ReferencesOption(fields[2]), links,
            ↪  WidthsOption(fields[3]));
            Assert.True(fields[4] == Hex(packet.ToBytes()), fields[1]);
            var read = LinksPacket.FromBytes(Unhex(fields[4]));
            Assert.Equal(Describe(packet), Describe(read));
            Assert.Equal(Describe(links), Describe(read.Links()));
            checkedVectors++;
        }
        Assert.True(checkedVectors >= 10, $"only {checkedVectors} links vectors");
    }

    [Fact]
    public void WidthTiersFollowTheNumberOfLinks()
    {
        Assert.Equal(1, LinksPacket.AddressTier(0, false));
        Assert.Equal(1, LinksPacket.AddressTier(255, false));
        Assert.Equal(2, LinksPacket.AddressTier(256, false));
        Assert.Equal(2, LinksPacket.AddressTier(65 535, false));
        Assert.Equal(4, LinksPacket.AddressTier(65 536, false));
        Assert.Equal(4, LinksPacket.AddressTier(uint.MaxValue, false));
        Assert.Equal(8, LinksPacket.AddressTier((ulong)uint.MaxValue + 1, false));
        // External references take the top bit, halving every range.
        Assert.Equal(1, LinksPacket.AddressTier(127, true));
        Assert.Equal(2, LinksPacket.AddressTier(128, true));
        Assert.Equal(2, LinksPacket.AddressTier(32 767, true));
        Assert.Equal(4, LinksPacket.AddressTier(32 768, true));
        Assert.Equal(8, LinksPacket.AddressTier(ulong.MaxValue, false));
        // No width holds an internal address in the external half.
        Assert.Equal(BinaryErrorKind.Unencodable, Assert.Throws<BinaryNotationException>(() =>
        ↪  LinksPacket.AddressTier(1UL << 63, true)).Kind);
        Assert.Equal(ulong.MaxValue, LinksPacket.InternalCapacity(8, false));
        Assert.Equal((ulong)long.MaxValue, LinksPacket.InternalCapacity(8, true));
        Assert.Equal(127UL, LinksPacket.ExternalCapacity(1));
    }

    [Fact]
```

```
    public void ExternalReferencesMatchPlatformDataHybrid()
    {
        foreach (var width in LinksPacket.Widths)
        {
            foreach (var value in new ulong[] { 0, 1, 2, 100, LinksPacket.ExternalCapacity(width) })
            {
                var raw = LinksPacket.EncodeExternal(value, width);
                Assert.True(LinksPacket.TryDecodeExternal(raw, width, out var decoded));
                Assert.Equal(value, decoded);
            }
            Assert.False(LinksPacket.TryDecodeExternal(LinksPacket.InternalCapacity(width, true), width, out var ignored));
        }
        Assert.Equal(0xFFUL, LinksPacket.EncodeExternal(1, 1));
        Assert.Equal(0x80UL, LinksPacket.EncodeExternal(0, 1));

    }

    private static string ManyLinks(int count) =>
        string.Join("\n", Enumerable.Range(0, count).Select(index => $"(n{index} m{index})"));

    private static byte[] SectionWidths(LinksPacket packet) => packet.Sections.Select(section => section.Width).ToArray();

    [Fact]
    public void UniformWidthsGrowPast256AddressesAndPackedWidthsStaySmall()
    {
        Assert.Equal(new byte[] { 1 }, SectionWidths(LinoMapping.EncodeDocument(Parse(ManyLinks(3)))));

        var document = Parse(ManyLinks(400));
        var uniform = LinoMapping.EncodeDocument(document);
        Assert.Equal(new byte[] { 2 }, SectionWidths(uniform));
        var uniformBytes = uniform.ToBytes();
        const int header = 1 + 2; // header byte + two-byte LEB128 count
        Assert.Equal(header + (long)uniform.LinkCount * 2 * 2, uniformBytes.Length);

        var packed = LinoMapping.EncodeDocument(document, new BinaryLinoOptions().WithPackedWidths());
        Assert.Equal(1, SectionWidths(packed)[0]);
        Assert.Contains((byte)2, SectionWidths(packed));
        var packedBytes = packed.ToBytes();
        Assert.True(packedBytes.Length < uniformBytes.Length);
        foreach (var bytes in new[] { uniformBytes, packedBytes })
        {
            Assert.Equal(document, LinoMapping.DecodeDocument(LinksPacket.FromBytes(bytes)));
        }
    }

    [Fact]
    public void PackedWidthsNeverTakeMoreBytesThanUniformOnes()
    {
        var corpus = BinaryLinoCodecTests.Corpus.Append(ManyLinks(200)).Append("(1000 1)");
        foreach (var text in corpus)
        {
            var document = Parse(text);
            foreach (var options in BinaryLinoCodecTests.AllOptions().Where(options => options.PackedWidths))
            {
                var packed = LinoMapping.EncodeDocument(document, options).ToBytes();
                var uniform = LinoMapping.EncodeDocument(document, options.WithPackedWidths(false)).ToBytes();
                Assert.True(packed.Length <= uniform.Length, $"{text} with {options}");
            }
        }
    }

    [Fact]
    public void LargeExternalValuesWidenOnlyTheirSection()
    {
        var options = new BinaryLinoOptions().WithExternalReferences().WithPackedWidths();
        Assert.Equal(new byte[] { 1 }, SectionWidths(LinoMapping.EncodeDocument(Parse("(100 1)"), options)));
        Assert.Equal(new byte[] { 4, 1 }, SectionWidths(LinoMapping.EncodeDocument(Parse("(70000 1)"), options)));
        // Beyond 63 bits the number falls back to in-band unary links.
        Assert.True(LinoMapping.EncodeDocument(Parse("18446744073709551615"), options).LinkCount > 60);
```

```
198     }
199
200     private static int[] LinkLengths(IReadOnlyList<LinoLink> document, BinaryLinoOptions options)
201     {
202         var packet = LinoMapping.EncodeDocument(document, options);
203         Assert.Equal(document, new BinaryLinoCodec().Decode(packet.ToBytes()));
204         return packet.Links().Select(link => link.References.Length).ToArray();
205     }
206
207     [Fact]
208     public void ArityOptionsChooseDoubletsTripletsOrAnyLength()
209     {
210         var document = Parse("(a b c)\n(d e)");
211         var external = new BinaryLinoOptions().WithExternalReferences();
212         Assert.All(LinkLengths(document, external), length => Assert.Equal(2, length));
213         // `(a b c)` is one triplet; the strings `(String code point)` stay doublets.
214         var triplets = LinkLengths(document, external.WithArity(ArityRange.Between(2, 3)));
215         Assert.Single(triplets, length => length == 3);
216         Assert.All(triplets, length => Assert.InRange(length, 2, 3));
217         Assert.Equal(triplets, LinkLengths(document, external.WithArity(ArityRange.AtLeast(1))));
218         // Only any length turns a one-item list, here the root, into one link.
219         Assert.Equal(new[] { 2, 2, 2, 2 }, LinkLengths(Parse("a"), external));
220         Assert.Equal(new[] { 2, 2, 1 }, LinkLengths(Parse("a"),
            ↪  external.WithArity(ArityRange.AtLeast(1))));
221         // Long lists only become single links when the range allows their length.
222         var longList = Parse("(a b c d e f g h)");
223         foreach (var (arity, singleLink) in new[]
224         {
225             (ArityRange.Between(2, 3), false),
226             (ArityRange.Between(2, 8), true),
227             (ArityRange.AtLeast(2), true),
228         })
229         {
230             Assert.True(LinkLengths(longList, external.WithArity(arity)).Max() == 8 ==
                ↪  singleLink, arity.ToString());
231         }
232     }
233
234     [Fact]
235     public void AritiesWithoutDoubletsAreUnencodable()
236     {
237         foreach (var arity in new[] { ArityRange.Exactly(3), ArityRange.Exactly(1),
            ↪  ArityRange.AtLeast(3) })
238         {
239             var error = Assert.Throws<BinaryNotationException>(
240                 () => LinoMapping.EncodeDocument(Parse("(a b)"), new
                    ↪  BinaryLinoOptions().WithArity(arity)));
241             Assert.Equal(BinaryErrorKind.Unencodable, error.Kind);
242         }
243     }
244
245     [Fact]
246     public void ArityRangesParseAndDisplay()
247     {
248         foreach (var (text, range) in new[]
249         {
250             ("2", ArityRange.Doublets),
251             ("2..3", ArityRange.Between(2, 3)),
252             ("1..", ArityRange.AtLeast(1)),
253             ("3..3", ArityRange.Exactly(3)),
254             ("+2", ArityRange.Doublets),
255         })
256         {
257             Assert.Equal(range, ArityRange.Parse(text));
258         }
259         Assert.Equal("3", ArityRange.Between(3, 3).ToString());
260         Assert.Equal("2..3", ArityRange.Between(2, 3).ToString());
261         Assert.Equal("1..", ArityRange.AtLeast(1).ToString());
262         Assert.Equal(ArityRange.Doublets, new BinaryLinoOptions().Arity);
263         Assert.Equal(default, new BinaryLinoOptions().WithArity(ArityRange.Doublets));
264         foreach (var invalid in new[] { "", "0", "0..2", "3..2", "a", "1..b", "..3", "-1", " 2",
            ↪  "2 ", "++2", null })
265         {
266             Assert.False(ArityRange.TryParse(invalid, out var ignored), invalid);
267         }
268         Assert.False(ArityRange.TryParse("3..2", out var invalidRange, out var error));
269         Assert.Equal("arity range 3..2 is empty", error);
270         Assert.False(ArityRange.TryParse("x", out invalidRange, out error));
271         Assert.Equal("invalid arity 'x': expected n, min..max or min..", error);
```

```
272         Assert.Equal("arity must be at least 1", Assert.Throws<FormatException>(() =>
            ↪ AnyRange.Parse("0")).Message);
273         Assert.True(AnyRange.AtLeast(2).Contains(1 000));
274         Assert.False(AnyRange.Between(2, 3).Contains(4));
275         Assert.False(AnyRange.Between(2, 3).Contains(1));
276         Assert.True(AnyRange.Exactly(2).IsFixed && !AnyRange.AtLeast(2).IsFixed);
277     }
278
279     [Fact]
280     public void ReplyOptionsFollowTheReceivedPacket()
281     {
282         var document = Parse("(a b c) 70000");
283         var options = new BinaryLinoOptions()
284             .WithExternalReferences()
285             .WithArity(AnyRange.Between(2, 3))
286             .WithPackedWidths();
287         Assert.Equal(options, BinaryLinoOptions.OfPacket(LinoMapping.EncodeDocument(document,
            ↪ options)));
288         Assert.Equal(default, BinaryLinoOptions.OfPacket(LinoMapping.EncodeDocument(document)));
289         Assert.Equal(default, BinaryLinoOptions.OfPacket(new LinksPacket()));
290         Assert.Equal("BinaryLinoOptions { ExternalReferences = True, Arity = 2..3, PackedWidths =
            ↪ True }", options.ToString());
291     }
292
293     private static Section DoubletSection(ulong gap, AnyRange arity, byte width, params (ulong
        ↪ Source, ulong Target)[] links)
294     {
295         var section = new Section { Gap = gap, Arity = arity, Width = width };
296         section.Links.AddRange(links.Select(link =>
297             new[] { PacketReference.Internal(link.Source), PacketReference.Internal(link.Target)
                ↪ }));
298         return section;
299     }
300
301     [Fact]
302     public void HandBuiltPacketsDecode()
303     {
304         // One doublet (One One) is 2^1 in unary, so `(Number 7)` is the number 2.
305         var links = new (ulong, ulong)[] { (1, 1), (2, 6), (7, 0), (4, 8) };
306         var compact = new LinksPacket();
307         compact.Sections.Add(DoubletSection(5, AnyRange.Doublets, 1, links));
308         var bytes = compact.ToBytes();
309         Assert.Equal("10 04 01 01 02 06 07 00 04 08", Hex(bytes));
310         Assert.Equal("2", LinoFormat.FormatDocument(new BinaryLinoCodec().Decode(bytes)));
311
312         // The same links as a variable section of width 2 use the explicit layout.
313         var explicitLayout = new LinksPacket();
314         explicitLayout.Sections.Add(DoubletSection(5, AnyRange.AtLeast(1), 2, links));
315         bytes = explicitLayout.ToBytes();
316         Assert.Equal("12 01 1d 05 00 04 01 01 00 01 00 01 02 00 06 00 01 07 00 00 00 01 04 00 08
            ↪ 00", Hex(bytes));
317         Assert.Equal("2", LinoFormat.FormatDocument(new BinaryLinoCodec().Decode(bytes)));
318         Assert.Equal("10 00", Hex(new LinksPacket().ToBytes()));
319     }
320
321     private static PacketReference[] Internals(params ulong[] addresses) =>
322         addresses.Select(PacketReference.Internal).ToArray();
323
324     [Fact]
325     public void PackingRejectsLinksItCannotLayOut()
326     {
327         void Unencodable(bool externalReferences, params (ulong, PacketReference[])[] links)
328         {
329             var error = Assert.Throws<BinaryNotationException>(() =>
                ↪ LinksPacket.Pack(externalReferences, links, true));
330             Assert.Equal(BinaryErrorKind.Unencodable, error.Kind);
331         }
332         Unencodable(false, (0, Internals(1))); // address 0 is null
333         Unencodable(false, (2, Internals(1)), (1, Internals(1))); // descending
334         Unencodable(false, (2, Internals(1)), (2, Internals(1))); // repeated
335         Unencodable(false, (1, Array.Empty<PacketReference>())); // no references
336         Unencodable(false, (1, new[] { PacketReference.External(1) }));
337         Unencodable(true, (1, new[] { PacketReference.External(1UL << 63) }));
338         Unencodable(true, (1, Internals(1UL << 63))); // the top bit marks externals
339         Assert.NotNull(LinksPacket.Pack(false, new[] { (1UL, Internals(ulong.MaxValue)) },
            ↪ true));
340     }
341
342     [Fact]
```

```
343     public void WritingRejectsSectionsThatDoNotHoldTheirLinks()
344     {
345         static void Unencodable(Section section)
346         {
347             var packet = new LinksPacket();
348             packet.Sections.Add(section);
349             var error = Assert.Throws<BinaryNotationException>(() => packet.ToBytes());
350             Assert.Equal(BinaryErrorKind.Unencodable, error.Kind);
351         }
352         Unencodable(DoubletSection(0, ArityRange.Exactly(3), 1, (1, 1)));
353         Unencodable(DoubletSection(0, ArityRange.Exactly(0), 1));
354         Unencodable(DoubletSection(0, ArityRange.Between(3, 2), 1));
355         Unencodable(DoubletSection(0, ArityRange.Exactly(1UL << 62), 1));
356         Unencodable(DoubletSection(0, ArityRange.Doublets, 3, (1, 1)));
357         Unencodable(DoubletSection(0, ArityRange.Doublets, 1, (256, 0)));
358         Unencodable(DoubletSection(ulong.MaxValue, ArityRange.Doublets, 1, (1, 1)));
359     }
360 
361     private static BinaryNotationException DecodeError(byte[] bytes, DecodeLimits? limits = null)
362     {
363         var protocol = new BinaryLinoCodec { Limits = limits ?? DecodeLimits.Default };
364         return Assert.Throws<BinaryNotationException>(() => protocol.Decode(bytes));
365     }
366 
367     /// <summary>Each packet breaks exactly one rule, the one its detail names.</summary>
368     [Theory]
369     [InlineData("", "empty input")]
370     [InlineData("20 00", "unsupported binary header byte 0x20")]
371     [InlineData("10 01 06 00", "link 6 refers to Internal(6), which is not an earlier link")]
372     [InlineData("10 01 07 00", "link 6 refers to Internal(7), which is not an earlier link")]
373     [InlineData("10 02 00", "unexpected end of packet")]
374     [InlineData("10 00 00", "trailing bytes after the packet")]
375     [InlineData("10 ff ff ff ff ff ff ff ff ff 7f", "LEB128 value overflows 64 bits")] // a tenth
        ↪  byte above 1
376     [InlineData("10 80 80 80 80 80 80 80 80 80 81", "LEB128 value overflows 64 bits")] // an
        ↪  eleventh byte
377     [InlineData("10 01 01 01", "the root link is not a list")]
378     [InlineData("10 02 00 02 04 06", "broken element chain")] // the chain ends in marker 2, not
        ↪  null
379     [InlineData("10 01 04 02", "broken element chain")]
380     [InlineData("10 02 02 00 04 06", "marker 2 used as a value")]
381     [InlineData("10 03 01 00 06 00 04 07", "marker 1 cannot start a typed value")]
382     [InlineData("12 02 14 05 01 20 02 02 06 00 04 07", "a number needs exactly one value")] //
        ↪  (Number) alone
383     [InlineData("10 03 05 00 06 00 04 07", "an identified link needs an id")]
384     [InlineData("10 03 02 03 06 00 04 07", "expected a unary number")] // (Number 3): marker 3 is
        ↪  no number
385     [InlineData("10 05 05 05 06 00 02 07 08 00 04 09", "expected a unary number")] // (Number 6)
        ↪  where link 6 is (5 5)
386     [InlineData("10 04 00 00 05 06 07 00 04 08", "a link id must be a reference")] // the id is
        ↪  ()
387     [InlineData("16 00", "the explicit layout keeps the header width bits clear")]
388     [InlineData("12 01 04 05 00", "arity must be at least 1")]
389     [InlineData("12 01 18 01 01 05 00", "link length outside the section arity 1..2")] // a link
        ↪  of 6
390     [InlineData("12 01 f8 ff ff ff ff ff ff ff ff 01 ff ff ff ff ff ff ff ff ff 01 00", "arity
        ↪  range overflows 64 bits")]
391     [InlineData("12 01 24 ff ff ff ff ff ff ff ff ff 01 01", "addresses overflow 64 bits")] //
        ↪  the gap
392     [InlineData("12 01 20 ff ff ff ff ff ff ff ff ff 01", "addresses overflow 64 bits")] // the
        ↪  count
393     [InlineData("12 01 24 06 01 01 01", "a LiNo packet stores its links contiguously from address
        ↪  6, found link 7 where 6 belongs")]
394     [InlineData("12 02 24 05 01 24 01 01 01 01 04 06", "a LiNo packet stores its links
        ↪  contiguously from address 6, found link 8 where 7 belongs")] // a hole
395     public void MalformedPacketsAreRejected(string hex, string detail)
396     {
397         var error = DecodeError(Unhex(hex));
398         Assert.Equal(BinaryErrorKind.Malformed, error.Kind);
399         Assert.Equal(detail, error.Detail);
400     }
401 
402     private static List<(ulong Address, PacketReference[] References)> FromAddress6(params
        ↪  PacketReference[][] links) =>
403         links.Select((references, index) => (6UL + (ulong)index, references)).ToList();
404 
405     [Fact]
406     public void InvalidCodePointsAreRejected()
407     {
```

```
408             // (String (0x110000)) is not a valid code point.
409             var links = FromAddress6(
410                 new[] { PacketReference.External(0x11 0000), PacketReference.Null },
411                 Internals(3, 6),
412                 new[] { PacketReference.Internal(7), PacketReference.Null },
413                 Internals(4, 8));
414             Assert.Equal("invalid code point 1114112", DecodeError(LinksPacket.Pack(true, links,
                ↪  false).ToBytes()).Detail);
415         }
416 
417         [Fact]
418         public void HostilePacketsHitLimits()
419         {
420             void Limited(string hex, DecodeLimits limits) =>
421                 Assert.True(BinaryErrorKind.LimitExceeded == DecodeError(Unhex(hex), limits).Kind,
                    ↪  hex);
422             var fewLinks = new DecodeLimits { MaxLinks = 8 };
423             Limited("10 09", fewLinks);
424             Limited("12 09", fewLinks); // more sections than links allowed
425             Limited("12 02 20 05 20 05", fewLinks); // 10 in all
426             var fewReferences = new DecodeLimits { MaxReferences = 5 };
427             Limited("10 03 01 01 01 01 01 01", fewReferences);
428             Limited("12 01 18 00 01 05", fewReferences); // one link of 6
429 
430             // A doubling chain `d(k) = (d(k-1) d(k-1))` expands to 2^k nodes.
431             var chain = new List<PacketReference[]> { new[] { PacketReference.Null,
                ↪  PacketReference.Null } };
432             for (ulong address = 6; address < 60; address++)
433             {
434                 chain.Add(Internals(address, address));
435             }
436             var last = 5 + (ulong)chain.Count;
437             chain.Add(new[] { PacketReference.Internal(last), PacketReference.Null });
438             chain.Add(Internals(4, last + 1));
439             var bomb = LinksPacket.Pack(false, FromAddress6(chain.ToArray()), false);
440             Assert.Equal("limit exceeded: too many LiNo nodes", DecodeError(bomb.ToBytes(), new
                ↪  DecodeLimits { MaxNodes = 1 << 12 }).Message);
441 
442             var deep = string.Concat(Enumerable.Repeat("(y ", 40)) + "x" + new string(')', 40);
443             var bytes = new BinaryLinoCodec().Encode(Parse(deep));
444             Assert.Equal("limit exceeded: nesting deeper than 10", DecodeError(bytes, new
                ↪  DecodeLimits { MaxDepth = 10 }).Message);
445 
446             var threeLinks = new BinaryLinoCodec().Encode(Parse("a b c"));
447             Assert.Equal("limit exceeded: chain too long", DecodeError(threeLinks, new DecodeLimits
                ↪  { MaxNodes = 2 }).Message);
448             Assert.Equal(ulong.MaxValue, DecodeLimits.Unlimited.MaxLinks);
449         }
450 }
```

1.31 ./Link.Foundation.Links.Notation.Tests/BinaryLinoCodecTests.cs

```
1   using System;
2   using System.Collections.Generic;
3   using System.IO;
4   using System.Linq;
5   using Xunit;
6   using Link.Foundation.Links.Notation.Binary;
7 
8   using LinoLink = Link.Foundation.Links.Notation.Link<string>;
9 
10  namespace Link.Foundation.Links.Notation.Tests.Binary;
11 
12  /// <summary>Binary and text LiNo protocol codecs (issue #105).</summary>
13  public sealed class BinaryLinoCodecTests
14  {
15      [Fact]
16      public void EncoderUsesConfiguredLimits()
17      {
18          var document = new[] { new LinoLink("abcdef", new[] { new LinoLink("value") }) };
19          foreach (var limits in new[]
20          {
21              new DecodeLimits { MaxNodes = 1 },
22              new DecodeLimits { MaxStringBytes = 2 },
23              new DecodeLimits { MaxDepth = 1 },
24              new DecodeLimits { MaxLinks = 1 },
25              new DecodeLimits { MaxReferences = 1 },
26          })
27          {
28              Assert.Throws<BinaryNotationException>(() => new BinaryLinoCodec { Limits = limits
                    ↪  }.Encode(document));
```

```
        }
        var deep = new LinoLink("a");
        for (var index = 0; index < 70; index++) deep = new LinoLink(null, new[] { deep });
        var model = new[] { deep };
        var codec = new BinaryLinoCodec { Limits = new DecodeLimits { MaxDepth = 80 } };
        Assert.Equal(model, codec.Decode(codec.Encode(model)));
    }

    internal static readonly string[] Corpus =
    {
        "() ((1 1))",
        "((1: 1 1)) ((1: 1 2))",
        "((1 1)) ()",
        "((1: 1 1)) ()",
        "(($i: $s $t)) (($i: $s $t))",
        "((($index: $source $target)) (($index: $target $source)))",
        "(a b c d)",
        "(name: 'with space' \"it's\")",
        "((a))",
        "(((a)))",
        "(a (b c) ((d)))",
        "1\n2\n3",
        "hello",
        "\U0001F600 привет \u4E16\u754C",
        "0",
        "007",
        "9223372036854775807",
        "9223372036854775808",
        "18446744073709551615",
        "18446744073709551616",
        "'multi\nline'",
        ".dot",
        "'a''b\"c`d'",
        "(1: (2: 3 4) 5)",
        "() ()",
        "(* *)",
        "(type: type type)",
    };

    internal static IEnumerable<BinaryLinoOptions> AllOptions()
    {
        foreach (var externalReferences in new[] { false, true })
        {
            foreach (var arity in new[] { ArityRange.Doublets, ArityRange.Between(2, 3),
            ↪  ArityRange.AtLeast(1) })
            {
                foreach (var packedWidths in new[] { false, true })
                {
                    yield return new BinaryLinoOptions
                    {
                        ExternalReferences = externalReferences,
                        Arity = arity,
                        PackedWidths = packedWidths,
                    };
                }
            }
        }
    }

    private static IReadOnlyList<LinoLink> Parse(string text) => LinoFormat.ParseDocument(text);

    [Fact]
    public void EveryOptionSetRoundTripsTheCorpus()
    {
        foreach (var text in Corpus)
        {
            var document = Parse(text);
            foreach (var options in AllOptions())
            {
                var binary = new BinaryLinoCodec(options);
                var bytes = binary.Encode(document);
                Assert.InRange(bytes[0], 0x10, 0x1F);
                Assert.True(document.SequenceEqual(binary.Decode(bytes)), $"{text} with
                ↪  {options}");
            }
        }
    }

    [Fact]
    public void CanonicalTextRoundTripsTheCorpus()
    {
```

```
108         foreach (var text in Corpus)
109         {
110             var document = Parse(text);
111             var canonical = LinoFormat.FormatDocument(document);
112             Assert.True(document.SequenceEqual(Parse(canonical)), $"{text} -> {canonical}");
113         }
114         Assert.Equal("() ((1 1))", LinoFormat.FormatDocument(Parse("(()((1 1)))")));
115         Assert.Equal("((1: 1 1)) ((1: 1 2))", LinoFormat.FormatDocument(Parse("((1: 1 1)) ((1: 1
    ↪  2))")));
116         // Only a binary message carries an id without values; links-notation reads its text back
    ↪  as the reference.
117         Assert.Equal("(a:)", LinoFormat.FormatDocument(new[] { LinoFormat.Link("a", new
    ↪  List<LinoLink>()) }));
118         Assert.True(Parse("(a:)").SequenceEqual(new[] { LinoFormat.Reference("a") }));
119     }
120
121     [Fact]
122     public void ParsedDocumentsMatchTheRustModel()
123     {
124         // The C# parser wraps single references in an extra group; the
125         // canonical model drops it so both ports decode to equal documents.
126         // These structures match the canonical model used by the shared vectors.
127         var a = LinoFormat.Reference("a");
128         LinoLink Group(LinoLink value) => LinoFormat.Link(null, new List<LinoLink> { value });
129         Assert.Equal(new[] { a }, Parse("a"));
130         Assert.Equal(new[] { a }, Parse("(a)"));
131         Assert.Equal(new[] { Group(a) }, Parse("((a))"));
132         Assert.Equal(new[] { Group(Group(a)) }, Parse("(((a)))"));
133         Assert.Equal(new[] { LinoFormat.Link("a", new List<LinoLink> { LinoFormat.Reference("b")
    ↪  }) }, Parse("(a: (b))"));
134         Assert.Equal(new[] { LinoFormat.Reference("1"), LinoFormat.Reference("2"),
    ↪  LinoFormat.Reference("3") }, Parse("1\n2\n3"));
135     }
136
137     [Fact]
138     public void ReferencesAreQuotedOnlyWhenNeeded()
139     {
140         Assert.Equal("plain", LinoFormat.FormatReference("plain"));
141         Assert.Equal("$x", LinoFormat.FormatReference("$x"));
142         Assert.Equal("~1{}", LinoFormat.FormatReference(""));
143         Assert.Equal("'a b'", LinoFormat.FormatReference("a b"));
144         Assert.Equal("\"it's\"", LinoFormat.FormatReference("it's"));
145         Assert.Equal("`'\"`", LinoFormat.FormatReference("'\""));
146         Assert.Equal("\"\"\"'\"`\"\"\"", LinoFormat.FormatReference("'\"`"));
147         foreach (var reference in new[]
148         {
149             "", "a b", "a:b", "(x)", "it's", "'\"", "'\"`", "tab\there", "a\nb", "'x", "x'",
    ↪  "''", "'''", "''''",
150             "\"'`x`'\"", "a''''b\"\"`", "`'\"\"\"'''``", "(a) ''",
151         })
152         {
153             var formatted = LinoFormat.FormatReference(reference);
154             var document = Parse(formatted);
155             Assert.Equal(formatted, LinoFormat.FormatDocument(document));
156             Assert.Equal(new[] { LinoFormat.Reference(reference) }, document);
157         }
158     }
159
160     [Fact]
161     public void EveryShortReferenceOverDelimitersRoundTrips()
162     {
163         var alphabet = new[] { '\'', '"', '`', 'a', ' ', '(', ')', ':' };
164         var stack = new Stack<string>();
165         stack.Push(string.Empty);
166         while (stack.TryPop(out var reference))
167         {
168             if (reference.Length > 0)
169             {
170                 var formatted = LinoFormat.FormatReference(reference);
171                 var document = Parse(formatted);
172                 Assert.True(
173                     document.Count == 1 && document[0].Equals(LinoFormat.Reference(reference)),
174                     $"{reference} as {formatted}");
175             }
176             if (reference.Length < 5)
177             {
178                 foreach (var character in alphabet)
179                 {
180                     stack.Push(reference + character);
```

```
                }
            }
        }
    }

    [Fact]
    public void CommentReferencesSurviveBinaryAndTextRoundTrips()
    {
        foreach (var reference in new[] { "#", "#tag", "# with space", "issue#1047" })
        {
            var document = new[] { LinoFormat.Reference(reference) };
            var codec = new BinaryLinoCodec();
            var decoded = codec.Decode(codec.Encode(document));
            Assert.Equal(document, decoded);
            Assert.Equal(document, Parse(LinoFormat.FormatDocument(decoded)));
        }
    }

    [Fact]
    public void TextHelpersPreserveUnicodeWhitespaceReferences()
    {
        foreach (var reference in new[] { "\u00a0", "\u2003", "\u2028" })
        {
            var document = new[] { LinoFormat.Reference(reference) };
            Assert.Equal(document, Parse(reference));
            Assert.Equal(document, Parse(LinoFormat.FormatDocument(document)));
        }
    }
    [Fact]
    public void NativeParserGroupsRoundTripWithoutCanonicalization()
    {
        foreach (var text in Corpus)
        {
            var document = new Parser().Parse(text).ToList();
            foreach (var options in AllOptions())
            {
                var codec = new BinaryLinoCodec(options);
                Assert.Equal(document, codec.Decode(codec.Encode(document)));
            }
        }
    }

    [Fact]
    public void InvalidEncoderModelsAreErrors()
    {
        var link = LinoFormat.Reference("a");
        for (var depth = 0; depth < DecodeLimits.Default.MaxDepth; depth++)
        {
            link = LinoFormat.Link(null, new List<LinoLink> { link });
        }
        Assert.Equal(BinaryErrorKind.Unencodable, Assert.Throws<BinaryNotationException>(() =>
        ↪  new BinaryLinoCodec().Encode(new[] { link })).Kind);
        var invalid = new BinaryLinoOptions().WithArity(ArityRange.Between(0, 2));
        Assert.Equal(BinaryErrorKind.Unencodable, Assert.Throws<BinaryNotationException>(() =>
        ↪  new BinaryLinoCodec(invalid).Encode(Array.Empty<LinoLink>())).Kind);
        Assert.Equal(BinaryErrorKind.Unencodable, Assert.Throws<BinaryNotationException>(() =>
        ↪  new BinaryLinoCodec().Encode(new[] { LinoFormat.Reference("\ud800") })).Kind);
    }

    [Fact]
    public void IdsDoNotAddAnExtraModelNestingLevel()
    {
        var link = LinoFormat.Link("id", new List<LinoLink>());
        for (var depth = 1; depth < DecodeLimits.Default.MaxDepth; depth++)
        {
            link = LinoFormat.Link(null, new List<LinoLink> { link });
        }
        var document = new[] { link };
        foreach (var options in AllOptions())
        {
            var codec = new BinaryLinoCodec(options);
            Assert.Equal(document, codec.Decode(codec.Encode(document)));
        }
    }

    [Fact]
    public void InMemoryPacketsObeyLimits()
    {
        var packet = LinoMapping.EncodeDocument(Parse("(a b)"));
```

```
257         foreach (var limits in new[] { new DecodeLimits { MaxLinks = 0 }, new DecodeLimits {
            ↪ MaxReferences = 0 } })
258         {
259             Assert.Equal(BinaryErrorKind.LimitExceeded,
                ↪ Assert.Throws<BinaryNotationException>(() => LinoMapping.DecodeDocument(packet,
                ↪ limits)).Kind);
260         }
261         var links = new List<(ulong, PacketReference[])> { (6, new[] { PacketReference.Null,
            ↪ PacketReference.Null }) };
262         for (ulong address = 7; address < 80; address++)
263         {
264             links.Add((address, new[] { PacketReference.Internal(address - 1),
                ↪ PacketReference.Null }));
265         }
266         var deep = LinksPacket.Pack(false, links, false);
267         Assert.Equal(BinaryErrorKind.LimitExceeded, Assert.Throws<BinaryNotationException>(() =>
            ↪ LinoMapping.DecodeDocument(deep)).Kind);
268         Assert.Equal(BinaryErrorKind.Unencodable, Assert.Throws<BinaryNotationException>(() =>
            ↪ LinksPacket.Pack(false, new[] { (ulong.MaxValue, new[] { PacketReference.Null }) },
            ↪ false)).Kind);
269     }
270
271     [Fact]
272     public void InMemoryPacketsRejectInvalidWireShapes()
273     {
274         foreach (var kind in new[] { "external", "width", "arity" })
275         {
276             var packet = LinksPacket.Pack(true, new[] { (6UL, new[] {
                ↪ PacketReference.External(3) }) }, false);
277             if (kind == "external") packet.ExternalReferences = false;
278             if (kind == "width") packet.Sections[0].Width = 3;
279             if (kind == "arity") packet.Sections[0].Arity = ArityRange.Doublets;
280             Assert.Equal(BinaryErrorKind.Malformed, Assert.Throws<BinaryNotationException>(() =>
                ↪ LinoMapping.DecodeDocument(packet)).Kind);
281         }
282     }
283
284     [Fact]
285     public void CompactAddressOverflowIsRejectedBeforeReadingLinks()
286     {
287         var bytes = new List<byte> { 0x10 };
288         LinksPacket.WriteLeb128(bytes, ulong.MaxValue);
289         var exception = Assert.Throws<BinaryNotationException>(() =>
            ↪ LinksPacket.FromBytes(bytes.ToArray(), DecodeLimits.Unlimited));
290         Assert.Equal(BinaryErrorKind.Malformed, exception.Kind);
291         Assert.Contains("addresses overflow", exception.Detail);
292     }
293
294     [Fact]
295     public void RepeatedReferencesObeyATotalUtf8ExpansionBudget()
296     {
297         foreach (var reference in new[] { "\U0001F600", "12" })
298         {
299             var document = new[] { LinoFormat.Reference(reference),
                ↪ LinoFormat.Reference(reference) };
300             foreach (var options in AllOptions())
301             {
302                 var packet = LinoMapping.EncodeDocument(document, options);
303                 var bytes = System.Text.Encoding.UTF8.GetByteCount(reference) * 2;
304                 Assert.Equal(document, LinoMapping.DecodeDocument(packet, new DecodeLimits {
                    ↪ MaxStringBytes = bytes }));
305                 Assert.Equal(BinaryErrorKind.LimitExceeded,
                    ↪ Assert.Throws<BinaryNotationException>(() =>
                    ↪ LinoMapping.DecodeDocument(packet, new DecodeLimits { MaxStringBytes = bytes
                    ↪ - 1 })).Kind);
306             }
307         }
308     }
309
310     [Fact]
311     public void StringBudgetStopsBeforeLaterScalars()
312     {
313         var bytes = Convert.FromHexString("13024605011001030000009fffffff9efffff0028fff06");
314         var codec = new BinaryLinoCodec { Limits = new DecodeLimits { MaxStringBytes = 1 } };
315         Assert.Equal(BinaryErrorKind.LimitExceeded, Assert.Throws<BinaryNotationException>(() =>
            ↪ codec.Decode(bytes)).Kind);
316     }
317
```

```
    [Fact]
    public void StreamPacketsAreReadOneAtATimeAndTruncationIsRejected()
    {
        var bytes = new BinaryLinoCodec().Encode(Parse("(a b)"));
        for (var end = 0; end < bytes.Length; end++)
        {
            Assert.Throws<BinaryNotationException>(() => LinksPacket.FromBytes(bytes[..end]));
        }
        var reader = PacketReader.FromBytes(bytes.Concat(bytes).ToArray());
        Assert.NotNull(LinksPacket.ReadFrom(reader));
        Assert.NotNull(LinksPacket.ReadFrom(reader));
        Assert.Null(LinksPacket.ReadFrom(reader));
    }
}
```

1.32 ./Link.Foundation.Links.Notation.Tests/CommentsTests.cs

```
using System.Linq;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    /// <summary>
    /// Conformance tests for line comments.
    ///
    /// https://github.com/link-foundation/links-notation/issues/301
    ///
    /// <c>#</c> starts a comment that runs to the end of the line, unless it sits
    /// inside a token or inside a delimited reference. Parsers read comments by
    /// default and can be told to treat <c>#</c> as an ordinary character again.
    /// The table below is shared with the Rust, JavaScript, Python, Go, Java and
    /// PHP suites, so a document written by one implementation reads the same in
    /// all of them.
    /// </summary>
    public static class CommentsTests
    {
        /// <summary>
        /// Renders a parsed node unambiguously: every reference is wrapped in
        /// angle brackets.
        /// </summary>
        private static string Render(Link<string> node)
        {
            if (node.Values == null || node.Values.Count == 0)
            {
                return "<" + (node.Id ?? "") + ">";
            }
            var head = node.Id == null ? "" : "<" + node.Id + ">: ";
            return "(" + head + string.Join(" ", node.Values.Select(Render)) + ")";
        }

        private static string RenderedWith(Parser parser, string source) =>
            string.Join("\n", parser.Parse(source).Select(Render));

        private static string Rendered(string source) => RenderedWith(new Parser(), source);

        private static void AssertParsesAs(string source, string expected)
        {
            Assert.Equal(expected, Rendered(source));
        }

        [Fact]
        public static void ALineThatStartsWithAHashIsAComment()
        {
            AssertParsesAs("# a b\n", "");
        }

        [Fact]
        public static void ACommentMayHoldAColon()
        {
            // The document from #301: prose with a colon used to be rejected.
            AssertParsesAs("# a: b\n", "");
        }

        [Fact]
        public static void ACommentMayHoldAnythingAtAll()
        {
            AssertParsesAs("# ) : ( \" ' ` #\n", "");
        }

        [Fact]
        public static void ACommentEndsAtTheEndOfItsLine()
        {
```

```
            AssertParsesAs("# note\na: b\n", "(<a>: <b>)");
        }

        [Fact]
        public static void ACommentMayFollowALink()
        {
            AssertParsesAs("a: b # why\n", "(<a>: <b>)");
        }

        [Fact]
        public static void ACommentMayFollowAGroup()
        {
            AssertParsesAs("(a b) # why\n", "(<a> <b>)");
        }

        [Fact]
        public static void ACommentNeedsNoClosingNewline()
        {
            AssertParsesAs("a: b # why", "(<a>: <b>)");
        }

        [Fact]
        public static void ACommentLineInsideAnIndentedBlockIsSkipped()
        {
            Assert.Equal(
                Rendered("parent\n  child\n"),
                Rendered("parent\n  # what the child is for\n  child\n"));
        }

        [Fact]
        public static void ACommentLineInsideAGroupIsSkipped()
        {
            Assert.Equal(
                Rendered("(\n  a\n  b\n)\n"),
                Rendered("(\n  a\n  # why\n  b\n)\n"));
        }

        [Fact]
        public static void ALineOfSpacesSeparatesLinksTheWayAnEmptyLineDoes()
        {
            // Blanking a comment leaves a line of spaces behind, so such a line
            // has to read the way an empty line does.
            Assert.Equal(Rendered("a\n\nb\n"), Rendered("a\n   \nb\n"));
        }

        [Fact]
        public static void ADocumentOfCommentsAloneHoldsNoLinks()
        {
            AssertParsesAs("# one\n# two\n", "");
        }

        [Fact]
        public static void AHashInsideATokenIsAnOrdinaryCharacter()
        {
            AssertParsesAs("issue#1047\n", "(<issue#1047>)");
        }

        [Fact]
        public static void AHashThatOpensATokenIsAnOrdinaryCharacter()
        {
            AssertParsesAs("a: b#c\n", "(<a>: <b#c>)");
        }

        [Fact]
        public static void AHashInsideADelimitedReferenceIsContent()
        {
            AssertParsesAs("\"# not a comment\" a\n", "(<# not a comment> <a>)");
        }

        [Fact]
        public static void ACommentMayFollowADelimitedReference()
        {
            AssertParsesAs("\"a\" # why\n", "(<a>)");
        }

        [Fact]
        public static void AHashInsideAMultiLineDelimitedReferenceIsContent()
        {
            AssertParsesAs("\"a # b\nc\" d\n", "(<a # b\nc> <d>)");
        }

```

```
        [Fact]
        public static void CommentsAreOnByDefault()
        {
            Assert.True(new Parser().Comments);
            AssertParsesAs("# a b\n", "");
        }

        [Fact]
        public static void AParserWithoutCommentsKeepsTheHash()
        {
            Assert.Equal("(<#> <a> <b>)", RenderedWith(new Parser(comments: false), "# a b\n"));
        }

        [Fact]
        public static void BlankingACommentKeepsTheLengthOfTheDocument()
        {
            Assert.Equal("a: b      \n", Comments.StripComments("a: b # why\n"));
            Assert.Equal("\"# kept\"\n", Comments.StripComments("\"# kept\"\n"));
            Assert.Equal("issue#1047\n", Comments.StripComments("issue#1047\n"));
            Assert.Equal("     \n", Comments.StripComments("# why\n"));
        }

        [Fact]
        public static void ACommentDoesNotMoveThePositionALaterErrorIsReportedAt()
        {
            var error = Assert.Throws<ParseException>(
                () => new Parser().Parse("# a comment\nstage: rust: nextest\n"));

            Assert.Equal(2, error.Line);
            Assert.Equal(12, error.Column);
        }

        [Fact]
        public static void AReferenceThatBeginsWithAHashIsWrittenQuoted()
        {
            // Without the quotes the document would read as `a` followed by a comment.
            var document = new Link<string>(new Link<string>("a"), new
            ↪  Link<string>("#tag")).ToString();

            Assert.Equal("(a '#tag')", document);
            Assert.Equal("(<a> <#tag>)", Rendered(document));
        }

        [Fact]
        public static void AHashThatCannotOpenACommentIsLeftUnquoted()
        {
            Assert.Equal("issue#1047", Link<string>.EscapeReference("issue#1047"));
            Assert.Equal("'#'", Link<string>.EscapeReference("#"));
            Assert.Equal("'#ff0000'", Link<string>.EscapeReference("#ff0000"));
        }

        // Issue #377: the comment reader and the grammar must agree on where every
        // delimited reference starts and how far it reaches, or a comment is blanked
        // inside what the grammar then reads as a reference.

        [Fact]
        public static void AHashInsideAReferenceThatFollowsAGroupIsContent()
        {
            AssertParsesAs("(a)\" #b\"\n", "((<a>) < #b>)");
        }

        [Fact]
        public static void AHashInsideAReferenceThatFollowsAnotherReferenceIsContent()
        {
            AssertParsesAs("\"x\"' #y'\n", "(<x> < #y>)");
            AssertParsesAs("```q```\"\" #c\n\"\"\"\n", "(<q> < #c\n\">)");
        }

        [Fact]
        public static void AHashInsideAReferenceThatFollowsALiteralIsContent()
        {
            AssertParsesAs("~1{62}\" #b\"\n", "(<b> < #b>)");
        }

        [Fact]
        public static void AReferenceIsReadInTheDocumentAsWrittenCommentsIncluded()
        {
            // The body of the first `""` straddles a parenthesis, so it is the empty
            // reference and the `#` after it opens a comment. Blanking that comment must
            // not turn the `""` into a reference that holds the blanked text.
```

```
            AssertParsesAs("\"\" a #)b\n\"\"\n", "(<> <a>)\n(<>)");
        }

        [Fact]
        public static void TheCommentReaderSkipsTheReferencesTheGrammarReads()
        {
            Assert.Equal("(a)\" #b\"\n", Comments.StripComments("(a)\" #b\"\n"));
            Assert.Equal("\"x\"' #y'\n", Comments.StripComments("\"x\"' #y'\n"));
            Assert.Equal("~1{62}\" #b\"\n", Comments.StripComments("~1{62}\" #b\"\n"));
            Assert.Equal("a\"b    \n", Comments.StripComments("a\"b #c\"\n"));
        }
    }
}
```

1.33 ./Link.Foundation.Links.Notation.Tests/CstTests.cs

```
using System.Linq;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    /// <summary>
    /// The lossless concrete syntax tree of a document (issue #364).
    /// </summary>
    public static class CstTests
    {
        private static string RoundTrip(string input) => new Parser().ParseCst(input).ToString();

        [Fact]
        public static void IsTheLosslessViewOfADocument()
        {
            const string input = "(t 1 '1')";
            var cst = new Parser().ParseCst(input);
            Assert.Equal(CstKind.Document, cst.Kind);
            Assert.False(cst.IsToken);
            Assert.Equal(input, cst.ToString());
            Assert.Equal(input, cst.Text);
            Assert.Equal(
                "(document (line (group open paren:\"(\" (line reference:\"t\" whitespace:\" \"
                ↪  reference:\"1\" whitespace:\" \" reference:\"'1'\") close paren:\")\")))",
                cst.ToSExpression());
        }

        [Fact]
        public static void EmptyDocumentHasNoLines()
        {
            var cst = new Parser().ParseCst("");
            Assert.Empty(cst.Children);
            Assert.Equal("(document)", cst.ToSExpression());
        }

        [Fact]
        public static void ReferenceTokensCarryTheirValueAndDelimiter()
        {
            var references = new Parser()
                .ParseCst("(a \"b c\" ~1{64} ''e''''f'')")
                .Tokens()
                .Where(token => token.Kind == CstKind.Reference)
                .Select(token => (token.Text, token.Value, token.Quote))
                .ToArray();
            Assert.Equal(
                new (string, string?, char?)[]
                {
                    ("a", "a", null),
                    ("\"b c\"", "b c", '"'),
                    ("~1{64}", "d", null),
                    ("''e''''f''", "e''f", '\''),
                },
                references);
        }

        [Fact]
        public static void TokensHaveNoChildrenAndOnlyReferencesHaveValues()
        {
            foreach (var token in new Parser().ParseCst("a: (b) # c\n").Tokens())
            {
                Assert.True(token.IsToken);
                Assert.Empty(token.Children);
                Assert.Equal(token.Kind == CstKind.Reference, token.Value != null);
                Assert.Equal(token.Text, token.ToString());
            }
```

```
        }

        [Theory]
        [InlineData("")]
        [InlineData(" \t ")]
        [InlineData("\n\n")]
        [InlineData("a\r\nb\rc\r\n")]
        [InlineData("a\t b  \n")]
        [InlineData("# head\na: b # tail\n  # inner\n  c\n# end")]
        [InlineData("(note: \"one\n  two\r\nthree\")")]
        [InlineData("(a (b\n  (c d)\n) e)\n")]
        [InlineData("root:\n  child:\n    leaf\n  tail")]
        [InlineData("(~1{} ~1{e6a0b9})")]
        [InlineData("('#a' \"b # c\" `#`)")]
        [InlineData("(\u6839: café \U0001F642)\n")]
        public static void RecoversTheDocumentExactly(string input) => Assert.Equal(input,
        ↪  RoundTrip(input));

        [Fact]
        public static void WithoutCommentSupportHashIsPartOfReferences()
        {
            const string input = "a #b";
            var cst = new Parser(comments: false).ParseCst(input);
            Assert.Equal(input, cst.ToString());
            Assert.Equal(
                new[] { CstKind.Reference, CstKind.Whitespace, CstKind.Reference },
                cst.Tokens().Select(token => token.Kind).ToArray());
            Assert.Equal(
                new[] { CstKind.Reference, CstKind.Whitespace, CstKind.Comment },
                new Parser().ParseCst(input).Tokens().Select(token => token.Kind).ToArray());
        }

        [Fact]
        public static void AGroupBelongsToTheLineItOpensOn()
        {
            var cst = new Parser().ParseCst("a (b\nc)\nd");
            Assert.Equal(new[] { "a (b\nc)\n", "d" }, cst.Children.Select(line =>
            ↪  line.ToString()).ToArray());
            var group = cst.Children[0].Children[2];
            Assert.Equal(CstKind.Group, group.Kind);
            Assert.Equal(
                new[] { CstKind.OpenParen, CstKind.Line, CstKind.Line, CstKind.CloseParen },
                group.Children.Select(child => child.Kind).ToArray());
        }

        [Fact]
        public static void UnclosedQuoteIsABareReference()
        {
            var references = new Parser().ParseCst("('x y)").Tokens()
                .Where(token => token.Kind == CstKind.Reference)
                .Select(token => (token.Text, token.Quote))
                .ToArray();
            Assert.Equal(new (string, char?)[] { ("'x", null), ("y", null) }, references);
        }

        [Fact]
        public static void KindsHaveTheSharedNames()
        {
            Assert.Equal(
                "document line group whitespace newline comment open paren close paren colon
                ↪  reference",
                string.Join(" ", System.Enum.GetValues<CstKind>().Select(kind =>
                ↪  kind.ToName())));
        }

        [Fact]
        public static void RejectsWhatParseRejects()
        {
            Assert.Throws<ParseException>(() => new Parser().ParseCst("(a"));
            Assert.Throws<ParseException>(() => new Parser().ParseCst("a)"));
            var tooDeep = Assert.Throws<ParseException>(() => new Parser(true,
            ↪  1).ParseCst("((a))"));
            Assert.Equal(1, tooDeep.MaxDepth);
        }
    }
}
```

1.34 ./Link.Foundation.Links.Notation.Tests/EdgeCaseParserTests.cs

```
using System;
using Xunit;
```

```
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class EdgeCaseParserTests
7      {
8          [Fact]
9          public static void EmptyLinkTest()
10         {
11             var source = @":";
12             var parser = new Parser();
13             // Standalone ':' is now forbidden and should throw an exception
14             Assert.Throws<ParseException>(() => parser.Parse(source));
15         }
16
17         [Fact]
18         public static void EmptyLinkWithParenthesesTest()
19         {
20             var source = @"()";
21             var target = @"()";
22             var parser = new Parser();
23             var links = parser.Parse(source);
24             var formattedLinks = links.Format();
25             Assert.Equal(target, formattedLinks);
26         }
27
28         [Fact]
29         public static void EmptyLinkWithEmptySelfReferenceTest()
30         {
31             var source = @"(:)";
32             var parser = new Parser();
33             // '(:)' is now forbidden and should throw an exception
34             Assert.Throws<ParseException>(() => parser.Parse(source));
35         }
36
37         [Fact]
38         public static void AllFeaturesTest()
39         {
40             // Test single-line link with id
41             var input = "id: value1 value2";
42             var result = new Parser().Parse(input);
43             Assert.NotEmpty(result);
44
45             // Test multi-line link with id
46             input = "(id: value1 value2)";
47             result = new Parser().Parse(input);
48             Assert.NotEmpty(result);
49
50             // Test link without id (single-line) - now forbidden
51             input = ": value1 value2";
52             Assert.Throws<ParseException>(() => new Parser().Parse(input));
53
54             // Test link without id (multi-line) - now forbidden
55             input = "(: value1 value2)";
56             Assert.Throws<ParseException>(() => new Parser().Parse(input));
57
58             // Test singlet link
59             input = "(singlet)";
60             result = new Parser().Parse(input);
61             Assert.Single(result);
62             Assert.Null(result[0].Id);
63             Assert.NotNull(result[0].Values);
64             Assert.Single(result[0].Values!);
65             Assert.Equal("singlet", result[0].Values?[0].Id);
66             Assert.Null(result[0].Values?[0].Values);
67
68             // Test value link
69             input = "(value1 value2 value3)";
70             result = new Parser().Parse(input);
71             Assert.NotEmpty(result);
72
73             // Test quoted references
74             input = @"(""id with spaces"": ""value with spaces"")";
75             result = new Parser().Parse(input);
76             Assert.NotEmpty(result);
77
78             // Test single-quoted references
79             input = "('id': 'value')";
80             result = new Parser().Parse(input);
81             Assert.NotEmpty(result);
82
83             // Test nested links
```

```
            input = "(outer: (inner: value))";
            result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void TestSingletLinks()
        {
            // Test singlet (1)
            var input = "(1)";
            var result = new Parser().Parse(input);
            Assert.Single(result);
            Assert.Null(result[0].Id);
            Assert.NotNull(result[0].Values);
            Assert.Single(result[0].Values!);
            Assert.Equal("1", result[0].Values?[0].Id);
            Assert.Null(result[0].Values?[0].Values);

            // Test (1 2)
            input = "(1 2)";
            result = new Parser().Parse(input);
            Assert.Single(result);
            Assert.Null(result[0].Id);
            Assert.NotNull(result[0].Values);
            Assert.Equal(2, result[0].Values?.Count);
            Assert.Equal("1", result[0].Values?[0].Id);
            Assert.Null(result[0].Values?[0].Values);
            Assert.Equal("2", result[0].Values?[1].Id);
            Assert.Null(result[0].Values?[1].Values);

            // Test (1 2 3)
            input = "(1 2 3)";
            result = new Parser().Parse(input);
            Assert.Single(result);
            Assert.Null(result[0].Id);
            Assert.NotNull(result[0].Values);
            Assert.Equal(3, result[0].Values?.Count);
            Assert.Equal("1", result[0].Values?[0].Id);
            Assert.Null(result[0].Values?[0].Values);
            Assert.Equal("2", result[0].Values?[1].Id);
            Assert.Null(result[0].Values?[1].Values);
            Assert.Equal("3", result[0].Values?[2].Id);
            Assert.Null(result[0].Values?[2].Values);

            // Test (1 2 3 4)
            input = "(1 2 3 4)";
            result = new Parser().Parse(input);
            Assert.Single(result);
            Assert.Null(result[0].Id);
            Assert.NotNull(result[0].Values);
            Assert.Equal(4, result[0].Values?.Count);
            Assert.Equal("1", result[0].Values?[0].Id);
            Assert.Null(result[0].Values?[0].Values);
            Assert.Equal("2", result[0].Values?[1].Id);
            Assert.Null(result[0].Values?[1].Values);
            Assert.Equal("3", result[0].Values?[2].Id);
            Assert.Null(result[0].Values?[2].Values);
            Assert.Equal("4", result[0].Values?[3].Id);
            Assert.Null(result[0].Values?[3].Values);
        }

        [Fact]
        public static void EmptyDocumentTest()
        {
            var input = "";
            // Empty document should return empty list
            var result = new Parser().Parse(input);
            Assert.Empty(result);
        }

        [Fact]
        public static void WhitespaceOnlyTest()
        {
            var input = "   \n   \n   ";
            // Whitespace-only document should return empty list (similar to empty document)
            var result = new Parser().Parse(input);
            Assert.Empty(result);
        }

        [Fact]
```

```
164         public static void EmptyLinksTest()
165         {
166             var input = "()";
167             var result = new Parser().Parse(input);
168             Assert.NotEmpty(result);
169 
170             // '(:)' is now forbidden
171             input = "(:)";
172             Assert.Throws<ParseException>(() => new Parser().Parse(input));
173 
174             input = "(id:)";
175             result = new Parser().Parse(input);
176             Assert.NotEmpty(result);
177         }
178 
179         [Fact]
180         public static void InvalidInputTest()
181         {
182             var input = "(invalid";
183             // Unclosed parentheses should throw an exception
184             Assert.Throws<ParseException>(() => new Parser().Parse(input));
185         }
186     }
187 }
```

1.35 ./Link.Foundation.Links.Notation.Tests/EmptyReferenceTests.cs

```
1  using System.Collections.Generic;
2  using System.Linq;
3  using Xunit;
4 
5  namespace Link.Foundation.Links.Notation.Tests
6  {
7      /// <summary>
8      /// Conformance tests for the empty reference.
9      ///
10     /// https://github.com/link-foundation/links-notation/issues/288
11     ///
12     /// A bare delimiter pair is the empty reference. The three delimiters
13     /// <c>""</c>, <c>''</c> and <c>``</c> behave identically, and every longer
14     /// n-quote run keeps the meaning it already had. The table below is shared
15     /// with the Rust, JavaScript, Python, Go, Java and PHP suites, so a document
16     /// written by one implementation reads the same in all of them.
17     /// </summary>
18     public static class EmptyReferenceTests
19     {
20         /// <summary>
21         /// Renders a parsed node unambiguously: every reference is wrapped in
22         /// angle brackets so an empty one is visible as &amp;lt;&amp;gt;.
23         /// </summary>
24         private static string Render(Link<string> node)
25         {
26             if (node.Values == null || node.Values.Count == 0)
27             {
28                 return "<" + (node.Id ?? "") + ">";
29             }
30             var head = node.Id == null ? "" : "<" + node.Id + ">: ";
31             return "(" + head + string.Join(" ", node.Values.Select(Render)) + ")";
32         }
33 
34         private static string Rendered(string source)
35         {
36             var links = new Parser().Parse(source);
37             return string.Join("\n", links.Select(Render));
38         }
39 
40         private static void AssertParsesAs(string source, string expected)
41         {
42             Assert.Equal(expected, Rendered(source));
43         }
44 
45         [Fact]
46         public static void BareDelimiterPairIsTheEmptyReference()
47         {
48             AssertParsesAs("(a \"\" b)", "(<a> <> <b>)");
49         }
50 
51         [Fact]
52         public static void EveryDelimiterStyleYieldsTheSameEmptyReference()
53         {
54             AssertParsesAs("(a \"\" b)", "(<a> <> <b>)");
55             AssertParsesAs("(a '' b)", "(<a> <> <b>)");
```

```
            AssertParsesAs("(a `` b)", "(<a> <> <b>)");
        }

        [Fact]
        public static void AdjacentEmptyReferencesStaySeparate()
        {
            AssertParsesAs("(a \"\" \"\" b)", "(<a> <> <> <b>)");
            AssertParsesAs("(a '' '' b)", "(<a> <> <> <b>)");
            AssertParsesAs("(a `` `` b)", "(<a> <> <> <b>)");
            AssertParsesAs("(a \"\"  \"\" b)", "(<a> <> <> <b>)");
        }

        [Fact]
        public static void NestedEmptyReferencesParse()
        {
            AssertParsesAs("(\"\" (\"\" 1))", "(<> (<> <1>))");
            AssertParsesAs("(\"\" ('' 1))", "(<> (<> <1>))");
            AssertParsesAs("(\"x\" (\"\" 1))", "(<x> (<> <1>))");
            AssertParsesAs("(\"\" (\"x\" 1))", "(<> (<x> <1>))");
            AssertParsesAs("(\"\" x (\"\" 1))", "(<> <x> (<> <1>))");
            AssertParsesAs("(\"\" 1 (\"\" 1))", "(<> <1> (<> <1>))");
        }

        [Fact]
        public static void EmptyReferenceIsValidAsAnId()
        {
            AssertParsesAs("(\"\": 1)", "(<>: <1>)");
            AssertParsesAs("(o: (\"\" (o: (\"\" 1))))", "(<o>: (<> (<o>: (<> <1>))))");
        }

        [Fact]
        public static void NQuoteDelimitedBodiesAreUnchanged()
        {
            // A run that encloses a substantive body keeps its n-quote meaning.
            AssertParsesAs("(a \"\"x\"\" b)", "(<a> <x> <b>)");
            AssertParsesAs("(x \"\" \" \"\")", "(<x> < \" >)");
            AssertParsesAs("(x ' \" ')", "(<x> < \" >)");
            // An n-quote-delimited empty is still empty.
            AssertParsesAs("(a \"\"\"\" b)", "(<a> <> <b>)");
        }

        [Fact]
        public static void ASingleSpaceStillReadsAsASpace()
        {
            AssertParsesAs("(a \" \" b)", "(<a> < > <b>)");
        }

        [Fact]
        public static void EmptyReferenceSurvivesARoundTrip()
        {
            var sources = new[]
            {
                "(a \"\" b)",
                "(a \"\" \"\" b)",
                "(\"\" (\"\" 1))",
                "(\"\": 1)",
                "(o: (\"\" (o: (\"\" 1))))",
            };
            foreach (var source in sources)
            {
                var formatted = ((IList<Link<string>>)new Parser().Parse(source)).Format();
                var reparsed = ((IList<Link<string>>)new Parser().Parse(formatted)).Format();
                Assert.Equal(formatted, reparsed);
            }
        }

        [Fact]
        public static void EmptyReferenceIsWrittenAsAVersionedLiteral()
        {
            var links = (IList<Link<string>>)new Parser().Parse("(a \"\" b)");
            Assert.Equal("(a ~1{} b)", links.Format());
        }
    }
}
```

1.36 ./Link.Foundation.Links.Notation.Tests/FormatConfigTests.cs

```
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
```

```
5       public class FormatConfigTests
6       {
7           [Fact]
8           public void FormatConfigBasicTest()
9           {
10              var config = new FormatConfig();
11              Assert.False(config.LessParentheses);
12              Assert.Equal(80, config.MaxLineLength);
13              Assert.False(config.IndentLongLines);
14          }
15
16          [Fact]
17          public void FormatWithLineLengthLimitTest()
18          {
19              // Create a config with line length limit
20              // The line would be 32+ chars, so use threshold of 30
21              var config = new FormatConfig
22              {
23                  IndentLongLines = true,
24                  MaxLineLength = 30,
25                  PreferInline = false
26              };
27
28              // Verify config is set correctly
29              Assert.True(config.IndentLongLines);
30              Assert.Equal(30, config.MaxLineLength);
31              // Note: Full formatting integration would test actual output here
32          }
33
34          [Fact]
35          public void FormatWithMaxInlineRefsTest()
36          {
37              // Create a config with MaxInlineRefs=3 (should trigger indentation with 4 refs)
38              var config = new FormatConfig
39              {
40                  MaxInlineRefs = 3,
41                  PreferInline = false
42              };
43
44              // Verify config is set correctly
45              Assert.Equal(3, config.MaxInlineRefs);
46              Assert.True(config.ShouldIndentByRefCount(4));
47              // Note: Full formatting integration would test actual output here
48          }
49
50          [Fact]
51          public void FormatWithConsecutiveGroupingTest()
52          {
53              // Create a config with consecutive grouping enabled
54              var config = new FormatConfig
55              {
56                  GroupConsecutive = true
57              };
58
59              // Verify config is set correctly
60              Assert.True(config.GroupConsecutive);
61              // Note: Full formatting integration would test grouping behavior here
62          }
63
64          [Fact]
65          public void FormatConfigCustomIndentTest()
66          {
67              var config = new FormatConfig
68              {
69                  MaxInlineRefs = 3,
70                  PreferInline = false,
71                  IndentString = "    " // 4 spaces
72              };
73
74              Assert.Equal("    ", config.IndentString);
75          }
76
77          [Fact]
78          public void FormatConfigLessParenthesesTest()
79          {
80              var config = new FormatConfig
81              {
82                  LessParentheses = true
83              };
84
85              Assert.True(config.LessParentheses);
86          }
```

```
        [Fact]
        public void RoundtripWithLineLengthFormattingTest()
        {
            // This test would require full format integration
            // For now, just verify config creation
            var config = new FormatConfig
            {
                MaxInlineRefs = 2,
                PreferInline = false
            };

            Assert.Equal(2, config.MaxInlineRefs);
        }

        [Fact]
        public void ShouldIndentByLengthTest()
        {
            var config = new FormatConfig
            {
                IndentLongLines = true,
                MaxLineLength = 80
            };

            Assert.False(config.ShouldIndentByLength("short"));
            Assert.True(config.ShouldIndentByLength(new string('a', 100)));
        }

        [Fact]
        public void ShouldIndentByRefCountTest()
        {
            var config = new FormatConfig
            {
                MaxInlineRefs = 3
            };

            Assert.False(config.ShouldIndentByRefCount(2));
            Assert.False(config.ShouldIndentByRefCount(3));
            Assert.True(config.ShouldIndentByRefCount(4));
        }
    }
}
```

1.37 ./Link.Foundation.Links.Notation.Tests/IndentationConsistencyTests.cs

```
using Xunit;
using System.Collections.Generic;

namespace Link.Foundation.Links.Notation.Tests
{
    public class IndentationConsistencyTests
    {
        [Fact]
        public void LeadingSpacesVsNoLeadingSpaces()
        {
            // Example with 2 leading spaces (from issue #135)
            var withLeading = @"  TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
  TELEGRAM ALLOWED CHATS:
    -1002975819706
    -1002861722681
  TELEGRAM HIVE OVERRIDES:
    --all-issues
    --once
  TELEGRAM BOT VERBOSE: true";

            // Example without leading spaces (from issue #135)
            var withoutLeading = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
TELEGRAM ALLOWED CHATS:
  -1002975819706
  -1002861722681
TELEGRAM HIVE OVERRIDES:
  --all-issues
  --once
TELEGRAM BOT VERBOSE: true";

            var resultWith = new Parser().Parse(withLeading);
            var resultWithout = new Parser().Parse(withoutLeading);

            // Compare the entire formatted output (complete round trip test)
            Assert.Equal(resultWithout.Format(), resultWith.Format());
        }

        [Fact]
        public void TwoSpacesVsFourSpacesIndentation()
```

```
        {
            // Example with 2 spaces per level
            var twoSpaces = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
TELEGRAM ALLOWED CHATS:
  -1002975819706
  -1002861722681
TELEGRAM HIVE OVERRIDES:
  --all-issues
  --once
  --auto-fork
  --skip-issues-with-prs
  --attach-logs
  --verbose
  --no-tool-check
TELEGRAM SOLVE OVERRIDES:
  --auto-fork
  --auto-continue
  --attach-logs
  --verbose
  --no-tool-check
TELEGRAM BOT VERBOSE: true";

            // Example with 4 spaces per level
            var fourSpaces = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
TELEGRAM ALLOWED CHATS:
    -1002975819706
    -1002861722681
TELEGRAM HIVE OVERRIDES:
    --all-issues
    --once
    --auto-fork
    --skip-issues-with-prs
    --attach-logs
    --verbose
    --no-tool-check
TELEGRAM SOLVE OVERRIDES:
    --auto-fork
    --auto-continue
    --attach-logs
    --verbose
    --no-tool-check
TELEGRAM BOT VERBOSE: true";

            var resultTwo = new Parser().Parse(twoSpaces);
            var resultFour = new Parser().Parse(fourSpaces);

            // Compare the entire formatted output (complete round trip test)
            Assert.Equal(resultFour.Format(), resultTwo.Format());
        }

        [Fact]
        public void SimpleTwoVsFourSpacesIndentation()
        {
            // Simple example with 2 spaces
            var twoSpaces = @"parent:
  child1
  child2";

            // Simple example with 4 spaces
            var fourSpaces = @"parent:
    child1
    child2";

            var resultTwo = new Parser().Parse(twoSpaces);
            var resultFour = new Parser().Parse(fourSpaces);

            // Compare the entire formatted output (complete round trip test)
            Assert.Equal(resultFour.Format(), resultTwo.Format());
        }

        [Fact]
        public void ThreeLevelNestingWithDifferentIndentation()
        {
            // Three levels with 2 spaces
            var twoSpaces = @"level1:
  level2:
    level3a
    level3b
  level2b";

            // Three levels with 4 spaces
            var fourSpaces = @"level1:
    level2:
        level3a
        level3b
```

```
    level2b";

            var resultTwo = new Parser().Parse(twoSpaces);
            var resultFour = new Parser().Parse(fourSpaces);

            // Compare the entire formatted output (complete round trip test)
            Assert.Equal(resultFour.Format(), resultTwo.Format());
        }
    }
}
```

1.38 ./Link.Foundation.Links.Notation.Tests/IndentedIdNestedValuesTests.cs

```
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public class IndentedIdNestedValuesTests
    {
        [Fact]
        public void IndentedIdRetainsNestedValues()
        {
            var cases = new[]
            {
                new[] { "outer:\n  inner:\n    value1\n    value2\n  value3", "(outer: (inner:
                ↪  value1 value2) value3)" },
                new[] { "statement:\n  subject:\n    I\n  verb:\n    love\n  object:\n    you:\n
                ↪  very\n      much", "(statement: (subject: I) (verb: love) (object: (you: very
                ↪  much)))" },
                new[] { "document:\n  (metadata: title author date)\n  content:\n    paragraph1\n
                ↪  (paragraph2: text (with: nested structure))\n    paragraph3", "(document:
                ↪  (metadata: title author date) (content: paragraph1 (paragraph2: text (with:
                ↪  nested structure)) paragraph3))" },
                new[] { "image:\n    file: .gitpod.Dockerfile\n    context: ./docker-content",
                ↪  "(image: (file: .gitpod.Dockerfile) (context: ./docker-content))" },
                new[] { "level1:\n  level2:\n    level3a\n    level3b\n  level2b", "(level1:
                ↪  (level2: level3a level3b) level2b)" },
                new[] { "root:\n  child1\n  child2\n    grandchild", "(root: child1 (child2
                ↪  grandchild))" },
            };

            var parser = new Parser();
            foreach (var testCase in cases)
            {
                Assert.Equal(testCase[1], parser.Parse(testCase[0]).Format());
            }
        }
    }
}
```

1.39 ./Link.Foundation.Links.Notation.Tests/IndentedIdSyntaxTests.cs

```
using System;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class IndentedIdSyntaxTests
    {
        [Fact]
        public static void BasicIndentedIdSyntaxTest()
        {
            var indentedSyntax = @"3:
  papa
  loves
  mama";
            var inlineSyntax = "(3: papa loves mama)";

            var parser = new Parser();
            var indentedResult = parser.Parse(indentedSyntax);
            var inlineResult = parser.Parse(inlineSyntax);

            var indentedFormatted = indentedResult.Format();
            var inlineFormatted = inlineResult.Format();

            Assert.Equal(inlineFormatted, indentedFormatted);
            Assert.Equal("(3: papa loves mama)", indentedFormatted);
        }

        [Fact]
        public static void IndentedIdSyntaxWithSingleValueTest()
        {
```

```
            var input = @"greeting:
  hello";

            var parser = new Parser();
            var result = parser.Parse(input);
            var formatted = result.Format();

            Assert.Equal("(greeting: hello)", formatted);
            Assert.Single(result);
            Assert.Equal("greeting", result[0].Id);
            Assert.Single(result[0].Values!);
            Assert.Equal("hello", result[0].Values![0].Id);
        }

        [Fact]
        public static void IndentedIdSyntaxWithMultipleValuesTest()
        {
            var input = @"action:
  run
  fast
  now";

            var parser = new Parser();
            var result = parser.Parse(input);
            var formatted = result.Format();

            Assert.Equal("(action: run fast now)", formatted);
            Assert.Single(result);
            Assert.Equal("action", result[0].Id);
            Assert.Equal(3, result[0].Values!.Count);
        }

        [Fact]
        public static void IndentedIdSyntaxWithNumericIdTest()
        {
            var input = @"42:
  answer
  to
  everything";

            var parser = new Parser();
            var result = parser.Parse(input);
            var formatted = result.Format();

            Assert.Equal("(42: answer to everything)", formatted);
        }

        [Fact]
        public static void IndentedIdSyntaxWithQuotedIdTest()
        {
            var input = @"""complex id"":
  value1
  value2";

            var parser = new Parser();
            var result = parser.Parse(input);
            var formatted = result.Format();

            Assert.Equal("('complex id': value1 value2)", formatted);
        }

        [Fact]
        public static void MultipleIndentedIdLinksTest()
        {
            var input = @"first:
  a
  b
second:
  c
  d";

            var parser = new Parser();
            var result = parser.Parse(input);
            var formatted = result.Format();

            Assert.Equal(2, result.Count);
            Assert.Contains("(first: a b)", formatted);
            Assert.Contains("(second: c d)", formatted);
        }

        [Fact]
        public static void MixedIndentedAndRegularSyntaxTest()
```

```
        {
            var input = @"first:
  a
  b
(second: c d)
third value";

            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.Equal(3, result.Count);

            var formatted = result.Format();
            Assert.Contains("(first: a b)", formatted);
            Assert.Contains("(second: c d)", formatted);
            Assert.Contains("third value", formatted);
        }

        [Fact]
        public static void UnsupportedColonOnlySyntaxShouldFailTest()
        {
            var input = @":
  papa
  loves
  mama";

            var parser = new Parser();
            Assert.Throws<ParseException>(() => parser.Parse(input));
        }

        [Fact]
        public static void EmptyIndentedIdShouldWorkTest()
        {
            var input = "empty:";

            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.Single(result);
            Assert.Equal("empty", result[0].Id);
            var emptyValues = result[0].Values;
            Assert.True(emptyValues == null || emptyValues.Count == 0);

            var formatted = result.Format();
            Assert.Equal("(empty)", formatted);
        }

        [Fact]
        public static void EquivalenceTestComprehensiveTest()
        {
            var testCases = new[]
            {
                new { Indented = "test:\n  one", Inline = "(test: one)" },
                new { Indented = "x:\n  a\n  b\n  c", Inline = "(x: a b c)" },
                new { Indented = "\"quoted\":\n  value", Inline = "(\"quoted\": value)" }
            };

            var parser = new Parser();

            foreach (var testCase in testCases)
            {
                var indentedResult = parser.Parse(testCase.Indented);
                var inlineResult = parser.Parse(testCase.Inline);

                var indentedFormatted = indentedResult.Format();
                var inlineFormatted = inlineResult.Format();

                Assert.Equal(inlineFormatted, indentedFormatted);
            }
        }

        [Fact]
        public static void IndentedIdWithDeeperNestingTest()
        {
            var input = @"root:
  child1
  child2
    grandchild";

            var parser = new Parser();
            var result = parser.Parse(input);

```

```
            Assert.NotEmpty(result);

            var rootLink = result[0];
            Assert.Equal("root", rootLink.Id);
            Assert.Equal(2, rootLink.Values!.Count);
            Assert.Equal("(root: child1 (child2 grandchild))", result.Format());
        }
    }
}
```

1.40 ./Link.Foundation.Links.Notation.Tests/LinkTests.cs

```
using System;
using Xunit;
using System.Collections.Generic;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class LinkTests
    {
        [Fact]
        public static void LinkConstructorWithIdOnlyTest()
        {
            var link = new Link<string>("test", null);
            Assert.Equal("test", link.Id);
            Assert.Null(link.Values);
        }

        [Fact]
        public static void LinkConstructorWithIdAndValuesTest()
        {
            var values = new List<Link<string>> { new Link<string>("value1"), new
            ↪  Link<string>("value2") };
            var link = new Link<string>("parent", values);
            Assert.Equal("parent", link.Id);
            Assert.Equal(2, link.Values?.Count);
        }

        [Fact]
        public static void LinkToStringWithIdOnlyTest()
        {
            var link = new Link<string>("test");
            Assert.Equal("(test)", link.ToString());
        }

        [Fact]
        public static void LinkToStringWithValuesOnlyTest()
        {
            var values = new List<Link<string>> { new Link<string>("value1"), new
            ↪  Link<string>("value2") };
            var link = new Link<string>(null, values);
            Assert.Equal("(value1 value2)", link.ToString());
        }

        [Fact]
        public static void LinkToStringWithIdAndValuesTest()
        {
            var values = new List<Link<string>> { new Link<string>("child1"), new
            ↪  Link<string>("child2") };
            var link = new Link<string>("parent", values);
            Assert.Equal("(parent: child1 child2)", link.ToString());
        }

        [Fact]
        public static void LinkEqualsTest()
        {
            var link1 = new Link<string>("test");
            var link2 = new Link<string>("test");
            var link3 = new Link<string>("other");

            Assert.Equal(link1, link2);
            Assert.NotEqual(link1, link3);
        }

        [Fact]
        public static void LinkCombineTest()
        {
            var link1 = new Link<string>("first");
            var link2 = new Link<string>("second");
            var combined = link1.Combine(link2);

```

```
            Assert.Null(combined.Id);
            Assert.Equal(2, combined.Values?.Count);
            Assert.Equal("first", combined.Values?[0].Id);
            Assert.Equal("second", combined.Values?[1].Id);
        }

        [Fact]
        public static void LinkEscapeReferenceSimpleTest()
        {
            Assert.Equal("test", Link<string>.EscapeReference("test"));
            Assert.Equal("test123", Link<string>.EscapeReference("test123"));
        }

        [Fact]
        public static void LinkEscapeReferenceWithSpecialCharactersTest()
        {
            Assert.Equal("'has space'", Link<string>.EscapeReference("has space"));
            Assert.Equal("'has:colon'", Link<string>.EscapeReference("has:colon"));
            Assert.Equal("'has(paren)'", Link<string>.EscapeReference("has(paren)"));
            Assert.Equal("'has)paren'", Link<string>.EscapeReference("has)paren"));
            Assert.Equal("\"has'quote\"", Link<string>.EscapeReference("has'quote"));
            Assert.Equal("'has\"quote'", Link<string>.EscapeReference("has\"quote"));
        }

        [Fact]
        public static void LinkSimplifyTest()
        {
            // Test simplify with empty values
            var link1 = new Link<string>("test", null);
            var simplified1 = link1.Simplify();
            Assert.Equal(link1, simplified1);

            // Test simplify with single value
            var link2 = new Link<string>(null, new List<Link<string>> { new
            ↪  Link<string>("single", null) });
            var simplified2 = link2.Simplify();
            Assert.Equal("single", simplified2.Id);
            Assert.Null(simplified2.Values);
        }
    }
}
```

1.41 ./Link.Foundation.Links.Notation.Tests/LinksGroupTests.cs

```
using System;
using System.Collections.Generic;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class LinksGroupTests
    {
        [Fact]
        public static void LinksGroupConstructorTest()
        {
            var element = new Link<string>("root", null);
            var groups = new List<LinksGroup<string>>
            {
                new LinksGroup<string>(new Link<string>("child1", null)),
                new LinksGroup<string>(new Link<string>("child2", null))
            };
            var group = new LinksGroup<string>(element, groups);

            Assert.Equal(element, group.Link);
            Assert.Equal(groups, group.Groups);
        }

        [Fact]
        public static void LinksGroupToListFlattensStructureTest()
        {
            var root = new Link<string>("root", null);
            var child1 = new Link<string>("child1", null);
            var child2 = new Link<string>("child2", null);
            var grandchild = new Link<string>("grandchild", null);

            var childGroup = new LinksGroup<string>(child2, new List<LinksGroup<string>>
            {
                new LinksGroup<string>(grandchild)
            });

            var group = new LinksGroup<string>(root, new List<LinksGroup<string>>
            {
```

```
                new LinksGroup<string>(child1),
                childGroup
            });

            var list = group.ToLinksList();
            // The C# implementation creates a hierarchical structure
            // root, root.Combine(child1), root.Combine(child2),
            ↪  root.Combine(child2).Combine(grandchild)
            Assert.Equal(4, list.Count);
            Assert.Equal(root, list[0]);
        }

        [Fact]
        public static void LinksGroupAppendToLinksListTest()
        {
            var element = new Link<string>("root", null);
            var children = new List<LinksGroup<string>>
            {
                new LinksGroup<string>(new Link<string>("child1", null)),
                new LinksGroup<string>(new Link<string>("child2", null))
            };
            var group = new LinksGroup<string>(element, children);

            var list = new List<Link<string>>();
            group.AppendToLinksList(list);

            Assert.Equal(3, list.Count);
            Assert.Equal(element, list[0]);
        }

        [Fact]
        public static void LinksGroupToStringTest()
        {
            var element = new Link<string>("root", null);
            var children = new List<LinksGroup<string>>
            {
                new LinksGroup<string>(new Link<string>("child1", null)),
                new LinksGroup<string>(new Link<string>("child2", null))
            };
            var group = new LinksGroup<string>(element, children);

            var str = group.ToString();
            Assert.Contains("root", str);
            Assert.Contains("child1", str);
            Assert.Contains("child2", str);
        }

        [Fact]
        public static void LinksGroupConstructorEquivalentTest()
        {
            // Test creating a LinksGroup structure in an equivalent way
            var root = new Link<string>("root", null);
            var children = new List<Link<string>>
            {
                new Link<string>("child1", null),
                new Link<string>("child2", null)
            };

            // Create a group with an id
            var groupElement = new Link<string>("group", null);
            var childGroups = new List<LinksGroup<string>>
            {
                new LinksGroup<string>(root),
                new LinksGroup<string>(children[0]),
                new LinksGroup<string>(children[1])
            };
            var group = new LinksGroup<string>(groupElement, childGroups);

            Assert.Equal("group", group.Link.Id);
            Assert.Equal(3, group.Groups!.Count);
            Assert.Equal(root, group.Groups[0].Link);
        }
    }
}
```

1.42 ./Link.Foundation.Links.Notation.Tests/MixedIndentationModesTests.cs

```
using System;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
```

```
  public static class MixedIndentationModesTests
  {
      [Fact]
      public static void HeroExampleMixedModesTest()
      {
          var input = @"empInfo
employees:
  (
    name (James Kirk)
    age 40
  )
  (
    name (Jean-Luc Picard)
    age 45
  )
  (
    name (Wesley Crusher)
    age 27
  )";

          var parser = new Parser();
          var result = parser.Parse(input);

          Assert.NotEmpty(result);
          var formatted = result.Format();
          Assert.Contains("empInfo", formatted);
          Assert.Contains("employees", formatted);
          Assert.Contains("James Kirk", formatted);
          Assert.Contains("Jean-Luc Picard", formatted);
          Assert.Contains("Wesley Crusher", formatted);
      }

      [Fact]
      public static void HeroExampleAlternativeFormatTest()
      {
          var input = @"empInfo
(
  employees:
    (
      name (James Kirk)
      age 40
    )
    (
      name (Jean-Luc Picard)
      age 45
    )
    (
      name (Wesley Crusher)
      age 27
    )
)";

          var parser = new Parser();
          var result = parser.Parse(input);

          Assert.NotEmpty(result);
          var formatted = result.Format();
          Assert.Contains("empInfo", formatted);
          Assert.Contains("employees:", formatted);
          Assert.Contains("James Kirk", formatted);
          Assert.Contains("Jean-Luc Picard", formatted);
          Assert.Contains("Wesley Crusher", formatted);
      }

      [Fact]
      public static void HeroExampleEquivalenceTest()
      {
          var version1 = @"empInfo
employees:
  (
    name (James Kirk)
    age 40
  )
  (
    name (Jean-Luc Picard)
    age 45
  )
  (
    name (Wesley Crusher)
    age 27
  )";
```

```
            var version2 = @"empInfo
    (
      employees:
        (
          name (James Kirk)
          age 40
        )
        (
          name (Jean-Luc Picard)
          age 45
        )
        (
          name (Wesley Crusher)
          age 27
        )
    )";

            var parser = new Parser();
            var result1 = parser.Parse(version1);
            var result2 = parser.Parse(version2);

            Assert.NotEmpty(result1);
            Assert.NotEmpty(result2);

            var formatted1 = result1.Format();
            var formatted2 = result2.Format();

            Assert.Equal(formatted1, formatted2);
        }

        [Fact]
        public static void SetContextWithoutColonTest()
        {
            var input = @"empInfo
    employees";

            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            var formatted = result.Format();
            Assert.Contains("empInfo", formatted);
            Assert.Contains("employees", formatted);
        }

        [Fact]
        public static void SequenceContextWithColonTest()
        {
            var input = @"employees:
    James Kirk
    Jean-Luc Picard
    Wesley Crusher";

            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            Assert.Single(result);
            var formatted = result.Format();
            Assert.Contains("employees:", formatted);
            Assert.Contains("James Kirk", formatted);
            Assert.Contains("Jean-Luc Picard", formatted);
            Assert.Contains("Wesley Crusher", formatted);
        }

        [Fact]
        public static void SequenceContextWithComplexValuesTest()
        {
            var input = @"employees:
    (
      name (James Kirk)
      age 40
    )
    (
      name (Jean-Luc Picard)
      age 45
    )";

            var parser = new Parser();
            var result = parser.Parse(input);

```

```
            Assert.NotEmpty(result);
            Assert.Single(result);
            var formatted = result.Format();
            Assert.Contains("employees:", formatted);
            Assert.Contains("James Kirk", formatted);
            Assert.Contains("Jean-Luc Picard", formatted);
        }

        [Fact]
        public static void NestedSetAndSequenceContextsTest()
        {
            var input = @"company
  departments:
    engineering
    sales
  employees:
    (name John)
    (name Jane)";

            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            var formatted = result.Format();
            Assert.Contains("company", formatted);
            Assert.Contains("departments:", formatted);
            Assert.Contains("employees:", formatted);
        }

        [Fact]
        public static void DeeplyNestedMixedModesTest()
        {
            var input = @"root
  level1
    level2:
      value1
      value2
    level2b
      level3";

            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            var formatted = result.Format();
            Assert.Contains("root", formatted);
            Assert.Contains("level2:", formatted);
        }
    }
}
```

1.43 ./Link.Foundation.Links.Notation.Tests/MultiQuoteParserTests.cs

```
using System;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class MultiQuoteParserTests
    {
        // Helper to extract single reference ID
        private static string? GetSingleRefId(System.Collections.Generic.IList<Link<string>>
        ↪  result)
        {
            if (result.Count == 1 && result[0].Id == null && result[0].Values?.Count == 1)
            {
                return result[0].Values![0].Id;
            }
            return result.Count == 1 ? result[0].Id : null;
        }

        // ============================================================================
        // Backtick Quote Tests (Single Backtick)
        // ============================================================================

        [Fact]
        public static void TestBacktickQuotedReference()
        {
            var parser = new Parser();
            var result = parser.Parse("`backtick quoted`");
            Assert.Equal("backtick quoted", GetSingleRefId(result));
        }

```

```
    [Fact]
    public static void TestBacktickQuotedWithSpaces()
    {
        var parser = new Parser();
        var result = parser.Parse("`text with spaces`");
        Assert.Equal("text with spaces", GetSingleRefId(result));
    }

    [Fact]
    public static void TestBacktickQuotedMultiline()
    {
        var parser = new Parser();
        var result = parser.Parse("(`line1\nline2`)");
        Assert.Single(result);
        Assert.NotNull(result[0].Values);
        Assert.Single(result[0].Values!);
        Assert.Equal("line1\nline2", result[0].Values![0].Id);
    }

    [Fact]
    public static void TestBacktickQuotedWithEscapedBacktick()
    {
        var parser = new Parser();
        var result = parser.Parse("`text with `` escaped backtick`");
        Assert.Equal("text with ` escaped backtick", GetSingleRefId(result));
    }

    // ========================================================================
    // Single Quote Tests (with escaping)
    // ========================================================================

    [Fact]
    public static void TestSingleQuoteWithEscapedSingleQuote()
    {
        var parser = new Parser();
        var result = parser.Parse("'text with '' escaped quote'");
        Assert.Equal("text with ' escaped quote", GetSingleRefId(result));
    }

    // ========================================================================
    // Double Quote Tests (with escaping)
    // ========================================================================

    [Fact]
    public static void TestDoubleQuoteWithEscapedDoubleQuote()
    {
        var parser = new Parser();
        var result = parser.Parse("\"text with \"\" escaped quote\"");
        Assert.Equal("text with \" escaped quote", GetSingleRefId(result));
    }

    // ========================================================================
    // Double Quotes (2 quote chars) Tests
    // ========================================================================

    [Fact]
    public static void TestDoubleDoubleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"double double quotes\"\"");
        Assert.Equal("double double quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestDoubleDoubleQuotesWithSingleQuoteInside()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"text with \" inside\"\"");
        Assert.Equal("text with \" inside", GetSingleRefId(result));
    }

    [Fact]
    public static void TestDoubleDoubleQuotesWithEscape()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"text with \"\"\"\" escaped double\"\"");
        Assert.Equal("text with \"\" escaped double", GetSingleRefId(result));
    }

    [Fact]
```

```
    public static void TestDoubleSingleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("''double single quotes''");
        Assert.Equal("double single quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestDoubleSingleQuotesWithSingleQuoteInside()
    {
        var parser = new Parser();
        var result = parser.Parse("''text with ' inside''");
        Assert.Equal("text with ' inside", GetSingleRefId(result));
    }

    [Fact]
    public static void TestDoubleSingleQuotesWithEscape()
    {
        var parser = new Parser();
        var result = parser.Parse("''text with '''' escaped single''");
        Assert.Equal("text with '' escaped single", GetSingleRefId(result));
    }

    [Fact]
    public static void TestDoubleBacktickQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("``double backtick quotes``");
        Assert.Equal("double backtick quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestDoubleBacktickQuotesWithBacktickInside()
    {
        var parser = new Parser();
        var result = parser.Parse("``text with ` inside``");
        Assert.Equal("text with ` inside", GetSingleRefId(result));
    }

    [Fact]
    public static void TestDoubleBacktickQuotesWithEscape()
    {
        var parser = new Parser();
        var result = parser.Parse("``text with ```` escaped backtick``");
        Assert.Equal("text with `` escaped backtick", GetSingleRefId(result));
    }

    // ================================================================================
    // Triple Quotes (3 quote chars) Tests
    // ================================================================================

    [Fact]
    public static void TestTripleDoubleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"\"triple double quotes\"\"\"");
        Assert.Equal("triple double quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestTripleDoubleQuotesWithDoubleQuoteInside()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"\"text with \"\" inside\"\"\"");
        Assert.Equal("text with \"\" inside", GetSingleRefId(result));
    }

    [Fact]
    public static void TestTripleDoubleQuotesWithEscape()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"\"text with \"\"\"\"\"\" escaped triple\"\"\"");
        Assert.Equal("text with \"\"\" escaped triple", GetSingleRefId(result));
    }

    [Fact]
    public static void TestTripleSingleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("'''triple single quotes'''");
```

```
        Assert.Equal("triple single quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestTripleBacktickQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("```triple backtick quotes```");
        Assert.Equal("triple backtick quotes", GetSingleRefId(result));
    }

    // ================================================================================
    // Quadruple Quotes (4 quote chars) Tests
    // ================================================================================

    [Fact]
    public static void TestQuadrupleDoubleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"\"\"quadruple double quotes\"\"\"\"");
        Assert.Equal("quadruple double quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestQuadrupleSingleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("''''quadruple single quotes''''");
        Assert.Equal("quadruple single quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestQuadrupleBacktickQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("````quadruple backtick quotes````");
        Assert.Equal("quadruple backtick quotes", GetSingleRefId(result));
    }

    // ================================================================================
    // Quintuple Quotes (5 quote chars) Tests
    // ================================================================================

    [Fact]
    public static void TestQuintupleDoubleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"\"\"\"quintuple double quotes\"\"\"\"\"");
        Assert.Equal("quintuple double quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestQuintupleSingleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("'''''quintuple single quotes'''''");
        Assert.Equal("quintuple single quotes", GetSingleRefId(result));
    }

    [Fact]
    public static void TestQuintupleBacktickQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("`````quintuple backtick quotes`````");
        Assert.Equal("quintuple backtick quotes", GetSingleRefId(result));
    }

    // ================================================================================
    // Complex Scenarios Tests
    // ================================================================================

    [Fact]
    public static void TestMixedQuotesInLink()
    {
        var parser = new Parser();
        var result = parser.Parse("(\"double\" 'single' `backtick`)");
        Assert.Single(result);
        Assert.NotNull(result[0].Values);
        Assert.Equal(3, result[0].Values!.Count);
        Assert.Equal("double", result[0].Values![0].Id);
```

```
        Assert.Equal("single", result[0].Values![1].Id);
        Assert.Equal("backtick", result[0].Values![2].Id);
    }

    [Fact]
    public static void TestBacktickAsIdInLink()
    {
        var parser = new Parser();
        var result = parser.Parse("(`myId`: value1 value2)");
        Assert.Single(result);
        Assert.Equal("myId", result[0].Id);
        Assert.NotNull(result[0].Values);
        Assert.Equal(2, result[0].Values!.Count);
    }

    [Fact]
    public static void TestCodeBlockLikeContent()
    {
        var parser = new Parser();
        var result = parser.Parse("```const x = 1;```");
        Assert.Equal("const x = 1;", GetSingleRefId(result));
    }

    [Fact]
    public static void TestNestedQuotesInMarkdown()
    {
        var parser = new Parser();
        var result = parser.Parse("``Use `code` in markdown``");
        Assert.Equal("Use `code` in markdown", GetSingleRefId(result));
    }

    [Fact]
    public static void TestJsonStringWithQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("\"\"{ \"key\": \"value\"}\"\"");
        Assert.Equal("{ \"key\": \"value\"}", GetSingleRefId(result));
    }

    // ========================================================================
    // Edge Cases
    // ========================================================================

    [Fact]
    public static void TestWhitespacePreservedInQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("\"  spaces  \"");
        Assert.Equal("  spaces  ", GetSingleRefId(result));
    }

    [Fact]
    public static void TestMultilineInDoubleDoubleQuotes()
    {
        var parser = new Parser();
        var result = parser.Parse("(\"\"line1\nline2\"\")");
        Assert.Single(result);
        Assert.NotNull(result[0].Values);
        Assert.Single(result[0].Values!);
        Assert.Equal("line1\nline2", result[0].Values![0].Id);
    }

    // ========================================================================
    // Unlimited Quotes (6+ quote chars) Tests
    // ========================================================================

    [Fact]
    public static void TestUnlimitedQuotes6()
    {
        // Test 6-quote strings
        var parser = new Parser();
        var result = parser.Parse("\"\"\"\"\"\"hello\"\"\"\"\"\"");
        Assert.Equal("hello", GetSingleRefId(result));
    }

    [Fact]
    public static void TestUnlimitedQuotes10()
    {
        // Test 10-quote strings
        var parser = new Parser();
```

```
350             var result = parser.Parse("\"\"\"\"\"\"\"\"\"very deeply
                ↪  quoted\"\"\"\"\"\"\"\"\"");
351             Assert.Equal("very deeply quoted", GetSingleRefId(result));
352         }
353 
354         [Fact]
355         public static void TestUnlimitedQuotes6WithInnerQuotes()
356         {
357             // Test 6-quote strings with inner 5-quote sequences
358             var parser = new Parser();
359             var result = parser.Parse("\"\"\"\"\"\"hello with \"\"\"\"\" five quotes
                ↪  inside\"\"\"\"\"\"");
360             Assert.Equal("hello with \"\"\"\"\" five quotes inside", GetSingleRefId(result));
361         }
362 
363         [Fact]
364         public static void TestUnlimitedSingleQuotes7()
365         {
366             // Test 7-quote single quote strings
367             var parser = new Parser();
368             var result = parser.Parse("'''''''seven single quotes'''''''");
369             Assert.Equal("seven single quotes", GetSingleRefId(result));
370         }
371 
372         [Fact]
373         public static void TestUnlimitedBackticks8()
374         {
375             // Test 8-quote backtick strings
376             var parser = new Parser();
377             var result = parser.Parse("````````eight backticks````````");
378             Assert.Equal("eight backticks", GetSingleRefId(result));
379         }
380     }
381 }
```

1.44 ./Link.Foundation.Links.Notation.Tests/MultilineParserTests.cs

```
1  using System;
2  using Xunit;
3  
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MultilineParserTests
7      {
8          [Fact]
9          public static void TwoLinksTest()
10         {
11             var source = @"(first: x y)
12 (second: a b)";
13             var parser = new Parser();
14             var links = parser.Parse(source);
15             var target = links.Format();
16             Assert.Equal(source, target);
17         }
18 
19         [Fact]
20         public static void ParseAndStringifyTest()
21         {
22             var source = @"(papa (lovesMama: loves mama))
23 (son lovesMama)
24 (daughter lovesMama)
25 (all (love mama))";
26             var parser = new Parser();
27             var links = parser.Parse(source);
28             var target = links.Format();
29             Assert.Equal(source, target);
30         }
31 
32         [Fact]
33         public static void ParseAndStringifyTest2()
34         {
35             var source = @"father (lovesMom: loves mom)
36 son lovesMom
37 daughter lovesMom
38 all (love mom)";
39             var links = new Parser().Parse(source);
40             var target = links.Format(lessParentheses: true);
41             Assert.Equal(source, target);
42         }
43 
44         [Fact]
```

```
        public static void ParseAndStringifyWithLessParenthesesTest()
        {
            var source = @"lovesMama: loves mama
papa lovesMama
son lovesMama
daughter lovesMama
all (love mama)";
            var parser = new Parser();
            var links = parser.Parse(source);
            var target = links.Format(lessParentheses: true);
            Assert.Equal(source, target);
        }

        [Fact]
        public static void DuplicateIdentifiersTest()
        {
            var source = @"(a: a b)
(a: b c)";
            var target = @"(a: a b)
(a: b c)";
            var parser = new Parser();
            var links = parser.Parse(source);
            var formattedLinks = links.Format();
            Assert.Equal(target, formattedLinks);
        }

        [Fact]
        public static void ComplexStructureTest()
        {
            var input = @"(Type: Type Type)
  Number
  String
  Array
  Value
    (property: name type)
    (method: name params return)";

            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void MixedFormatsTest()
        {
            // Mix of single-line and multi-line formats
            var input = @"id1: value1
(id2: value2 value3)
simple ref
(complex:
  nested1
  nested2
)";

            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void MultilineWithIdTest()
        {
            // Test multi-line link with id
            var input = "(id: value1 value2)";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void MultipleTopLevelElementsTest()
        {
            // Test multiple top-level elements
            var input = "(elem1: val1)\n(elem2: val2)";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void MultilineSimpleLinksTest()
        {
            var input = "(1: 1 1)\n(2: 2 2)";
            var parser = new Parser();
            var parsed = parser.Parse(input);
```

```
            Assert.NotEmpty(parsed);

            // Validate regular formatting
            var output = parsed.Format();
            Assert.Contains("(1: 1 1)", output);
            Assert.Contains("(2: 2 2)", output);

            // Validate alternate formatting matches input
            var outputAlternate = parsed.Format();
            Assert.Equal(input, outputAlternate);
        }

        [Fact]
        public static void IndentedChildrenTest()
        {
            var input = "parent\n  child1\n  child2";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // The parsed structure should have parent with children
            Assert.NotEmpty(parsed);
        }
    }
}
```

1.45 ./Link.Foundation.Links.Notation.Tests/MultilineQuotedStringTests.cs

```
using System;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class MultilineQuotedStringTests
    {
        [Fact]
        public static void TestMultilineDoubleQuotedReference()
        {
            var input = @"(
  ""long
string literal representing
the reference""

  'another
long string literal
as another reference'
)";
            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            Assert.Single(result);

            var link = result[0];
            Assert.Null(link.Id);
            Assert.NotNull(link.Values);
            Assert.Equal(2, link.Values.Count);

            // The two references sit on separate lines inside the parentheses, so the
            // nested context turns each of them into its own link.
            Assert.Equal(@"long
string literal representing
the reference", link.Values![0].Values![0].Id);

            Assert.Equal(@"another
long string literal
as another reference", link.Values![1].Values![0].Id);
        }

        [Fact]
        public static void TestSimpleMultilineDoubleQuoted()
        {
            var input = @(""line1
line2"")";
            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            Assert.Single(result);

            var link = result[0];
            Assert.Null(link.Id);
            Assert.NotNull(link.Values);
            Assert.Single(link.Values);
```

```
            Assert.Equal("line1\nline2", link.Values[0].Id);
        }

        [Fact]
        public static void TestSimpleMultilineSingleQuoted()
        {
            var input = @"('line1
line2')";
            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            Assert.Single(result);

            var link = result[0];
            Assert.Null(link.Id);
            Assert.NotNull(link.Values);
            Assert.Single(link.Values);
            Assert.Equal("line1\nline2", link.Values[0].Id);
        }

        [Fact]
        public static void TestMultilineQuotedAsId()
        {
            var input = @(""multi
line
id"": value1 value2)";
            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.NotEmpty(result);
            Assert.Single(result);

            var link = result[0];
            Assert.Equal("multi\nline\nid", link.Id);
            Assert.NotNull(link.Values);
            Assert.Equal(2, link.Values.Count);
        }
    }
}
```

1.46 ./Link.Foundation.Links.Notation.Tests/NestedIndentationTests.cs

```
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    /// <summary>
    /// Tests for indentation inside parentheses.
    ///
    /// See https://github.com/link-foundation/links-notation/issues/282
    ///
    /// Indentation is structural at the root, so it must be structural inside parentheses too:
    /// a parenthesized group opens a nested context that starts fresh at indentation level zero
    /// and follows exactly the root's rules.
    /// </summary>
    public class NestedIndentationTests
    {
        private static string FormatSource(string source) => new Parser().Parse(source).Format();

        private static void AssertFormat(string source, string expected) =>
        ↪  Assert.Equal(expected, FormatSource(source));

        [Fact]
        public void ParenthesesReproduceRootIndentation()
        {
            AssertFormat("a\n  b\nc\n  d", "(a)\n((a) (b))\n(c)\n((c) (d))");

            // The same lines inside parentheses keep the same structure, nested under
            // the link the group belongs to.
            AssertFormat("array (\n  a\n    b\n  c\n    d\n)", "(array ((a) ((a) (b)) (c) ((c)
            ↪  (d))))");
        }

        [Fact]
        public void ParenthesesKeepRecordBoundaries()
        {
            var source = "value (\n  id \"1\"\n  label \"one\"\n)";
            AssertFormat(source, "(value ((id 1) (label one)))");

            var links = new Parser().Parse(source);
            Assert.Single(links);
```

```

            var group = links[0].Values![1];
            Assert.Null(group.Id);
            Assert.Equal(2, group.Values!.Count);
            Assert.Equal("id", group.Values[0].Values![0].Id);
            Assert.Equal("1", group.Values[0].Values![1].Id);
            Assert.Equal("label", group.Values[1].Values![0].Id);
            Assert.Equal("one", group.Values[1].Values![1].Id);
        }

        [Fact]
        public void ParenthesesKeepSeveralRecordsSeparate()
        {
            AssertFormat(
                "value (\n  (id \"1\" label \"one\")\n  (id \"2\" label \"two\")\n)",
                "(value ((id 1 label one) (id 2 label two)))");
        }

        [Fact]
        public void ParenthesesNestDeeply()
        {
            AssertFormat("outer (\n  inner (\n    x 1\n    y 2\n  )\n  z 3\n)", "(outer ((inner
            ↪  ((x 1) (y 2))) (z 3)))");
        }

        [Fact]
        public void SingleLineParenthesesAreUnchanged()
        {
            AssertFormat("(a b c)", "(a b c)");
            AssertFormat("(1: 2 3)", "(1: 2 3)");
            AssertFormat("(a: b c)", "(a: b c)");
            AssertFormat("((a b))", "((a b))");
            AssertFormat("(a)", "(a)");
            AssertFormat("()", "()");
        }

        [Fact]
        public void ParenthesesWithIndentedIdSyntax()
        {
            AssertFormat("(\n  a:\n    b\n    c\n)", "(a: b c)");
        }

        [Fact]
        public void EmployeeRecordsKeepTheirFields()
        {
            var source = "empInfo\n  employees:\n    (\n      name (James Kirk)\n      age 40\n
            ↪  )\n"
                + "    (\n      name (Jean-Luc Picard)\n      age 45\n    )";
            var expected = "(empInfo)\n((empInfo) (employees: ((name (James Kirk)) (age 40))"
                + " ((name (Jean-Luc Picard)) (age 45))))";
            AssertFormat(source, expected);
        }
    }
}
```

1.47 ./Link.Foundation.Links.Notation.Tests/NestedParserTests.cs

```
using System;
using System.Linq;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class NestedParserTests
    {
        [Fact]
        public static void SignificantWhitespaceTest()
        {
            var source = @"
users
    user1
        id
            43
        name
            first
                John
            last
                Williams
        location
            New York
        age
            23
```

```
    user2
        id
            56
        name
            first
                Igor
            middle
                Petrovich
            last
                Ivanov
        location
            Moscow
        age
            20";
            var target = @"(users)
((users) (user1))
(((users) (user1)) (id))
((((users) (user1)) (id)) (43))
(((users) (user1)) (name))
((((users) (user1)) (name)) (first))
(((((users) (user1)) (name)) (first)) (John))
((((users) (user1)) (name)) (last))
(((((users) (user1)) (name)) (last)) (Williams))
(((users) (user1)) (location))
((((users) (user1)) (location)) (New York))
(((users) (user1)) (age))
((((users) (user1)) (age)) (23))
((users) (user2))
(((users) (user2)) (id))
((((users) (user2)) (id)) (56))
(((users) (user2)) (name))
((((users) (user2)) (name)) (first))
(((((users) (user2)) (name)) (first)) (Igor))
((((users) (user2)) (name)) (middle))
(((((users) (user2)) (name)) (middle)) (Petrovich))
((((users) (user2)) (name)) (last))
(((((users) (user2)) (name)) (last)) (Ivanov))
(((users) (user2)) (location))
((((users) (user2)) (location)) (Moscow))
(((users) (user2)) (age))
((((users) (user2)) (age)) (20))";
            var parser = new Parser();
            var links = parser.Parse(source);
            var formattedLinks = links.Format();
            Assert.Equal(target, formattedLinks);
        }

        [Fact]
        public static void SimpleSignificantWhitespaceTest()
        {
            var source = @"a
    b
    c";
            var target = @"(a)
((a) (b))
((a) (c))";
            var parser = new Parser();
            var links = parser.Parse(source);
            var formattedLinks = links.Format();
            Assert.Equal(target, formattedLinks);
        }

        [Fact]
        public static void TwoSpacesSizedWhitespaceTest()
        {
            var source = @"
users
  user1";
            var target = @"(users)
((users) (user1))";
            var parser = new Parser();
            var links = parser.Parse(source);
            var formattedLinks = links.Format();
            Assert.Equal(target, formattedLinks);
        }

        [Fact]
        public static void ParseNestedStructureWithIndentationTest()
        {
            var source = @"parent
  child1
  child2";
```

```
            var parser = new Parser();
            var links = parser.Parse(source);
            Assert.NotNull(links);
            // Should create 3 links: (parent), (parent child1), (parent child2)
            Assert.Equal(3, links.Count);
        }

        [Fact]
        public static void IndentationConsistencyTest()
        {
            // Test that indentation must be consistent
            var input = @"parent
    child1
     child2"; // Inconsistent indentation
            var result = new Parser().Parse(input);
            // This should parse but child2 won't be a child of parent due to different
            ↪  indentation
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void IndentationBasedChildrenTest()
        {
            var input = @"parent
    child1
    child2
      grandchild";
            var parser = new Parser();
            var result = parser.Parse(input);
            Assert.NotNull(result);
            // Expected flattened links:
            // (parent), (parent child1), (parent child2), ((parent child2) grandchild)
            Assert.Equal(4, result.Count);
        }

        [Fact]
        public static void ComplexIndentationTest()
        {
            var input = @"root
    level1a
      level2a
      level2b
    level1b
      level2c";
            var parser = new Parser();
            var result = parser.Parse(input);
            Assert.NotNull(result);
            // Expected flattened links:
            // (root), (root level1a), ((root level1a) level2a), ((root level1a) level2b),
            // (root level1b), ((root level1b) level2c)
            Assert.Equal(6, result.Count);
        }

        [Fact]
        public static void NestedLinksTest()
        {
            var input = "(1: (2: (3: 3)))";
            var parser = new Parser();
            var parsed = parser.Parse(input);
            Assert.NotEmpty(parsed);

            // Validate regular formatting
            var output = parsed.Format();
            Assert.NotEmpty(output);

            // Validate that the structure is properly nested
            Assert.Single(parsed);
        }

        [Fact]
        public static void IndentationParserTest()
        {
            var input = "parent\n  child1\n  child2";
            var parser = new Parser();
            var result = parser.Parse(input);
            Assert.NotEmpty(result);
            // Should have parent link
            var parentLink = result.FirstOrDefault(l => l.Id == "parent");
        }

        [Fact]
```

```
        public static void NestedIndentationParserTest()
        {
            var input = "parent\n  child\n    grandchild";
            var parser = new Parser();
            var result = parser.Parse(input);
            Assert.NotEmpty(result);
            // Should create nested structure with proper hierarchy
            Assert.True(result.Count >= 1);
        }

        [Fact]
        public static void ThreeLevelNestingRoundtripTest()
        {
            var input = "(1: (2: (3: 3)))";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // Validate round-trip
            var output = parsed.Format();
            Assert.Equal(input, output);
        }

        [Fact]
        public static void DeepNestedStructureRoundtripTest()
        {
            var input = "(a: (b: (c: (d: d))))";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // Validate round-trip
            var output = parsed.Format();
            Assert.Equal(input, output);
        }

        [Fact]
        public static void MultipleNestedLinksRoundtripTest()
        {
            var input = "(parent: (child1: value1) (child2: value2))";
            var parser = new Parser();
            var parsed = parser.Parse(input);

            // Validate round-trip
            var output = parsed.Format();
            Assert.Equal(input, output);
        }
    }
}
```

1.48 ./Link.Foundation.Links.Notation.Tests/NestedSelfReferenceTests.cs

```
using System;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class NestedSelfReferenceTests
    {
        [Fact]
        public static void NestedSelfReferencedObjectInPairValueTest()
        {
            // Test case from PARSER BUG.md
            // This should parse a dict with two pairs, where the second pair's value
            // is itself a self-referenced dict definition (obj 1: dict ...)
            var notation = "(obj 0: dict ((str bmFtZQ==) (str ZGljdDE=)) ((str b3RoZXI=) (obj 1:
            ↪  dict ((str bmFtZQ==) (str ZGljdDI=)) ((str b3RoZXI=) obj 0))))";

            var parser = new Parser();
            var links = parser.Parse(notation);

            // Should parse exactly one top-level link
            Assert.Single(links);

            var link = links[0];

            // Top-level link should have ID "obj 0"
            Assert.Equal("obj 0", link.Id);

            // Should have: type marker + 2 pairs = 3 values
            Assert.Equal(3, link.Values!.Count);

            // First value is the type marker "dict"
            Assert.Equal("dict", link.Values[0].Id);
```

```
32
33              // Second and third values are the two pairs
34              var pair1 = link.Values[1];
35              var pair2 = link.Values[2];
36
37              // Pair 1: ((str bmFtZQ==) (str ZGljdDE=))
38              // This is a parenthesized expression containing two sub-expressions
39              Assert.Null(pair1.Id);
40              Assert.Equal(2, pair1.Values!.Count);
41
42              // First element of pair1: (str bmFtZQ==)
43              Assert.Null(pair1.Values[0].Id);
44              Assert.Equal(2, pair1.Values![0].Values!.Count);
45              Assert.Equal("str", pair1.Values![0].Values![0].Id);
46              Assert.Equal("bmFtZQ==", pair1.Values![0].Values![1].Id);
47
48              // Second element of pair1: (str ZGljdDE=)
49              Assert.Null(pair1.Values[1].Id);
50              Assert.Equal(2, pair1.Values![1].Values!.Count);
51              Assert.Equal("str", pair1.Values![1].Values![0].Id);
52              Assert.Equal("ZGljdDE=", pair1.Values![1].Values![1].Id);
53
54              // Pair 2: ((str b3RoZXI=) (obj 1: dict ...))
55              // This is a parenthesized expression containing two sub-expressions
56              Assert.Null(pair2.Id);
57              Assert.Equal(2, pair2.Values!.Count);
58
59              // First element of pair2: (str b3RoZXI=)
60              Assert.Null(pair2.Values[0].Id);
61              Assert.Equal(2, pair2.Values![0].Values!.Count);
62              Assert.Equal("str", pair2.Values![0].Values![0].Id);
63              Assert.Equal("b3RoZXI=", pair2.Values![0].Values![1].Id);
64
65              // Second element of pair2: (obj 1: dict ((str bmFtZQ==) (str ZGljdDI=)) ((str
                ↪  b3RoZXI=) obj 0))
66              // THIS IS THE KEY TEST - obj 1 should have its ID preserved
67              var obj1 = pair2.Values[1];
68              Assert.Equal("obj 1", obj1.Id);
69              Assert.NotNull(obj1.Values);
70              Assert.Equal(3, obj1.Values.Count); // type marker + 2 pairs
71
72              // obj 1's type marker
73              Assert.Equal("dict", obj1.Values[0].Id);
74
75              // obj 1's first pair: ((str bmFtZQ==) (str ZGljdDI=))
76              var obj1Pair1 = obj1.Values[1];
77              Assert.Equal(2, obj1Pair1.Values!.Count);
78              Assert.Null(obj1Pair1.Values[0].Id);
79              Assert.Equal(2, obj1Pair1.Values![0].Values!.Count);
80              Assert.Equal("str", obj1Pair1.Values![0].Values![0].Id);
81              Assert.Equal("bmFtZQ==", obj1Pair1.Values![0].Values![1].Id);
82              Assert.Null(obj1Pair1.Values[1].Id);
83              Assert.Equal(2, obj1Pair1.Values![1].Values!.Count);
84              Assert.Equal("str", obj1Pair1.Values![1].Values![0].Id);
85              Assert.Equal("ZGljdDI=", obj1Pair1.Values![1].Values![1].Id);
86
87              // obj 1's second pair: ((str b3RoZXI=) obj 0) - reference back to obj 0
88              var obj1Pair2 = obj1.Values[2];
89              Assert.Equal(2, obj1Pair2.Values!.Count);
90              Assert.Null(obj1Pair2.Values[0].Id);
91              Assert.Equal(2, obj1Pair2.Values![0].Values!.Count);
92              Assert.Equal("str", obj1Pair2.Values![0].Values![0].Id);
93              Assert.Equal("b3RoZXI=", obj1Pair2.Values![0].Values![1].Id);
94              Assert.Equal("obj 0", obj1Pair2.Values[1].Id);
95              Assert.Null(obj1Pair2.Values[1].Values); // Just a reference, no nested values
96          }
97
98          [Fact]
99          public static void SelfReferenceAsDirectChildWorksCorrectlyTest()
100         {
101             // This pattern should work (and did work before)
102             var notation = "(obj 0: list (int 1) (int 2) (obj 1: list (int 3) (int 4) obj 0))";
103
104             var parser = new Parser();
105             var links = parser.Parse(notation);
106
107             Assert.Single(links);
108             Assert.Equal("obj 0", links[0].Id);
109             Assert.Equal(4, links[0].Values!.Count); // list + 1 + 2 + obj 1
110
```

```
            // The fourth value should be obj 1 with a self-reference
            var obj1 = links[0].Values![3];
            Assert.Equal("obj 1", obj1.Id);
            Assert.Equal(4, obj1.Values!.Count); // list + 3 + 4 + obj 0
            Assert.Equal("obj 0", obj1.Values[3].Id); // Reference to obj 0
        }
    }
}
```

1.49 ./Link.Foundation.Links.Notation.Tests/NestingDepthTests.cs

```
using System;
using System.Linq;
using System.Threading;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    /// <summary>
    /// Reading nested groups must take time that grows with the size of the document,
    /// not with two to the power of its nesting depth
    /// (<see href="https://github.com/link-foundation/links-notation/issues/314">#314</see>).
    /// Before the fix, every level of nesting doubled the work: a document nested six
    /// levels deep took seconds. Each case runs on its own thread and fails when it has
    /// not finished within the budget, so a regression fails instead of hanging.
    /// </summary>
    public class NestingDepthTests
    {
        /// Generous enough for a debug build on a slow machine, and far below what the
        /// exponential reading took at the depths used here.
        private static readonly TimeSpan Budget = TimeSpan.FromSeconds(5);

        /// The depth used for the shapes that used to take exponential time.
        private const int Deep = 32;

        /// A depth that shows the reading stays linear well past the shallow cases.
        private const int VeryDeep = 256;

        private static string Closed(int depth) => new string('(', depth) + "a" + new
        ↪  string(')', depth);

        private static string ValueAfter(int depth) => new string('(', depth) + "a" +
        ↪  string.Concat(Enumerable.Repeat(") b", depth));

        private static string Unclosed(int depth) => new string('(', depth) + "a";

        /// Unclosed groups on lines that are each indented one space deeper.
        private static string IndentedUnclosed(int depth) =>
            string.Join("\n", Enumerable.Range(0, depth).Select(level => new string(' ', level)
            ↪  + "(a"));

        /// Formats the parsed document, or names the exception parsing raised.
        private static string ReadWithinBudget(string what, string source)
        {
            string? outcome = null;
            var reader = new Thread(() =>
            {
                try
                {
                    // The documents nest deeper than the default limit on purpose; only the
                    // time they take is measured here.
                    outcome = new Parser(true, int.MaxValue).Parse(source).Format();
                }
                catch (Exception error)
                {
                    outcome = error.GetType().Name;
                }
            }, 512 * 1024 * 1024)
            {
                // A reader that runs past the budget must not keep the test run alive.
                IsBackground = true,
            };
            reader.Start();
            Assert.True(reader.Join(Budget), $"{what} did not finish within {Budget}");
            return outcome!;
        }

        [Fact]
        public void ClosedGroupsReadInLinearTime()
        {
            var source = Closed(Deep);
            Assert.Equal(source, ReadWithinBudget("closed groups", source));
```

```
        }

        [Fact]
        public void ValuesAfterGroupsReadInLinearTime()
        {
            var source = ValueAfter(Deep);
            Assert.Equal($"({source})", ReadWithinBudget("values after groups", source));
        }

        [Fact]
        public void UnclosedGroupsFailInLinearTime()
        {
            Assert.Equal(nameof(ParseException), ReadWithinBudget("unclosed groups",
            ↪  Unclosed(Deep)));
        }

        [Fact]
        public void UnclosedGroupsOnIndentedLinesFailInLinearTime()
        {
            Assert.Equal(nameof(ParseException), ReadWithinBudget("unclosed indented groups",
            ↪  IndentedUnclosed(Deep)));
        }

        [Fact]
        public void ValuesAfterVeryDeepGroupsReadInLinearTime()
        {
            var source = ValueAfter(VeryDeep);
            Assert.Equal($"({source})", ReadWithinBudget("very deep values after groups",
            ↪  source));
        }

        [Fact]
        public void GroupFollowedByValuesKeepsItsStructure()
        {
            var cases = new[]
            {
                new[] { "(a) b", "((a) b)" },
                new[] { "(a) (b) c", "((a) (b) c)" },
                new[] { "((a) b) c", "(((a) b) c)" },
                new[] { "(a: b) c", "((a: b) c)" },
                new[] { "(a)\n(b) c", "(a)\n((b) c)" },
                new[] { "x\n  (a) b\n  (c)", "(x)\n((x) ((a) b))\n((x) (c))" },
            };

            var parser = new Parser();
            foreach (var testCase in cases)
            {
                Assert.Equal(testCase[1], parser.Parse(testCase[0]).Format());
            }
        }
    }
}
```

1.50 ./Link.Foundation.Links.Notation.Tests/NestingLimitTests.cs

```
using System;
using System.Linq;
using System.Threading;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    /// <summary>
    /// Links nested too deeply are refused with an error rather than recursed into until
    /// the stack overflows, which ends the process where no catch block sees it
    /// (https://github.com/link-foundation/links-notation/issues/315).
    /// </summary>
    /// <remarks>
    /// Every parenthesized group and every indentation level is one level, and the lines of
    /// a document start at level 0. The positions asserted here are the ones the Rust and
    /// JavaScript ports report for the same input.
    /// </remarks>
    public static class NestingLimitTests
    {
        private static string Parens(int depth) => new string('(', depth) + "a" + new
        ↪  string(')', depth);

        private static string Values(int depth) =>
            string.Concat(Enumerable.Repeat("(a ", depth)) + "b" + new string(')', depth);

        private static string Indentation(int depth) =>
```

```
        string.Concat(Enumerable.Range(0, depth + 1).Select(level => new string(' ', level)
        ↪  + "a\n"));

    private static ParseException TooDeep(string document, int maxDepth)
    {
        var error = Assert.Throws<ParseException>(() => new Parser(true,
        ↪  maxDepth).Parse(document));
        Assert.Equal(maxDepth, error.MaxDepth);
        return error;
    }

    private static bool Accepted(string document, int maxDepth) =>
        new Parser(true, maxDepth).Parse(document).Count > 0;

    [Fact]
    public static void DefaultLimitIsSharedByEveryImplementationTest()
    {
        Assert.Equal(64, Parser.DefaultMaxDepth);
        Assert.Equal(Parser.DefaultMaxDepth, new Parser().MaxDepth);
        Assert.Equal(Parser.DefaultMaxDepth, new Parser(comments: false).MaxDepth);
        Assert.Equal(3, new Parser { MaxDepth = 3 }.MaxDepth);
    }

    [Fact]
    public static void NegativeLimitIsRejectedTest()
    {
        Assert.Throws<ArgumentOutOfRangeException>(() => new Parser(true, -1));
        Assert.Throws<ArgumentOutOfRangeException>(() => new Parser { MaxDepth = -1 });
    }

    [Fact]
    public static void ParenthesesUpToTheLimitAreAcceptedTest()
    {
        Assert.True(Accepted(Parens(3), 3));
        Assert.True(Accepted(Values(3), 3));
        Assert.NotEmpty(new Parser().Parse(Parens(Parser.DefaultMaxDepth)));
        Assert.NotEmpty(new Parser().Parse(Values(Parser.DefaultMaxDepth)));
    }

    [Fact]
    public static void EveryGroupAndEveryIndentationLevelIsOneLevelTest()
    {
        Assert.True(Accepted("(a)", 1));
        Assert.True(Accepted("a b c", 0));
        TooDeep("(a)", 0);
        TooDeep("((a))", 1);
        TooDeep("(a (b))", 1);
        TooDeep("a\n  (b)\n", 1);
    }

    [Fact]
    public static void ParenthesesPastTheLimitAreRefusedAtTheGroupThatIsTooDeepTest()
    {
        var error = TooDeep(Parens(4), 3);

        Assert.Equal((1, 4, 3), (error.Line, error.Column, error.Offset));
        Assert.Equal("line 1, column 4: nesting depth exceeds the maximum of 3",
        ↪  error.Summary);
        Assert.Equal("1 | ((((a))))\n  |    ^", error.Snippet);
        Assert.Equal(
            "Nesting too deep at line 1, column 4: nesting depth exceeds the maximum of 3\n"
            ↪  +
            "1 | ((((a))))\n" +
            "  |    ^",
            error.Message);
    }

    [Fact]
    public static void GroupsInValuePositionCountLikeAnyOtherGroupTest()
    {
        var error = TooDeep(Values(4), 3);

        Assert.Equal((1, 10), (error.Line, error.Column));
    }

    [Fact]
    public static void IndentationUpToTheLimitIsAcceptedTest()
    {
        Assert.True(Accepted(Indentation(3), 3));
        Assert.NotEmpty(new Parser().Parse(Indentation(Parser.DefaultMaxDepth)));
```

```
    }

    [Fact]
    public static void IndentationPastTheLimitIsRefusedAtTheLineThatIsTooDeepTest()
    {
        var error = TooDeep(Indentation(4), 3);

        Assert.Equal((5, 5), (error.Line, error.Column));
        Assert.Equal("    a", error.LineText);
    }

    [Fact]
    public static void GroupsAndIndentationAddUpTest()
    {
        // `(b)` on the line indented once is at level 2.
        Assert.True(Accepted("a\n  (b)\n", 2));
        var error = TooDeep("a\n  (b)\n", 1);

        Assert.Equal((2, 3), (error.Line, error.Column));
    }

    [Fact]
    public static void IndentationInsideAGroupCountsOnTopOfTheGroupTest()
    {
        // `b` is inside one group and indented once within it.
        Assert.True(Accepted("(a\n  b)\n", 2));
        var error = TooDeep("(a\n  b)\n", 1);

        Assert.Equal((2, 3), (error.Line, error.Column));
    }

    [Fact]
    public static void TrailingSpacesOnADeepLineAreNotADeeperLineTest()
    {
        Assert.True(Accepted("a\n  b\n    c   \n", 2));
        Assert.True(Accepted("a\n  b\n    c\n      ", 2));
    }

    [Fact]
    public static void SyntaxErrorIsNotMistakenForNestingThatIsTooDeepTest()
    {
        var error = Assert.Throws<ParseException>(() => new Parser().Parse("a: b: c"));

        Assert.Null(error.MaxDepth);
        Assert.StartsWith("Syntax error at ", error.Message);
    }

    [Fact]
    public static void RefusesADocumentFarPastTheLimitWithoutOverflowingTheStackTest()
    {
        // Before the limit existed each of these overflowed the stack, which ends the
        // process. The documents are parsed on a 1 MiB stack, smaller than any thread
        // .NET starts by default.
        string?[] messages = new string?[3];
        Exception? unexpected = null;
        var thread = new Thread(() =>
        {
            try
            {
                var documents = new[] { Parens(100 000), Values(100 000), Indentation(2 000)
                ↪  };
                for (var index = 0; index < documents.Length; index++)
                {
                    var error = Assert.Throws<ParseException>(() => new
                    ↪  Parser().Parse(documents[index]));
                    Assert.Equal(Parser.DefaultMaxDepth, error.MaxDepth);
                    messages[index] = error.Message;
                }
            }
            catch (Exception error)
            {
                unexpected = error;
            }
        }, 1 << 20);
        thread.Start();
        thread.Join();

        Assert.Null(unexpected);
        Assert.All(messages, message => Assert.StartsWith("Nesting too deep at ", message));
    }

```

```
        [Fact]
        public static void StreamParserReportsWhereTheNestingIsTooDeepTest()
        {
            var stream = new StreamParser(new Parser(true, 1));
            stream.Write("a\nb ((c))\n");

            var error = Assert.Throws<StreamParseException>(() => stream.Finish());

            Assert.Equal(1, error.ParseError.MaxDepth);
            Assert.Equal((2, 4, 5), (error.Line, error.Column, error.Offset));
        }
    }
}
```

1.51 ./Link.Foundation.Links.Notation.Tests/ParseErrorPositionTests.cs

```
using System;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    /// <summary>
    /// A parse error has to say where the document stopped making sense.
    /// </summary>
    /// <remarks>
    /// The positions asserted here are the ones the JavaScript and Rust ports report for
    /// the same input, so the implementations can be held to the same contract
    /// (https://github.com/link-foundation/links-notation/issues/302).
    /// </remarks>
    public static class ParseErrorPositionTests
    {
        private static ParseException SyntaxError(string document) =>
            Assert.Throws<ParseException>(() => new Parser().Parse(document));

        [Fact]
        public static void ReportsTheLineAndColumnOfTheDefectTest()
        {
            // The example from the issue: the defect is the second colon on line 2,
            // and the two lines after it are fine.
            var error = SyntaxError("ci gate x\nstage: rust: nextest\nnext stage\n  clippy");

            Assert.Equal(2, error.Line);
            Assert.Equal(12, error.Column);
            Assert.Equal(':', error.Found);
        }

        [Fact]
        public static void OffsetAgreesWithTheOtherImplementationsTest()
        {
            // JavaScript and Rust report offset 21, line 2, column 12 for this document.
            var error = SyntaxError("ci gate x\nstage: rust: nextest\n");

            Assert.Equal(21, error.Offset);
            Assert.Equal(2, error.Line);
            Assert.Equal(12, error.Column);
        }

        [Fact]
        public static void ReportsTheLineALateDefectIsOnTest()
        {
            var error = SyntaxError("a\nb\nc\nd\ne: f: g\nh\n");

            Assert.Equal(5, error.Line);
            Assert.Equal(5, error.Column);
            Assert.Equal("e: f: g", error.LineText);
        }

        [Fact]
        public static void ReportsTheEndOfTheDocumentWhenAGroupIsNeverClosedTest()
        {
            var error = SyntaxError("a (b\n");

            Assert.Equal(5, error.Offset);
            Assert.Equal(2, error.Line);
            Assert.Equal(1, error.Column);
            Assert.Null(error.Found);
            Assert.Contains("end of input", error.Message);
        }

        [Fact]
        public static void ReportsAnUnmatchedClosingParenthesisTest()
        {
```

```
            var error = SyntaxError("a b)\n");

            Assert.Equal(3, error.Offset);
            Assert.Equal(1, error.Line);
            Assert.Equal(4, error.Column);
            Assert.Equal(')', error.Found);
        }

        [Fact]
        public static void MessagePointsACaretAtTheOffendingCharacterTest()
        {
            var error = SyntaxError("ci gate x\nstage: rust: nextest\n");

            Assert.Equal("line 2, column 12: unexpected \":\"", error.Summary);
            Assert.Equal("2 | stage: rust: nextest\n  |            ^", error.Snippet);
            Assert.Equal(
                "Syntax error at line 2, column 12: unexpected \":\"\n2 | stage: rust: nextest\n  |            ^",
                error.Message);
        }

        [Fact]
        public static void MessageQuotesOneLineRatherThanTheRestOfTheDocumentTest()
        {
            var document = "ci gate x\nstage: rust: nextest\n"
                + string.Concat(System.Linq.Enumerable.Repeat("trailing line\n", 500));

            var error = SyntaxError(document);

            Assert.Contains("line 2, column 12", error.Message);
            Assert.DoesNotContain("trailing line", error.Message);
            Assert.True(error.Message.Length < 200, $"message is {error.Message.Length} characters");
        }

        [Fact]
        public static void MessageOfALongLineStaysAMessageTest()
        {
            var document = new string('a', 400) + ": " + new string('b', 400) + ": c";

            var error = SyntaxError(document);

            Assert.Equal(1, error.Line);
            Assert.Equal(803, error.Column);
            Assert.Contains("...", error.Message);
            Assert.True(error.Message.Length < 300, $"message is {error.Message.Length} characters");
        }

        [Fact]
        public static void KeepsCatchingCodeThatExpectsAFormatExceptionWorkingTest()
        {
            // The generated parser used to raise a bare FormatException; code that catches
            // it keeps working, and now reads a position off the exception it catches.
            var error = Assert.Throws<ParseException>(() => new Parser().Parse("a: b: c"));

            Assert.IsAssignableFrom<FormatException>(error);
            Assert.Equal(4, error.Offset);
        }

        [Fact]
        public static void DoesNotMentionTheGrammarInternalsTest()
        {
            var error = SyntaxError("a: b: c");

            Assert.DoesNotContain("Failed to parse", error.Message);
            Assert.DoesNotContain("document", error.Message);
        }

        [Fact]
        public static void ADocumentThatParsesReportsNothingTest()
        {
            var links = new Parser().Parse("a: b\n  c: d\n");

            Assert.Equal(2, links.Count);
        }
    }
}
```

1.52 ./Link.Foundation.Links.Notation.Tests/ReferenceFidelityTests.cs

```
using System;
```

```
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Text;
using Link.Foundation.Links.Notation.Binary;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests;

public static class ReferenceFidelityTests
{
    private static IEnumerable<string> Fixtures() =>
        File.ReadLines(Path.Combine(AppContext.BaseDirectory, "Protocol",
        ↪ "reference-literals.txt"))
            .Where(line => !line.StartsWith('#'))
            .Select(line => line == "-" ? "" :
            ↪ Encoding.UTF8.GetString(Convert.FromHexString(line)));

    [Fact]
    public static void NativeReferenceFidelity()
    {
        foreach (var reference in Fixtures())
        {
            var original = new[] { new Link<string>(reference, new[] { new
            ↪ Link<string>(reference), new Link<string>("nested", new[] { new
            ↪ Link<string>(reference) }) }) };
            Assert.Equal(original, new Parser().Parse(original[0].ToString()));
            Assert.Equal(original, new Parser().Parse(original.Format(true)));
            Assert.Equal(original, new Parser().Parse(original.Format(new FormatConfig {
            ↪ MaxInlineRefs = 1, PreferInline = false })));
        }
    }

    [Fact(Timeout = 30000)]
    public static void ReferenceLiteralsAtEveryChunkSplit()
    {
        foreach (var reference in Fixtures())
        {
            Assert.Equal(reference, ReferenceLiteral.Decode(ReferenceLiteral.Encode(reference)));
            foreach (var literal in new[] { Link<string>.EscapeReference(reference),
            ↪ LinoFormat.FormatReference(reference), ReferenceLiteral.Encode(reference) })
            {
                var text = $"(root: {literal})\n({literal}: fixture)";
                var expected = new[] { new Link<string>("root", new[] { new
                ↪ Link<string>(reference) }), new Link<string>(reference, new[] { new
                ↪ Link<string>("fixture") }) };
                Assert.Equal(expected, new Parser().Parse(text));
                for (var split = 0; split <= text.Length; split++)
                {
                    TestContext.Current.CancellationToken.ThrowIfCancellationRequested();
                    var stream = new StreamParser();
                    stream.Write(text[..split]);
                    stream.Write(text[split..]);
                    Assert.Equal(expected, stream.Finish());
                }
            }
        }
    }

    [Fact]
    public static void BinaryReferenceFidelity()
    {
        var codec = new BinaryLinoCodec();
        foreach (var reference in Fixtures())
        {
            var original = new[] { new Link<string>(reference, new[] { new
            ↪ Link<string>(reference) }) };
            var decoded = codec.Decode(codec.Encode(original));
            Assert.Equal(original, decoded);
            Assert.Equal(original, LinoFormat.ParseDocument(LinoFormat.FormatDocument(decoded)));
        }
    }

    [Fact]
    public static void MalformedReferenceLiteralsAreRejected()
    {
        foreach (var literal in File.ReadLines(Path.Combine(AppContext.BaseDirectory,
        ↪ "Protocol", "invalid-reference-literals.txt")).Where(line => !line.StartsWith('#')))
        {
```

```
            Assert.Throws<FormatException>(() => ReferenceLiteral.Decode(literal));
            Assert.ThrowsAny<Exception>(() => new Parser().Parse(literal));
        }
        Assert.Equal("é", ReferenceLiteral.Decode("~1{C3A9}"));
        Assert.Throws<EncoderFallbackException>(() => ReferenceLiteral.Encode("\ud800"));
        Assert.Throws<EncoderFallbackException>(() => Link<string>.EscapeReference("\ud800"));
    }
}
```

1.53 ./Link.Foundation.Links.Notation.Tests/SharedSyntaxParityTests.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Text.Json;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests;

/// <summary>
/// Runs the syntax fixtures every Links Notation implementation shares (issue #364): the
/// raw text and delimiter recorded for each parsed reference, and the concrete syntax
/// tree, which must give back every shared input exactly.
/// </summary>
public static class SharedSyntaxParityTests
{
    private static JsonDocument Fixtures(string name) =>
        JsonDocument.Parse(File.ReadAllText(Path.Combine(AppContext.BaseDirectory, "Protocol",
        ↪ name)));

    private static object[] Tree(IEnumerable<Link<string>> links) => links
        .Select(link => (object)new
        {
            id = link.Id,
            raw = link.RawId,
            quote = link.Quote?.ToString(),
            values = Tree(link.Values ?? Array.Empty<Link<string>>()),
        }).ToArray();

    private static void AssertTree(JsonElement expected, IEnumerable<Link<string>> links, string
    ↪ id) =>
        Assert.True(
            JsonElement.DeepEquals(expected, JsonSerializer.SerializeToElement(Tree(links))),
            $"{id}: {JsonSerializer.Serialize(Tree(links))}");

    [Fact(Timeout = 30000)]
    public static void SharedSyntaxParity()
    {
        using var document = Fixtures("syntax-fixtures.json");
        Assert.NotEmpty(document.RootElement.EnumerateArray());
        foreach (var fixture in document.RootElement.EnumerateArray())
        {
            TestContext.Current.CancellationToken.ThrowIfCancellationRequested();
            var id = fixture.GetProperty("id").GetString()!;
            var input = fixture.GetProperty("input").GetString()!;
            var expected = fixture.GetProperty("links");
            var parser = new Parser();
            AssertTree(expected, parser.Parse(input), id);
            var cst = parser.ParseCst(input);
            Assert.Equal(fixture.GetProperty("cst").GetString(), cst.ToSExpression());
            Assert.Equal(input, cst.ToString());
            var stream = new StreamParser();
            foreach (var character in input) stream.Write(character.ToString());
            AssertTree(expected, stream.Finish(), id);
        }
    }

    [Fact(Timeout = 30000)]
    public static void EverySharedInputIsRecoveredFromItsConcreteSyntaxTree()
    {
        using var text = Fixtures("text-fixtures.json");
        using var syntax = Fixtures("syntax-fixtures.json");
        var fixtures = text.RootElement.EnumerateArray().Concat(syntax.RootElement.EnumerateArra
        ↪ y()).ToList();
        Assert.NotEmpty(fixtures);
        foreach (var fixture in fixtures)
        {
            TestContext.Current.CancellationToken.ThrowIfCancellationRequested();
            var input = fixture.GetProperty("input").GetString()!;
            var cst = new Parser().ParseCst(input);
```

```
            Assert.Equal(input, cst.ToString());
            Assert.Equal(input, string.Concat(cst.Tokens().Select(token => token.Text)));
        }
    }
}
```

1.54 ./Link.Foundation.Links.Notation.Tests/SharedTextParityTests.cs

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Text.Json;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests;

public static class SharedTextParityTests
{
    private static object[] Tree(IEnumerable<Link<string>> links) => links
        .Select(link => (object)new { id = link.Id, values = Tree(link.Values ??
        ↪ Array.Empty<Link<string>>()) }).ToArray();

    [Fact(Timeout = 30000)]
    public static void SharedTextParity()
    {
        using var document =
        ↪ JsonDocument.Parse(File.ReadAllText(Path.Combine(AppContext.BaseDirectory,
        ↪ "Protocol", "text-fixtures.json")));
        Assert.NotEmpty(document.RootElement.EnumerateArray());
        foreach (var fixture in document.RootElement.EnumerateArray())
        {
            TestContext.Current.CancellationToken.ThrowIfCancellationRequested();
            var input = fixture.GetProperty("input").GetString()!;
            var formatted = fixture.GetProperty("format").GetString()!;
            var expected = fixture.GetProperty("expected");
            var parsed = new Parser().Parse(input);
            Assert.True(JsonElement.DeepEquals(expected,
            ↪ JsonSerializer.SerializeToElement(Tree(parsed))));
            Assert.Equal(formatted, string.Concat(parsed.Select(link => link.ToString())));
            Assert.True(JsonElement.DeepEquals(expected,
            ↪ JsonSerializer.SerializeToElement(Tree(new Parser().Parse(formatted)))));
            var stream = new StreamParser();
            foreach (var character in input) stream.Write(character.ToString());
            Assert.True(JsonElement.DeepEquals(expected,
            ↪ JsonSerializer.SerializeToElement(Tree(stream.Finish()))));
        }
    }

    [Fact(Timeout = 30000)]
    public static void SharedLessParenthesesRoundTrip()
    {
        using var document =
        ↪ JsonDocument.Parse(File.ReadAllText(Path.Combine(AppContext.BaseDirectory,
        ↪ "Protocol", "text-fixtures.json")));
        Assert.NotEmpty(document.RootElement.EnumerateArray());
        foreach (var fixture in document.RootElement.EnumerateArray())
        {
            TestContext.Current.CancellationToken.ThrowIfCancellationRequested();
            var lessParentheses = fixture.GetProperty("lessParentheses").GetString()!;
            var expected = fixture.GetProperty("expected");
            var parsed = new Parser().Parse(fixture.GetProperty("input").GetString()!);
            Assert.Equal(lessParentheses, parsed.Format(new FormatOptions { LessParentheses =
            ↪ true }));
            Assert.True(JsonElement.DeepEquals(expected,
            ↪ JsonSerializer.SerializeToElement(Tree(new Parser().Parse(lessParentheses)))));
            // The boolean overload trims ToString(), which writes some text differently but must
            ↪ read back the same.
            Assert.True(JsonElement.DeepEquals(expected,
            ↪ JsonSerializer.SerializeToElement(Tree(new
            ↪ Parser().Parse(parsed.Format(lessParentheses: true))))));
        }
    }
}
```

1.55 ./Link.Foundation.Links.Notation.Tests/SingleLineParserTests.cs

```
using System;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
```

```
{
    public static class SingleLineParserTests
    {
        [Fact]
        public static void SingleLinkTest()
        {
            var source = @"(address: source target)";
            var parser = new Parser();
            var links = parser.Parse(source);
            var target = links.Format();
            Assert.Equal(source, target);
        }

        [Fact]
        public static void TripletSingleLinkTest()
        {
            var source = @"(papa has car)";
            var parser = new Parser();
            var links = parser.Parse(source);
            var target = links.Format();
            Assert.Equal(source, target);
        }

        [Fact]
        public static void BugTest1()
        {
            var source = @"(ignore conan-center-index repository)";
            var links = new Parser().Parse(source);
            var target = links.Format();
            Assert.Equal(source, target);
        }

        [Fact]
        public static void QuotedReferencesTest()
        {
            var source = @"(a: 'b' ""c"")";
            var target = @"(a: b c)";
            var parser = new Parser();
            var links = parser.Parse(source);
            var formattedLinks = links.Format();
            Assert.Equal(target, formattedLinks);
        }

        [Fact]
        public static void QuotedReferencesWithSpacesTest()
        {
            var source = @"('a a': 'b b' ""c c"")";
            var target = @"('a a': 'b b' 'c c')";
            var parser = new Parser();
            var links = parser.Parse(source);
            var formattedLinks = links.Format();
            Assert.Equal(target, formattedLinks);
        }

        [Fact]
        public static void ParseSimpleReferenceTest()
        {
            var source = "test";
            var parser = new Parser();
            var links = parser.Parse(source);
            Assert.NotNull(links);
            Assert.Single(links);
            // Simple reference creates a singlet link with null Id and one value
            Assert.Null(links[0].Id);
            Assert.NotNull(links[0].Values);
            Assert.Single(links[0].Values!);
            Assert.Equal("test", links[0].Values?[0].Id);
            Assert.Null(links[0].Values?[0].Values);
        }

        [Fact]
        public static void ParseReferenceWithColonAndValuesTest()
        {
            var source = "parent: child1 child2";
            var parser = new Parser();
            var links = parser.Parse(source);
            Assert.NotNull(links);
            Assert.Single(links);
            Assert.Equal("parent", links[0].Id);
            Assert.NotNull(links[0].Values);
```

```
            Assert.Equal(2, links[0].Values!.Count);
            Assert.Equal("child1", links[0].Values![0].Id);
            Assert.Equal("child2", links[0].Values![1].Id);
        }

        [Fact]
        public static void ParseMultilineLinkTest()
        {
            var source = "(parent: child1 child2)";
            var parser = new Parser();
            var links = parser.Parse(source);
            Assert.NotNull(links);
            Assert.Single(links);
            Assert.Equal("parent", links[0].Id);
            Assert.NotNull(links[0].Values);
            Assert.Equal(2, links[0].Values!.Count);
        }

        [Fact]
        public static void ParseValuesOnlyTest()
        {
            var source = ": value1 value2";
            var parser = new Parser();
            // Standalone ':' is now forbidden and should throw an exception
            Assert.Throws<ParseException>(() => parser.Parse(source));
        }

        [Fact]
        public static void SingletLinkTest()
        {
            // Test singlet link
            var input = "(singlet)";
            var result = new Parser().Parse(input);
            Assert.Single(result);
            Assert.Null(result[0].Id);
            Assert.NotNull(result[0].Values);
            Assert.Single(result[0].Values!);
            Assert.Equal("singlet", result[0].Values?[0].Id);
            Assert.Null(result[0].Values?[0].Values);
        }

        [Fact]
        public static void ValueLinkTest()
        {
            // Test value link
            var input = "(value1 value2 value3)";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void QuotedReferencesWithSpecialCharsTest()
        {
            // Test quoted references
            var input = @"(""id with spaces"": ""value with spaces"")";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void SingleQuotedReferencesTest()
        {
            // Test single-quoted references
            var input = "('id': 'value')";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void ParseQuotedReferencesValuesOnlyTest()
        {
            var source = "\"has space\" 'has:colon'";
            var parser = new Parser();
            var links = parser.Parse(source);
            Assert.NotNull(links);
            Assert.Single(links);
            Assert.Null(links[0].Id);
            Assert.NotNull(links[0].Values);
            Assert.Equal(2, links[0].Values!.Count);
            Assert.Equal("has space", links[0].Values![0].Id);
```

```
            Assert.Equal("has:colon", links[0].Values![1].Id);
            var formatted = links.Format();
            Assert.Equal("('has space' 'has:colon')", formatted);
        }

        [Fact]
        public static void NestedLinksTest()
        {
            // Test nested links
            var input = "(outer: (inner: value))";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void HyphenatedIdentifiersTest()
        {
            // Test support for hyphenated identifiers like in BugTest1
            var source = @"(conan-center-index: repository info)";
            var parser = new Parser();
            var links = parser.Parse(source);
            var target = links.Format();
            Assert.Equal(source, target);
        }

        [Fact]
        public static void MultipleWordsInQuotesTest()
        {
            var source = @"(""New York"": city state)";
            var parser = new Parser();
            var links = parser.Parse(source);
            var target = links.Format();
            // Should preserve quotes for multi-word references
            Assert.Contains("New York", target);
        }

        [Fact]
        public static void SpecialCharactersInQuotesTest()
        {
            var input = @"(""key:with:colons"": ""value(with)parens"")";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);

            input = @"('key with spaces': 'value: with special chars')";
            result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void DeeplyNestedTest()
        {
            var input = "(a: (b: (c: (d: (e: value)))))";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void SingleLineWithIdTest()
        {
            // Test single-line link with id
            var input = "id: value1 value2";
            var result = new Parser().Parse(input);
            Assert.NotEmpty(result);
        }

        [Fact]
        public static void SingleLineWithoutIdTest()
        {
            // Test link without id (single-line) - now forbidden
            var input = ": value1 value2";
            Assert.Throws<ParseException>(() => new Parser().Parse(input));
        }

        [Fact]
        public static void MultilineWithoutIdTest()
        {
            // Test link without id (multi-line) - now forbidden
            var input = "(: value1 value2)";
            Assert.Throws<ParseException>(() => new Parser().Parse(input));
        }

```

```
246         [Fact]
247         public static void SimpleRefTest()
248         {
249             var input = "simple ref";
250             var result = new Parser().Parse(input);
251             Assert.NotEmpty(result);
252         }
253 
254         [Fact]
255         public static void MultiLineLinkWithIdTest()
256         {
257             var input = "(id: value1 value2)";
258             var result = new Parser().Parse(input);
259             Assert.NotEmpty(result);
260         }
261 
262         [Fact]
263         public static void LinkWithoutIdMultiLineTest()
264         {
265             var input = "(: value1 value2)";
266             Assert.Throws<ParseException>(() => new Parser().Parse(input));
267         }
268 
269         [Fact]
270         public static void SimpleReferenceParserTest()
271         {
272             var input = "hello";
273             var result = new Parser().Parse(input);
274             Assert.Single(result);
275             Assert.Null(result[0].Id);
276             Assert.NotNull(result[0].Values);
277             Assert.Single(result[0].Values!);
278             Assert.Equal("hello", result[0].Values?[0].Id);
279         }
280 
281         [Fact]
282         public static void QuotedReferenceParserTest()
283         {
284             var input = "\"hello world\"";
285             var result = new Parser().Parse(input);
286             Assert.Single(result);
287             Assert.Null(result[0].Id);
288             Assert.NotNull(result[0].Values);
289             Assert.Single(result[0].Values!);
290             Assert.Equal("hello world", result[0].Values?[0].Id);
291         }
292 
293         [Fact]
294         public static void SingletLinkParserTest()
295         {
296             var input = "(singlet)";
297             var result = new Parser().Parse(input);
298             Assert.Single(result);
299             Assert.Null(result[0].Id);
300             Assert.NotNull(result[0].Values);
301             Assert.Single(result[0].Values!);
302             Assert.Equal("singlet", result[0].Values?[0].Id);
303             Assert.Null(result[0].Values?[0].Values);
304         }
305 
306         [Fact]
307         public static void ValueLinkParserTest()
308         {
309             var input = "(a b c)";
310             var result = new Parser().Parse(input);
311             Assert.Single(result);
312             Assert.Null(result[0].Id);
313             Assert.Equal(3, result[0].Values?.Count);
314         }
315 
316         [Fact]
317         public static void ParseQuotedReferencesTest()
318         {
319             var input = "(\"quoted id\": \"value with spaces\")";
320             var parser = new Parser();
321             var result = parser.Parse(input);
322 
323             Assert.NotEmpty(result);
324             var formatted = result.Format();
325             Assert.Contains("quoted id", formatted);
```

```
        Assert.Contains("value with spaces", formatted);
    }

    [Fact]
    public static void LinkWithoutIdSingleLineTest()
    {
        // Test that standalone ':' is forbidden and should throw
        var input = ": value1 value2";
        var parser = new Parser();

        // C# parser forbids this syntax (like JS/Rust)
        Assert.Throws<ParseException>(() => parser.Parse(input));
    }

    [Fact]
    public static void SingleLineLinkWithIdTest()
    {
        var input = "id: value1 value2";
        var parser = new Parser();
        var result = parser.Parse(input);

        Assert.NotEmpty(result);
        Assert.NotNull(result[0].Id);
        Assert.Equal("id", result[0].Id);
    }

    [Fact]
    public static void LinkWithIdTest()
    {
        var input = "(id: a b c)";
        var parser = new Parser();
        var result = parser.Parse(input);

        Assert.Single(result);
        Assert.Equal("id", result[0].Id);
        Assert.Equal(3, result[0].Values?.Count);
    }

    [Fact]
    public static void QuotedReferenceTest()
    {
        var input = "\"quoted value\"";
        var parser = new Parser();
        var result = parser.Parse(input);

        Assert.NotEmpty(result);
        Assert.NotNull(result[0].Values);
        Assert.Single(result[0].Values!);
        Assert.Equal("quoted value", result[0].Values![0].Id);
    }

    [Fact]
    public static void SimpleReferenceTest()
    {
        var input = "simplereference";
        var parser = new Parser();
        var result = parser.Parse(input);

        Assert.NotEmpty(result);
    }

    [Fact]
    public static void SingleLineLinkTest()
    {
        var input = "myid: value1 value2";
        var parser = new Parser();
        var result = parser.Parse(input);

        Assert.Single(result);
        Assert.Equal("myid", result[0].Id);
        Assert.Equal(2, result[0].Values?.Count);
    }

    [Fact]
    public static void ValuesOnlyInParenthesesTest()
    {
        var input = "(value1 value2)";
        var parser = new Parser();
        var result = parser.Parse(input);

        Assert.Single(result);
```

```
            Assert.Null(result[0].Id);
            Assert.Equal(2, result[0].Values?.Count);
        }

        [Fact]
        public static void ParseValuesOnlyStandaloneColonTest()
        {
            // Test that standalone ':' is forbidden and should throw
            var input = ": value1 value2";
            var parser = new Parser();

            // C# parser forbids this syntax (like JS/Rust)
            Assert.Throws<ParseException>(() => parser.Parse(input));
        }

        [Fact]
        public static void QuotedReferencesWithSpacesInLinkTest()
        {
            var input = "(id: \"value with spaces\")";
            var parser = new Parser();
            var result = parser.Parse(input);

            Assert.Single(result);
            Assert.Equal("id", result[0].Id);
            Assert.Single(result[0].Values!);
            Assert.Equal("value with spaces", result[0].Values?[0].Id);
        }
    }
}
```

1.56 ./Link.Foundation.Links.Notation.Tests/StreamParserTests.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Threading.Tasks;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    public static class StreamParserTests
    {
        private const string Document =
            "first loves data\n" +
            "# streamed comment\n" +
            "profile:\n" +
            "  name Ada\n" +
            "  note \"line one\nline two\"\n" +
            "(nested:\n  child value)\n" +
            "last sees first";

        private static string[] Render(IEnumerable<Link<string>> links) =>
            links.Select(link => link.ToString()).ToArray();

        [Fact]
        public static void MatchesCanonicalParserOneSymbolAtATime()
        {
            var stream = new StreamParser();
            foreach (var character in Document)
            {
                stream.Write(character.ToString());
            }

            Assert.Equal(Render(new Parser().Parse(Document)), Render(stream.Finish()));
        }

        [Fact]
        public static void EmitsOnlyCompleteRecords()
        {
            var seen = new List<Link<string>>();
            var stream = new StreamParser();
            stream.LinkParsed += seen.Add;

            Assert.Empty(stream.Write("profile:\n  name Ada\n  note \"first\nsecond\"\n"));
            var emitted = stream.Write("n");

            Assert.Single(emitted);
            Assert.Equal(emitted, seen);
        }

        [Fact]
        public static void SupportsLineChunksFinalRecordAndPosition()
```

```
        {
            var stream = new StreamParser();
            foreach (var line in Document.SplitAfter('\n'))
            {
                stream.Write(line);
            }

            Assert.Equal(Render(new Parser().Parse(Document)), Render(stream.Finish()));
            Assert.Equal(Document.Length, stream.Position.Offset);
            Assert.Equal(0, stream.Position.Buffered);
        }

        [Fact]
        public static void SupportsDrainResetAndBoundedMemory()
        {
            var seen = new List<Link<string>>();
            var stream = new StreamParser { Collect = false, MaxBufferSize = 8 };
            stream.LinkParsed += seen.Add;
            for (var index = 0; index < 100; index++) stream.Write("a\n");

            Assert.Empty(stream.Drain());
            Assert.Single(stream.Finish());
            Assert.Equal(100, seen.Count);

            stream.Reset();
            Assert.Throws<ArgumentException>(() => { stream.Write("123456789"); });
        }

        [Fact]
        public static void ProvidesLazyAdapters()
        {
            var actual = StreamParser.ParseChunks(new[] { "one link\n", "two link" });

            Assert.Equal(new[] { "(one link)", "(two link)" }, Render(actual));
        }

        [Fact]
        public static void RejectsWritesAfterFinish()
        {
            var stream = new StreamParser();
            stream.Finish("one");

            Assert.Throws<InvalidOperationException>(() => stream.Write("two"));
        }

        [Fact]
        public static async Task ProvidesAnAsyncEnumerableAdapter()
        {
            static async IAsyncEnumerable<string> Chunks()
            {
                yield return "one link\n";
                await Task.Yield();
                yield return "two link";
            }

            var actual = new List<Link<string>>();
            await foreach (var link in StreamParser.ParseChunksAsync(
                Chunks(),
                cancellationToken: TestContext.Current.CancellationToken)) actual.Add(link);

            Assert.Equal(new[] { "(one link)", "(two link)" }, Render(actual));
        }

        [Fact]
        public static void WaitsForTheLineThatClosesAnEvenRunOfQuotes()
        {
            // `""` is the empty reference only when nothing closes it, and here the next
            // line does, so the first line cannot be emitted on its own (issue #377).
            const string input = "a \"\" b\n\"\" c\n";
            var stream = new StreamParser();
            foreach (var character in input)
            {
                Assert.Empty(stream.Write(character.ToString()));
            }

            Assert.Equal(Render(new Parser().Parse(input)), Render(stream.Finish()));
        }
    }
}

internal static class StreamingTestStringExtensions
```

```
{
    public static IEnumerable<string> SplitAfter(this string value, char delimiter)
    {
        var start = 0;
        for (var index = 0; index < value.Length; index++)
        {
            if (value[index] != delimiter) continue;
            yield return value.Substring(start, index - start + 1);
            start = index + 1;
        }
        if (start < value.Length) yield return value.Substring(start);
    }
}
```

1.57 ./Link.Foundation.Links.Notation.Tests/SyntaxMetadataTests.cs

```
using System.Linq;
using Xunit;

namespace Link.Foundation.Links.Notation.Tests
{
    /// <summary>
    /// Parsed links record how their IDs were written (issue #364), without changing
    /// equality, hashing or formatting.
    /// </summary>
    public static class SyntaxMetadataTests
    {
        [Fact]
        public static void BareAndQuotedReferencesAreEqualButKeepHowTheyWereWritten()
        {
            var link = new Parser().Parse("(t 1 '1')").Single();
            var bare = link.Values![1];
            var quoted = link.Values[2];
            Assert.Equal("1", bare.Id);
            Assert.Equal("1", quoted.Id);
            Assert.Equal("1", bare.RawId);
            Assert.Equal("'1'", quoted.RawId);
            Assert.Null(bare.Quote);
            Assert.Equal('\'', quoted.Quote);
            // Equality, hashing and formatting ignore the metadata.
            Assert.Equal(bare, quoted);
            Assert.True(bare == quoted);
            Assert.Equal(bare.GetHashCode(), quoted.GetHashCode());
            Assert.Equal("(t 1 1)", link.ToString());
            Assert.Equal("(t 1 1)", new[] { link }.Format());
        }

        [Fact]
        public static void RecordsEveryQuoteKindAndRunLength()
        {
            var link = new Parser().Parse("(\"a\": 'b' `c` \"\"d\"\" ~1{65} f)").Single();
            Assert.Equal("\"a\"", link.RawId);
            Assert.Equal('"', link.Quote);
            Assert.Equal(
                new (string?, char?)[] { ("'b'", '\''), ("`c`", '`'), ("\"\"d\"\"", '"'),
                 ↪  ("~1{65}", null), ("f", null) },
                link.Values!.Select(value => (value.RawId, value.Quote)).ToArray());
        }

        [Fact]
        public static void KeepsRawTextFromTheSourceNotFromCommentStripping()
        {
            var link = new Parser().Parse("(a 'b # c') # d").Single();
            Assert.Equal("'b # c'", link.Values![1].RawId);
        }

        [Fact]
        public static void LeavesLinksWithoutWrittenIdsWithoutRawText()
        {
            var link = new Parser().Parse("(a b)").Single();
            Assert.Null(link.RawId);
            Assert.Null(link.Quote);
            Assert.Null(new Link<string>("x").RawId);
            Assert.Null(new Link<string>("x").Quote);
        }

        [Fact]
        public static void IndentedPathCombinationsHaveNoRawText()
        {
            // "a" and "(a) (b)": the second link is made by combining the two lines.
            var links = new Parser().Parse("a\n  b");
```

```
            Assert.Null(links[0].RawId);
            Assert.Equal("a", links[0].Values![0].RawId);
            Assert.Null(links[1].RawId);
            Assert.Null(links[1].Values![0].RawId);
            Assert.Equal("a", links[1].Values![0].Values![0].RawId);
            Assert.Equal("b", links[1].Values![1].Values![0].RawId);
        }

        [Fact]
        public static void DoesNotChangeEqualityOrFormatting()
        {
            var quoted = new Parser().Parse("('a': 'b')");
            var bare = new Parser().Parse("(a: b)");
            Assert.Equal(bare, quoted);
            Assert.Equal(bare.Format(), quoted.Format());
            Assert.Equal(bare[0].ToString(), quoted[0].ToString());
            Assert.Equal(bare[0].GetHashCode(), quoted[0].GetHashCode());
        }

        [Fact]
        public static void AstModeDropsTheMetadata()
        {
            var parser = new Parser { Syntax = false };
            var link = parser.Parse("('a': 'b')").Single();
            Assert.Null(link.RawId);
            Assert.Null(link.Values![0].RawId);
            Assert.Null(link.Values[0].Quote);
            Assert.Equal(new Parser().Parse("('a': 'b')").Single(), link);
            Assert.True(new Parser().Syntax);
            Assert.False(parser.Syntax);
            Assert.Null(new Parser(false) { Syntax = false }.Parse("'root':\n
            ↪  'child'").Single().RawId);
        }

        [Fact]
        public static void IndentedIdsAndStreamParserKeepTheMetadata()
        {
            var link = new Parser().Parse("'root':\n  \"child\"").Single();
            Assert.Equal("'root'", link.RawId);
            Assert.Equal("\"child\"", link.Values![0].RawId);
            var stream = new StreamParser();
            stream.Write("(a '");
            stream.Write("b')");
            var streamed = stream.Finish().Single();
            Assert.Equal("'b'", streamed.Values![1].RawId);
            Assert.Equal('\'', streamed.Values[1].Quote);
        }

        [Fact]
        public static void RawIdCanBeSetOnLinksBuiltInCode()
        {
            var link = new Link<string>("a") { RawId = "\"a\"" };
            Assert.Equal('"', link.Quote);
            Assert.Equal(new Link<string>("a"), link);
            var bare = link.WithRawId(null);
            Assert.Null(bare.Quote);
            Assert.Equal("\"a\"", link.RawId);
            Assert.Equal("`a`", bare.WithRawId("`a`").RawId);
        }

        [Fact]
        public static void ReferenceQuoteReadsTheDelimiterOfOneCompleteReference()
        {
            Assert.Equal('\'', Link<string>.ReferenceQuote("'1'"));
            Assert.Equal('"', Link<string>.ReferenceQuote("\"\"a b\"\""));
            Assert.Equal('`', Link<string>.ReferenceQuote("`x`"));
            Assert.Null(Link<string>.ReferenceQuote("1"));
            Assert.Null(Link<string>.ReferenceQuote("~1{31}"));
            Assert.Null(Link<string>.ReferenceQuote("'unclosed"));
            Assert.Null(Link<string>.ReferenceQuote("'a' b"));
            Assert.Null(Link<string>.ReferenceQuote(""));
            Assert.Null(Link<string>.ReferenceQuote(null));
        }
    }
}
```

1.58 ./Link.Foundation.Links.Notation.Tests/TupleTests.cs

```
using System.Collections.Generic;
using Xunit;
using LinkType = Link.Foundation.Links.Notation.Link<string>;
```

```
using id = Link.Foundation.Links.Notation.id<string>;
using   = Link.Foundation.Links.Notation. <string>;

namespace Link.Foundation.Links.Notation.Tests
{
    public class TupleTests
    {
        [Fact]
        public void TupleToLinkTest()
        {
            var source = @"(papa (lovesMama: loves mama))
(son lovesMama)
(daughter lovesMama)
(all (love mama))";
            var parser = new Parser();
            var links = parser.Parse(source);
            var targetFromString = links.Format();

            IList<LinkType> constructedLinks = new List<LinkType>()
            {
                ("papa", ( )("lovesMama", "loves", "mama")),
                ("son", "lovesMama"),
                ("daughter", "lovesMama"),
                ("all", ("love", "mama")),
            };
            var targetFromTuples = constructedLinks.Format();
            Assert.Equal(targetFromString, targetFromTuples);
        }

        [Fact]
        public void NamedTupleToLinkTest()
        {
            var source = @"(papa (lovesMama: loves mama))
(son lovesMama)
(daughter lovesMama)
(all (love mama))";
            var parser = new Parser();
            var links = parser.Parse(source);
            var targetFromString = links.Format();

            IList<LinkType> constructedLinks = new List<LinkType>()
            {
                ("papa", ((id)"lovesMama", "loves", "mama")),
                ("son", "lovesMama"),
                ("daughter", "lovesMama"),
                ("all", ("love", "mama")),
            };
            var targetFromTuples = constructedLinks.Format();
            Assert.Equal(targetFromString, targetFromTuples);
        }
    }
}
```

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